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INGENIEURSWESE

M&M Skripsie Topics

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1 Mechanical and/or Mechatronic Topics at MM

1.1 Design and Mechatronics Division

Design, build and test of a PV support structure vibration exciter

Reference	:	JB1
Supervisor	:	Mr Johann Bredell
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
In the structural design of utility-scale PV support structures, it is important to know the modal frequency and damping properties. These characteristics are used in wind studies to determine wind load coefficients and evaluate aeroelastic stability. It is required to design a portable dynamic exciter that can be attached to the structure and dynamically excite the structure at a range of frequencies. The measured response can be used to determine dynamic properties. The exciter must be powered from a 12 V power supply, like a battery or a car. A control interface must be provided to select a single frequency or perform a frequency sweep. (Mechanical or Mechatronic)

Key modules (Scientific and Engineering Knowledge):
Electrical Drive Systems (324) | Machine Design A (314) | Vibration and Noise (354)

Solar PV single-axis tracker truss structure

Reference	:	JB2
Supervisor	:	Mr Johann Bredell
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Finite Element Methods (FEM)

Description:
Solar PV single-axis trackers provide increased energy yield compared to fixed-tilt systems by actively tracking the sun throughout the day. A drawback of trackers is their sensitivity to dynamic wind excitation. This is due to the low stiffness of the main supporting member, the torque tube. Torque tubes are typically tubular, square or polygonal sections. This study requires the comprehensive engineering and feasibility study of an alternative tracker design that replaces the tubular section with a high-stiffness truss structure. The truss layout, members and connections must be designed. (Mechanical)

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Machine Design B (344) | Strength of Materials W (334)

Design of a re-deployable tensile structure for solar PV

Reference	:	JB3
Supervisor	:	Mr Johann Bredell
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
The design of a re-deployable, semi-permanent fixed-tilt structure for supporting PV modules is required. Current practice relies on steel frames for such structures. Alternative concepts based on tensile structural principles, including steel wire rope and/or inflatable systems, must be investigated. In addition, methods of non-permanent anchoring are to be evaluated. Prototype structures are to be fabricated to demonstrate functionality. (Mechanical)

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Machine Design B (344) | Strength of Materials W (334)

Software tool for slew drive sizing in PV single-axis trackers

Reference	:	JB4
Supervisor	:	Mr Johann Bredell
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Single-axis tracker (SAT) structures are widely used in utility-scale solar PV plants to improve energy yield relative to fixed-tilt systems. However, SATs are susceptible to aeroelastic instabilities and must move to a safe stow position to prevent divergent dynamic response once the wind speed exceeds a certain threshold. EPC companies require a software tool for the preliminary sizing of the high-cost slew drives and DC motors. The tool must account for both normal tracking operation over the design life and emergency stow scenarios, where the tracker must stow while exposed to wind loads. A slew drive may actuate a single tracker row or two rows via a mechanical linkage. An industry partner will provide representative input parameters. In addition to the report, a concise user manual which explains assumptions and calculation procedures must be written. This project involves developing and demonstrating such a tool.

Key modules (Scientific and Engineering Knowledge):
Electrical Drive Systems (324) | Machine Design A (314) | Modelling (334)

Voice activated motorcycle indicators and observing blind spots

Reference	:	CC1
Supervisor	:	Prof Corne Coetzee
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Different to motor cars, the indicator lights on a motorcycle must be cancelled (switched off) by the rider. Due to counter steering and small corrections to the handlebar to maintain balance, tracking the direction of turn by monitoring the handlebar is not successful to cancel the indicators.
The aim of this project is to design a voice activated indicator system which is mounted to the rider's helmet and interfacing with bike mounted hardware wirelessly. Speaking into a microphone, the indicators should be activated. To deactivate it, various options should be investigated. The rider could either deactivate it via a voice command, or it could be automatically timed out, an audio reminder can be played in the rider's ear, or a head-up display can be shown as a reminder to deactivate it. These and any other possible options should be implemented and tested for ease of use, efficiency, safety, etc. Appropriate filters should be used to filter out wind noise. This can be tested in the wind tunnel to simulate the flow of air over a helmet. Additionally, a camera should be mounted on the helmet to observe the rider's blind spots for other road users. When the rider activates the indicators, the blind spot should be automatically checked for any other vehicle which might be too close to safely change lane or overtake a vehicle. A warning, either audio or a heads-up display, should be given to the rider if an unsafe condition is detected. Video / image processing should be done to effectively identify any other road vehicles and to estimate their distance.
This project is ideal for student interested in hardware, software programming, image/video processing and ideally has a motorbike license (this is not compulsory).

Key modules (Scientific and Engineering Knowledge):
Electronics (245) | Electronics (315) | Mechatronics (424)

An automated draw-down and shear box test

Reference	:	CC2
Supervisor	:	Prof Corne Coetzee
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The Granular Materials Research Group (GMRG) focusses on the numerical modelling of granular materials with applications in the mining and agricultural sectors. Examples include the storage, transportation, and processing of all bulk granular materials: flow on a conveyor belt; flow and storage from silos, hoppers, and bins; the interaction between tillage tools (ploughs) and the soil, horticultural produce (fruit and vegetables) and minimising the damage due to bruising.
For this purpose, the Discrete Element Method (DEM) is used as a design tool to model and simulate the material and processes. However, accurate models require accurate input parameters such as material properties. GMRG specialises in designing and testing of new equipment used for material calibration. One such experiment is called the draw-down test where granular material flows from an upper compartment to a lower compartment through a trapdoor. In the shear test, a box is filled with material and one wall then removed, allowing the material to flow out of the box.
The experiments need to be repeated a large number of times to get representative statistical data. Also, videos and images are used to measure the material flow. In this project, a new draw-down and shear box test setup should be designed that allows for fast and easy testing, with automated trapdoors/walls. Material should be easily loaded into the compartments and discharged materials should be collected. Video images should be taken and then used together with particle image velocimetry (PIV) to analyse the material flow.

Key modules (Scientific and Engineering Knowledge):
Machine Design B (344) | Mechatronics (424) | Experimental Techniques (414)

Granular Column Collapse: Experiments and Simulation

Reference	:	CC3
Supervisor	:	Prof Corne Coetzee
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
A column collapse of granular material is closely related to slope failure and avalanches. In this project, a simple experiment must be designed to measure and observe the collapse of a column of granular material. Materials can include sand, grains, and any other granules or pellets. A vertical column of material must be kept in position by walls, the walls must then be quickly removed to allow the column of material to slump down under the action of gravity.
The collapse of the column occurs over a few seconds, and high-speed video cameras must be used to capture the material flow and the maximum runout. Particle image velocimetry (PIV) must then be used to further analyse the flow based on the high-speed images.
Discrete Element Method (DEM) is a numerical method where the individual particles are modelled. DEM must be used to model the collapse of the column, and the results must be compared to the experimental measurements.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Machine Design B (344) | Experimental Techniques (414)

Automatic Measurement of Chilling Injury in Citrus Fruit

Reference	:	CC4
Supervisor	:	Prof Corne Coetzee
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Citrus, like most other fruit, is refrigerated during export transportation. Citrus chilling injury (CI) is a physiological disorder of the rind (skin) that occurs when the fruit is exposed to prolonged cold temperatures below its tolerance threshold. It manifests as symptoms like pitting and discolouration, often becoming more severe after the fruit is removed from cold storage. Certain varieties, such as some grapefruit and lemons, are particularly susceptible.
The university is actively researching chilling injury and performing laboratory trials to identify the factors that influence chilling injury. A large number of fruit is tested, and currently each fruit must be manually checked for marks and discolouration on its skin. This is a tedious and subjective process.
For this project, an automated system must be designed. This will include cameras to capture the whole outside surface of the fruit, followed by image processing to automatically identify the spots of colouration. A measure of injury or discolouration must then be calculated based on the area of colouration as a ratio of the total fruit surface area. Secondly, a chute or conveyor system must be designed to effectively measure a large number of fruit (>100) with minimal human input.
Students will work closely with industry (Citrus Research International), and there is the opportunity for further post-graduate studies in the exciting field of agricultural engineering.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Mechatronics (424) | Experimental Techniques (414)

Abrasive damage to citrus fruit

Reference	:	CC5
Supervisor	:	Prof Corne Coetzee
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Citrus fruit is exposed to abrasive damage during harvest, transportation, and on the packing line. The damage can lead to discolouration of the rind (skin), scarring, and increased susceptibility to fungal infections. To better understand how the damage can influence the fruit, a laboratory experiment must be designed to subject fruit to various levels of abrasive damage. The damage must be representative of what the fruit will experience in practice, as observed and measured on an industrial packaging line. A series of experiments must then be conducted, subjecting fruit to various levels of abrasive damage, and the effects on the fruit must be quantified. This project is ideal for students interested in mechanical design and experimental techniques.

Key modules (Scientific and Engineering Knowledge):
Machine Design B (344) | Strength of Materials W (334) | Experimental Techniques (414)

Small wind tunnels for testing the effects of coolling on individual fruit

Reference	:	CC6
Supervisor	:	Prof Corne Coetzee
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
Citrus, like most other fruit, is refrigerated during export transportation. Citrus chilling injury (CI) is a physiological disorder of the rind (skin) that occurs when the fruit is exposed to prolonged cold temperatures below its tolerance threshold. It manifests as symptoms like pitting and discolouration, often becoming more severe after the fruit is removed from cold storage. Certain varieties, such as some grapefruit and lemons, are particularly susceptible.
The university is actively researching chilling injury and performing laboratory trials to identify the factors that influence chilling injury. Two of these factors include temperature and humidity. For this purpose, an environmental chamber which accurately controls the temperature and humidity is available.
For this project, a small windtunnel must be designed which can be placed inside the 1 m x 1 m environmental chamber. One fruit will be placed inside the tunnel, and the airflow across it should be accurately controlled. With a higher flow rate, it is expected that the fruit will experience a higher cooling rate, for example. A series of tests must then be conducted, varying the temperature and humidity in the environmental chamber, and varying the air flow rate across the fruit inside the wind tunnel. The effect of these variables on CI must then be identified.
Students will work closely with industry (Citrus Research International), and there is the opportunity for further post-graduate studies in the exciting field of agricultural engineering.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344)

Dynamic monitoring of railway pantograph-contact wire interface

Reference	:	PC1
Supervisor	:	Mr Pieter Conradie
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
In electric trains, power is transmitted from the overhead contact wire to the train through a pantograph. This pantograph-contact wire interface is critical and a deeper understanding of the dynamic behaviour is required to understand panto-bounce, accelerated wear, pantograph damage or even pantograph hookups.
With a growing trend toward equipping trains with advanced technology for in-service inspection, this project aims to develop a roof-mounted onboard system to monitor the pantograph-contact wire interface. The system will be tested on the X'Trapolis Mega trainset in the Cape Town maintenance depot. The proposed approach includes: - Installing cameras, accelerometers and GPS sensors on the roof and pantograph. - Detecting misalignment, irregularities, panto-bounce, and other contact wire defects. - Correlating these measurements with data from the train's onboard CFM log.
This system will provide real-time insights to enhance maintenance strategies and validate CFM data. The installation will be designed to withstand environmental conditions, electromagnetic interference and train vibrations.
The Gibela Engineering Research Chair, based in the Department of Mechanical and Mechatronic Engineering, will support the project by facilitating access to Gibela depots and trains and providing funding for prototyping. Students involved will benefit from being part of a collaborative team, gaining industry exposure, participating in additional development opportunities, and attending colloquia.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Control Systems (354) | Vibration and Noise (354)

Vision-Assisted Railway Inspection System on HO Model Track

Reference	:	PC2
Supervisor	:	Mr Pieter Conradie
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
To ensure the success of railway operations, it is essential to detect track defects accurately and non-intrusively. A modern approach is to equip scheduled passenger trains with advanced technology to inspect tracks during normal operation.
This project aims to simulate real-world track defects using an existing HO model train set at the Faculty of Engineering. The model train is equipped with a camera for fault and obstacle detection, linked to an onboard microcontroller for data processing. Building on previous research, the system offers several opportunities for improvement, such as enhancing video quality and optimising data velocity, reducing sensitivity to train speed, integrating a dedicated lightning system, improving detection accuracy, onboard processing and real-time streaming.
The Gibela Engineering Research Chair, based in the Department of Mechanical and Mechatronic Engineering, will support the project by providing logistical support, facilitating access to Gibela depots and trains and providing funding for prototyping. Students involved will benefit from being part of a collaborative team, gaining industry exposure, participating in additional development opportunities, and attending colloquia.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Design (E) (314) | Electronics (245)

Design of a PedalRail bicycle adaptor for track inspections

Reference	:	PC3
Supervisor	:	Mr Pieter Conradie
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Maintenance and inspection teams in the railway industry are transported to their work sites using motorised vehicles. However, once on site they still require a quick, flexible, and safe method to move short distances between work areas.
This project involves the development of a PedalRail adaptor system that enables a standard bicycle to operate safely on South African narrow gauge railway tracks. The initial phase will develop a single bicycle attachment, with the second phase extending the design to support a two-bicycle coupled configuration, maintaining full bicycle functionality.
The student must address key engineering challenges including wheel guidance on the rail, lateral stability on uneven track, tolerance compensation, clearance over rail components, torque balancing between bicycles, and load distribution for traction and control. Additional considerations include user safety, stability, ergonomics, modularity and the ability to mount lightweight inspection tools or sensors.
The Gibela Engineering Research Chair, based in the Department of Mechanical and Mechatronic Engineering, will support the project by facilitating access to rail track and providing funding for prototyping. Students involved will benefit from being part of a collaborative team, gaining industry exposure, participating in additional development opportunities, and attending colloquia.

Key modules (Scientific and Engineering Knowledge):
Machine Design B (344) | Experimental Techniques (414)

Smart Detection of rail infrastructure hazards with train-mounted sensors

Reference	:	PC4
Supervisor	:	Mr Pieter Conradie
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Railway safety and infrastructure maintenance are essential for ensuring reliable train operations. Traditional inspection methods are often manual, time-consuming, and may miss temporary of unexpected hazards. With advances in computer vision, it is now possible to automatically detect infrastructure issues and unsafe situations using video feeds from train-mounted cameras.
This project will utilise existing video feed from the X'Trapolis Mega to identify infrastructure anomalies and detect unsafe scenarios near the track (like people, vehicles or animals approaching). One focus will be on training an object detection model (e.g., YOLO) as effectively as possible using a combination of real and synthetic data. Additionally, the quality and capability of the current camera feed will be benchmarked against commercially available cameras, and the potential for data fusion with other technologies (like LiDAR) will be explored.
The Gibela Engineering Research Chair, based in the Department of Mechanical and Mechatronic Engineering, will support the project by facilitating access to trainsets and rail depots and providing funding for prototyping. Students involved will benefit from being part of a collaborative team, gaining industry exposure, participating in additional development opportunities, and attending colloquia.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Data Analytics (Eng) (344)

Automated High-Temperature Instrumented Indentation System

Reference	: LC1
Supervisor	: Mr Llewellyn Cupido
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
This project proposes the design, development, and validation of an automated, high-temperature, instrumented indentation system capable of measuring the hardness and elastic modulus of materials at temperatures up to 500°C. Unlike traditional microhardness testing that relies solely on optical measurement of indentation size, this system will employ the Oliver-Pharr method. The method extracts mechanical properties from continuous load-displacement curves recorded during indentation.
The Oliver-Pharr method represents a significant advancement over conventional hardness testing. By precisely measuring both the applied load and indenter displacement throughout the loading and unloading cycle, the technique enables determination of elastic modulus from the unloading curve stiffness without requiring optical measurement of the residual impression. This approach is particularly valuable for high-temperature testing where optical access may be limited and where both hardness and modulus changes are critical indicators of phase transformations.
Instrumented indentation offers fundamental advantages over traditional mechanical testing approaches: - Localised measurement: Small indentation volumes minimise thermal gradient effects and eliminate system-wide thermal stress problems - Simplified structural requirements: Static indentation eliminates the need for complex load frames; thermal expansion of the support structure does not affect measurement - Dual property measurement: Simultaneous hardness and modulus determination provides comprehensive characterisation - Established high-temperature capability: Commercial high-temperature nanoindentation systems demonstrate feasibility up to 800°C

Key modules (Scientific and Engineering Knowledge):
Electrical Drive Systems (324) | Electronics (245) | Machine Design A (314)

Inert Gas Chamber Development for High-Temperature Materials Testing

Reference	:	LC2
Supervisor	:	Mr Llewellyn Cupido
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Finite Element Methods (FEM)

Description:
This project involves designing and analysing an inert-gas-controlled-atmosphere chamber for high-temperature materials testing. The chamber will enable micro-hardness testing and other materials characterisation at elevated temperatures (up to 1000°C) while protecting specimens from oxidation through an argon atmosphere. Building on analytical and numerical foundations established in 2025, this continuation project addresses the complete engineering challenge: thermal management, atmospheric control, structural integrity, and device integration.
Core Engineering Activities: - Thermal-Structural Analysis: Validate and extend computational models. Investigate thermal fields, structural deflections, and innovative thermal management approaches, including magnetic field effects on heat transfer. - Atmospheric Control Design: Design complete inert gas handling systems, including supply, purging, flow control, oxygen monitoring, and safety interlocks.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Heat Transfer A (414) | Machine Design A (314)

Advanced Multi-Sensor Precision Temperature Control System

Reference	:	LC3
Supervisor	:	Mr Llewellyn Cupido
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Develop an advanced precision temperature control system that integrates multi-sensor state estimation with coordinated multi-zone heating to achieve improved thermal stability and spatial uniformity for high-temperature material testing.
Key Technical Areas 1. Multi-Sensor State Estimation Design an intelligent sensing system providing spatial temperature awareness: - Sensor placement optimisation - Sensor fusion algorithms - Spatial reconstruction - Uncertainty quantification - Fault detection & isolation
2. Multi-Zone Temperature Control Implement coordinated control of independent heating zones: - Zone architecture - Control strategy - Thermal coupling - Multi-objective optimisation - Real-time implementation

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Electronics (245)

Virtual stress measurement for assessing global fatigue on structures subjected to dynamic loading

Reference	:	GD1
Supervisor	:	Mr Gerhard Durandt
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Finite Element Methods (FEM)

Description:
Large industrial structures under dynamic loading conditions, such as ships or offshore platforms, have limited lifespans due to cyclic stress cycles from wind and waves which cause failure due to fatigue. Digital twin methodologies have the potential to prolong the lifespan of these expensive structures by providing insight into the fatigue state over time. This will enable improved decision making in terms of the operation or maintenance of the structure.
Structural strain is typically measured with strain gauges. However, on large structures the installation of strain measurement devices is labour intensive and does not necessarily represent the global stress state of the structure. An improved and more cost-effective solution is required where the global stress state is determined from alternative sensors by combining measured data and modelling techniques.
The scope of the project is to develop a virtual sensing methodology for the structural response of a laboratory scale specimen and validated the performance of the virtual sensor experimentally. Knowledge of finite element modelling is advantageous.

Key modules (Scientific and Engineering Knowledge):
Finite Element Methods (414) | Modelling (334) | Vibration and Noise (354)

Design of a cost-effective ventilator system for patients that require basic respiratory support

Reference	:	GD2
Supervisor	:	Mr Gerhard Durandt
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Healthcare practitioners may prescribe patients to be treated with ventilator support equipment if they suffer from medical conditions that impede respiratory function. Ventilators are medical devices that assist a patient to breath artificially.
The scope of this project is to design and build a portable and cost-effective ventilator system that is suitable for use in rural areas in South Africa. This entails the design and implementation of the mechanism which pumps oxygen to the patient's lungs, a control system to control breathing rate, and all additional functions required to showcase a working product. The ventilator should adhere to the applicable ISO specifications to order to receive approval from the South African Health Products Regulatory Authority (SAHPRA).

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Design of a braking system with automatic brake wear compensation for an underground mining locomotive

Reference	:	GD3
Supervisor	:	Mr Gerhard Durandt
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Locomotives are critical machines used for underground mining activities to move materials from the mining area to the processing plant where the valuable resources are extracted. The underground mining environment can be dangerous, especially if heavy vehicles are on the move or start moving unintentionally. Braking systems need to ensure locomotives stop effectively or remain stationary if not used. A significant problem in practice is that brake pads wear over time, which reduces the braking capacity of the locomotive. As a result, technicians must conduct regular inspections and manually re-adjust the brake pads to ensure a sufficient braking force is maintained. This is a time-consuming activity that reduces the overall productivity of a mine.
The scope of this project is to design, manufacture a test a small-scale custom braking system for an underground locomotive application. The system must be fail-safe and ensure that a locomotive is stopped in the event of a power interruption. In addition, the braking system must account for brake pad wear and automatically adjust its position to ensure a constant braking force is achieved.

Key modules (Scientific and Engineering Knowledge):
Machine Design B (344) | Mechatronics (424) | Experimental Techniques (414)

Large ACC Fan blade and tip deformation

Reference	:	DNJE1
Supervisor	:	Dr Daniel Els
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The tips of working (rotating) large fans are displaced by the aerodynamic lift and drag forces. Unsteady inlet flow conditions can cause the blade to vibrate. The standard method of measuring the blade deformations and vibrations is to place strain gauges at the root of the blade and the correlating the strain gauge output with static deformation. There is a large uncertainty between calibration the and dynamic displacements, because of effects such as centrifugal stiffening and more complex blade deformations.
The objective of this project is to measure the blade deformations of an ACC fan under operational conditions by optical or any other methods and correlate it with the strain gauge measurements. A simplified dynamic analytical model must be constructed to correlate the experimental values with the theoretical values.
Self study - Advanced vibration dynamics - Image processing and computer vision - Data aquisition systems. - Data analytics and optimization.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Mechatronics (424) | Vibration and Noise (354)

Large ACC Fan rotation velocity measurement

Reference	:	DNJE2
Supervisor	:	Dr Daniel Els
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
There is a need to accurately measure the rotation speed of the large fans in Air Cooled Condensers (ACC) in transient and steady state conditions. This speed data is used for dynamic modelling of the total fan system and analysing start-up and shut down shock loads.
The objective of this project is to measure the fan blades rotational speed under operational conditions to sub rotation accuracy. Develop numerical methods to calculate the rotation speed under transient and steady state conditions. Find the effects of torsional vibrations on the fan speed.
The system needs to be scalable to be employed on other ACC fans.
Selfstudy: - Fast acting optical and magnetic type pulse measurments. - Data aquisition systems. - Data analytics and optimization. - Precision manufacturing

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Electrical Drive Systems (324) | Mechatronics (424)

Tranquilizer dart test bench

Reference	:	DNJE3
Supervisor	:	Dr Daniel Els
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
A tranquilizer dart gun is a non-lethal gun that shoots a ballistic syringe filled with chemicals that sedates (or medicates) animals.
The objective of this study is to develop, build and test a dart gun test bench to safely fire and test tranquilizer darts in a laboratory environment: • The full weapon system with CO2 gas up to 20 bar. • Refine an accurate gas dynamics model to predict the internal ballistics of the gun. • Develop and build an accurate muzzle velocity control system by means of a programmable flow control system. • set an accurate muzzle velocity measurement system.
https://en.wikipedia.org/wiki/Tranquillizer_gun
Selfstudy: • Internal and external ballistics of darts gun system • Pneumatic and CO2 valves and other hardware components • Micro controlers • Data acquisition systems .

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Heat Transfer A (414) | Mechatronics (424)

Small linear direct shear tester for granular material

Reference	:	DNJE4
Supervisor	:	Dr Daniel Els
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
A direct shear tester is used to determine the shear strength of granular materials and the friction forces between the granular material and container walls. The test data are used as calibration test for Discrete Element Method (DEM) to determine micro parameters such as inter particle and particle-wall friction.
• Design and build shear box • Design and build or select linear actuators with control systems. • Design and build force vs displacement measurement system. • Perform a full set of shear tests on a range of granular materials
https://www.geoengineer.org/education/laboratory-testing/direct-shear-test https://en.wikipedia.org/wiki/Discrete_element_method https://www.itascacg.com/software/pfc
Selfstudy: • Granular materials • Custom force and displacement sensor design and built • Custom linear actuators with speed control.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Spring winding machine

Reference	:	DNJE5
Supervisor	:	Dr Daniel Els
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
The M&M workshop has a need for the design and manufacture of small coil springs. The main objectives of this project are:
• Develop a computer program for the design of springs to obtain the required geometry and force-displacement characteristics. • Develop a computer program to calculate the spring manufacturing parameters such as bending diameter, winding pitch, number of turns, wire length, etc. • Design and build a tool set that can be fitted to lathes or milling machines to manufacture the springs. • Develop a test procedure and design and build a test rig to test the springs.
https://www.youtube.com/shorts/fmAKIRShP-I https://www.youtube.com/shorts/XzEaa3TPiHU

Key modules (Scientific and Engineering Knowledge):
Finite Element Methods (414) | Machine Design B (344) | Strength of Materials W (334)

Adaptive Slip Ratio Control for Electric Vehicles Using Real-Time Friction Estimation

Reference	: DNJE6
Supervisor	: Dr Daniel Els
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: 26256347
Direction	: Mechatronic
Required elective	: None

Description:
Conventional traction control systems in EVs rely on fixed torque limits tuned for nominal road conditions, leading to either unnecessary performance reduction or wheel spin when friction (μ) changes, such as from dry to wet asphalt. This results in suboptimal traction, energy inefficiency, and potential instability. While high-end systems use expensive sensors for μ estimation, there is a gap in low-cost, model-based approaches that leverage standard EV signals (wheel speed, acceleration, torque) to adapt slip-ratio control online. Demonstrating this on a simulation-focused platform with optional simple hardware addresses the need for affordable solutions in emerging EV markets.
Objectives: 1. Develop a longitudinal EV model incorporating tire slip and variable friction dynamics 2. Design a real-time μ-estimation algorithm using Recursive Least Squares (RLS) or Extended Kalman Filter (EKF) 3. Formulate an adaptive slip-ratio controller (e.g., gain-scheduled PID) that updates based on estimated μ 4. Simulate and compare the adaptive controller against a fixed-gain baseline in various road scenarios 5. Optionally implement and test on a single-wheel hardware rig for real-time validation 6. Evaluate performance metrics like slip regulation, robustness to disturbances, and energy efficiency

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Mechatronics (424) | Modelling (334)

Design of voice activated baby pram

Reference	:	LG1
Supervisor	:	Mrs Liora Ginsberg
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The student needs to design a baby pram that will allow the user to jog independently of the pram. The student needs to research various available voice activation / voice recognition techniques as well as the specifications required for baby prams. The student needs to design a jogging baby pram including the physical structure and full movement (forward, backward, turning) as well as full voice recognition / voice activation of the pram, specifically to include the necessary specifications and safety aspects in the design. The design needs to be built and tested. This is a continuation of a previous skripsie topic.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Mechatronics (424)

Design with control of temperature and humidity a neonatal incubator

Reference	:	LG2
Supervisor	:	Mrs Liora Ginsberg
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The student must do research into literature as well as current modern neonatal incubators. The student must design a cost-effective, off-the-grid transportable incubator which allows for precise temperature and humidity control. The student must build and test the incubator to the prescribed standards. This is a continuation of a past skripsie topic.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Design of neonatal nebulizer

Reference	:	LG3
Supervisor	:	Mrs Liora Ginsberg
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The student needs to research current neonatal nebulizer designs and appropriate sensors to detect breathing patterns. The student needs to design a nebulizer suitable for a neonate with an appropriate sensor/s and interfaces to provide a specific dose of surfactant for neonates using appropriate sensors to activate the breathing mechanism. The nebulizer should provide droplet sizes below 10 μm to provide optimum delivery of the drug to the neonate. The physical design of the nebuliser needs to be designed, incorporating the sensor system and interfaces, and must be built and tested (no testing on neonates). This is continuation of past skripsie topics.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Modelling blood and lymph flow using Tesla valves

Reference	:	LG4
Supervisor	:	Mrs Liora Ginsberg
Can be completed in first semester	:	Yes
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
The student need to research blood flow and valves in the blood and lymph systems of the human body, and the modelling and experiments of the blood/lymph flow through valves The student must investigate the feasibility of using Tesla valves for modelling blood/lymph flow through valves. The student needs to validate the results with CFD such as Fluent and experimental work.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Heat Transfer A (414)

Design of water-saving showering device

Reference	:	LG5
Supervisor	:	Mrs Liora Ginsberg
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The student needs to investigate various water saving devices as well as various voice recognition techniques. The student must design, build and test a temperature controlled, water-saving device for a shower, using voice recognition to operate the system.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Design of an electronic monitoring system for a greenhouse

Reference	:	LG6
Supervisor	:	Mrs Liora Ginsberg
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The student needs to research greenhouse monitoring systems, current greenhouse techniques and the use of solar panels. The student needs to design the greenhouse itself, small enough to fit on the engineering building roof. The student needs to design the monitoring system required to record data, control and adjust the greenhouse conditions, which includes temperature, humidity, soil water content etc, in order to provide optimum conditions for plant growth. A cost analysis of the system needs to be done, in comparison with standard greenhouse designs. A prototype for the design needs to be designed, built and the system tested.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Design a retrofitted wheelchair stair assist

Reference	:	LG7
Supervisor	:	Mrs Liora Ginsberg
Can be completed in first semester	:	Yes
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The student needs to research available literature as well as the current options available for wheelchair uses with regards to using stairs. The student needs to design a adaptable device that can be attached to a standard wheelchair that would allow a wheelchair bound person to climb or descend a set of stairs. This can be with or without the aid of an able-bodied helper. The design would need to be built and tested.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Design of a neonatal incubator

Reference	:	LG8
Supervisor	:	Mrs Liora Ginsberg
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
The student must do research into literature as well as current modern neonatal incubators. The student must design a cost-effective, off-the-grid transportable incubator which allows for temperature and humidity optimisation. Heat transfer / CFD on the hood to verify temperature and flow distribution. The student must build and test the incubator to the prescribed standards. This is a continuation of a past skripsie topic.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Heat Transfer A (414) | Numerical Fluid Dynamics (414)

Design of a Declination Drive Sub-System for a Satellite Ground Station (2 students: 1 mechanical, 1 mechatronic))

Reference	:	NM1
Supervisor	:	Prof Nawaz Mahomed
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
A satellite ground station has been successfully built and operated. The SGS uses an AC motor with VSD to position the antenna in the azimuth plane. The control system consists of a Rasp Pi with already-developed Python code for converting satellite position data (TLE) to observer coordinates in terms of azimuth and declination. A GUI running on the Pi can be used to select the satellite to be tracked. The ground station uses a parabolic dish antenna designed to operate at 1.7 GHz. Satellites communicating in this band have already been successfully tracked and packets downloaded (and decoded). Currently, it can only track in the azimuth plane.
The ground station can be deployed in TLE (two-line-element) or signal strength mode.
The aim now is to add a declination axis (subsystem) to the existing ground station. The system will likely comprise a servomotor powered from the existing control box. The software needs to be developed further to allow for the declination positioning. The project will involve development of Python code, design of the declination motor structure, the design of any electronic components for interfacing the declination drive to the Rasp Pi.
This project requires the detailed design and software development for the declination drive. The build part depends on available time beyond the scope of the project.
In addition, the project will investigate the difference in position of the satellite based on the TLE data (which uses a propagator model) and signal strength (which is assumed to be the actual position). This will give a "residual" of the position vector, which will allow further understanding of the perturbations experienced by a satellite (such as atmospheric drag, solar radiation and gravitational fields).
The project entails some learning in the fields of orbital mechanics and RF engineering. The latter will involve calibration tests to be conducted on the antenna to establish radio frequency (RF) parameters, using a spectrum generator and spectrum analyser.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Control Systems (354) | Machine Design A (314)

Meteoritic Fragmentation using Light Curve Analysis (1 student)

Reference	:	NM2
Supervisor	:	Prof Nawaz Mahomed
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
From optical observation images of meteors, the light intensity / luminosity can be measured as a function of time, giving what is called a light curve. A light curve provides important information regarding the fragmentation / breakup and ablation (mass loss due to viscous forces) of a meteoroid. FLARE (Fireball Luminosity & Ablation Reconstruction Engine) was developed in 2025 to convert meteor captured image stacks into light curves, and to identify the type of fragmentation undergone by a meteor. This project aims to further improve the prediction accuracy of FLARE, which will be made available for public use.
The project aims to generate light curves for a number of meteors / fireballs captured through the allsky camera network in the Western Cape, as well as cameras linked to the Global Meteor Network. The light curve analysis (LCA) will be aimed at identifying the type of fragmentation and ablation (based on well-established classifications) experienced by different meteoroids. Using such data, the project will attempt to quantify the size of a meteoroid, based on images from multiple cameras.

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Experimental Techniques (414)

Fragmentation of Heterogeneous Granular Composite Materials under Thermal Shock (2 students)

Reference	: NM3
Supervisor	: Prof Nawaz Mahomed
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
Meteoroids entering the earth's atmosphere experience breakup / fragmentation at very low stagnation pressures. This phenomenon cannot be explained, and scientists currently only use scaling laws or empirical models to model fragmentation during atmospheric flight.
To arrive at a scientifically sound basis for this phenomenon, it is proposed that meteoroids are modelled as composite granular materials, a collection of discrete particles that may be a mixture of different materials or a combination of a granular solid within a matrix. Such materials behave as quasi-brittle materials, and the stress/strain relationship can then be modelled using Coulomb-Mohr Theory. Since atmospheric entry at hyper-velocity (Mach 20 upwards) results in a rapid change in temperature due to viscous drag and radiation, the thermal stress induced needs to be included in the stress/strain model. This project will attempt to model the thermal stress / strain field due to thermal shock, which leads to (rapid) crack propagation in the granular material, hence leading to breakup / fragmentation.
This project is divided into two distinct components. The first involves the experimental reconstruction of artificial meteoritic materials, and using such samples to undertake compression tests coupled to Digital Image Correlation (DIC), to analyse the volumetric strain fields and crack growth. This requires insight into stress/strain relationships and fracture mechanics. This work also includes the use of 3D scanning data on actual meteorite samples.
The second part deals with the modelling and simulation of the stress-strain fields using suitable computational simulation tools. This requires the development of a representative volume element (RVE) using data from the first part of the project.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Strength of Materials W (334) | Thermofluid Dynamics (344)

3D object ranging and orientation technology (2 students: 1 mechanical, 1 mechatronic)

Reference	:	NM4
Supervisor	:	Prof Nawaz Mahomed
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Light detection and ranging methods have rapidly developed over the past decade to precisely measure distances and create 3D images of objects. However, these technologies, such as those used in autonomous vehicles, require high power and onboard processing capacity, and are therefore not suitable for certain applications.
This project aims to develop a low-power technology capable of creating a 3D spectral signature of an object, which can be used to determine the range and orientation of an identified object. The idea is to use the solar radiation reflectivity of metallic materials (such as aluminium) to identify the target body, and to further use the intensity/magnitude/brightness of reflected light to determine orientation. The latter will generate a "light curve" (magnitude vs orientation in 3D) as a spectral signature of the object.
The project will require the design and build of a two degree-of-freedom (rotation along two axes) system that will support and move the object into different orientations with respect to the observer. A light source, emulating that of solar radiation, will be directed at the object. A photodetector at the observer will initially scan the environment and then lock onto the target based on the target material's spectral properties. The photodetector will then measure the magnitude of the reflected light to generate a 3D map of the object's orientation, i.e. the 3D spectral image.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Vibration and Noise (354)

Modelling and Design of a Robotic End-effector to Perform Complex Maintenance Activities in the Manufacturing Industry.

Reference	:	BR1
Supervisor	:	Dr Boitumelo Ramatsetse
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Maintenance plays a crucial role for transport manufacturing industry. Most maintenance activities are still performed manually, thus resulting in safety concerns and inconsistencies due to human errors. Thus, more intelligent, and automated type of maintenance solutions are required to guarantee safety of operations as well as continuous availability of equipment during operations. This project aims to model and simulate a robotic end-effector to perform automated maintenance operations in the transport manufacturing industry. The Computer Aided Design (CAD) will be carried out using CAD design software's under different loading conditions. Furthermore, kinematics models of the designed end-effector will be modelled in MATLAB environment, including the kinematic motion and trajectory of the robotic end-effector along the X-Y and Z coordinates.

Key modules (Scientific and Engineering Knowledge):
Machine Design B (344) | Mechatronics (424) | Modelling (334)

Design of a Reconfigurable Electric Press Brake Machine for Customised Sheet Metal Bending Operations

Reference	:	BR2
Supervisor	:	Dr Boitumelo Ramatsetse
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Sheet metal manufacturing is a fundamental part of modern industry, transforming raw metal into essential components for a wide range of products. Traditional press brake machines often lack the flexibility to adapt to varied and complex bending tasks required in modern manufacturing industries. To address this limitation, this project aims to design a reconfigurable electric press brake machine tailored for customized sheet metal bending operations. The proposed machine will incorporate reconfigurable manufacturing systems (RMS) principles to enable operators to adjust its configuration based on specific bending requirements. This design will use electric actuation system to enhance energy efficiency, precision, and environmental sustainability as compared to hydraulic counterparts. This will include control algorithms are integrated to ensure consistent bending angles, reduced spring back effects, and high repeatability. In addition, this design should be able to achieve quick tooling changes and process optimization, making it suitable for low-volume, high-mix production environments. CAD software will be used to model the machine, MATLAB Simulink will be used to design and simulate the control system. A simulation model of the proposed solution will also be developed to demonstrate its capabilities in bending different sheet metal geometries.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design B (344) | Mechatronics (424)

Design of a Modular Open Architecture Controller for a Reconfigurable Machine Tool

Reference	:	BR3
Supervisor	:	Dr Boitumelo Ramatsetse
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Controllers for dedicated machines, while effective for specific tasks, often face several limitations that hinder their adaptability and performance in modern manufacturing environments. These controllers are not suitable for reconfigurable manufacturing systems (RMS). RMS demands highly adaptable and modular controllers capable of integrating new hardware and software modules without significant reprogramming or downtime. This project aims to design a modular open-architecture controller (MOAC) for a reconfigurable machine tool (RMT) to enable seamless plug-and-play functionality. The proposed MOAC will integrate modular design and open communication protocols to ensure compatibility with various machine tool configurations and components. The controller will be designed with a layered architecture comprising hardware abstraction, communication, and application layers, ensuring scalability and ease of integration. Plug-and-play functionality will be achieved through standardized interfaces, automatic detection of modules, and dynamic reconfiguration capabilities. The controller will be validated through simulations as well as implementation on the existing RMT prototype machine.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (214) | Machine Design B (344) | Mechatronics (424)

Design and Simulation of a Reconfigurable Drilling Machine for Complex Part Geometries

Reference	: BR4
Supervisor	: Dr Boitumelo Ramatsetse
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Traditional drilling machines are often designed for specific operations or part geometries, limiting their adaptability to changing production needs. When product specifications or hole configurations vary, conventional machines require significant retooling or replacement, resulting in increased downtime, higher costs, and reduced productivity. Such dedicated systems are therefore unsuitable for modern manufacturing environments that demand flexibility and rapid responsiveness to design variations.
To address these challenges, this project proposes the design and simulation of a Reconfigurable Drilling Machine capable of adapting to various drilling tasks, workpiece sizes, and hole patterns with minimal reconfiguration time. The proposed system integrates modular components and adjustable tool heads to enable quick structural and functional changes, aligning with the principles of Reconfigurable Manufacturing Systems (RMS).
The design process will involve Computer-Aided Design (CAD) modeling using modern engineering software to simulate and optimize different machine configurations. The reconfigurable drilling machine aims to enhance operational flexibility, reduce setup time, and minimize the need for multiple dedicated drilling units.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Design of a Reconfigurable Sorting and Inspection Machine in Automotive Manufacturing Industries

Reference	: BR5
Supervisor	: Dr Boitumelo Ramatsetse
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
In automotive manufacturing industries, sorting and inspection systems play a critical role in ensuring product quality and consistency. Conventional sorting and inspection machines are often designed for specific component types or inspection parameters, making them inflexible to accommodate part variations or design updates. When new models or parts are introduced, existing systems require complete redesign or replacement, leading to increased downtime, high operational costs, and reduced production efficiency.
To address these limitations, this project proposes the design and simulation of a reconfigurable sorting and inspection machine capable of adapting to different automotive components and quality inspection requirements. The reconfigurable design concept aims to enable rapid adjustment of sensors, conveyors, and inspection modules to handle multiple part families without extensive mechanical or software reconfiguration. This flexibility enhances the system's adaptability to dynamic production environments and supports the automotive industry's demand for modular, scalable, and cost-efficient solutions.
The Computer-Aided Design (CAD) of the proposed machine will be developed using advanced CAD tools to model various reconfiguration scenarios for part sorting and defect detection. Simulation studies will be performed to evaluate the system's sorting accuracy, inspection speed, and reconfiguration time under different operational conditions.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Design of a Reconfigurable Moulding Tool for Complex Part Geometries

Reference	:	BR6
Supervisor	:	Dr Boitumelo Ramatsetse
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Conventional moulding tools are typically designed for a single product geometry, making them inflexible and costly to modify when part designs change. In industries where product variety and customization are increasingly demanded, dedicated moulds lead to high retooling costs, increased downtime, and wasted material resources. Each new part geometry often requires an entirely new mould, which limits production agility and delays product development cycles.
To address these challenges, this project proposes the design and simulation of a Reconfigurable Moulding Tool capable of adapting to complex and varying part geometries with minimal retooling effort. The system will incorporate modular and adjustable components that can be reconfigured to suit different shapes, sizes, and contours of moulded parts. This flexibility enhances the responsiveness of manufacturing systems, allowing for efficient production of multiple product variants using a single adaptable tool.
The Computer-Aided Design (CAD) of the reconfigurable moulding tool will be developed using advanced CAD software to simulate its functionality and evaluate its ability to form various part families. The tool's modular design will integrate reconfigurable inserts, adjustable cavity sections, and quick-change mechanisms to accommodate complex geometries without compromising precision or surface quality.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Optimisation and Performance Evaluation of a Semi-Automated Reconfigurable Welding Jig for Automotive Component Manufacturing

Reference	: BR7
Supervisor	: Dr Boitumelo Ramatsetse
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Welding is a critical process in automotive component manufacturing, where high dimensional accuracy, repeatability, and production flexibility are essential. Traditional welding jigs, however, are typically dedicated fixtures designed for a single component geometry. This lack of flexibility becomes a major limitation in modern automotive environments where frequent design updates, model variations, and low-volume/high-mix production demand adaptable fixturing solutions. To address these challenges, this project aims to optimise and evaluate a semi-automated reconfigurable welding jig tailored for diverse automotive components. The jig integrates adjustable locating pins, modular clamping units, and sliding fixture plates to allow operators to reconfigure the setup with minimal tooling and reduced downtime. Furthermore, the jig has semi-automation features, including sensors, pneumatic or electric clamping mechanisms as well as positioning and safety systems. A control subsystem is integrated to manage key functions such as clamp actuation, component positioning feedback, and sequence control. MATLAB/Simulink will be used to optimise and validate the control logic, particularly for clamping timing, position sensing, and safety interlocks. The project will involve the fabrication of a fully functional prototype of the reconfigurable welding jig. Performance evaluation will then be conducted by testing the jig with different automotive component geometries, assessing parameters such as setup time, repositioning accuracy, clamping repeatability, weld quality consistency, and operator usability. A simulation model will also be developed to demonstrate the jig's capability to handle various component designs and welding configurations.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Machine Design A (314) | Mechatronics (424)

Design and Integration of a Load-Cell-Based Force Monitoring System for a Modular Railway Wheel-Set Test Rig

Reference	: BR8
Supervisor	: Dr Boitumelo Ramatsetse
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
This project focuses on enhancing a modular, small-scale railway wheel-set test rig through the design and integration of a load-cell-based force monitoring system. The original rig, developed to simulate wheel–rail interaction under controlled vibration, incorporates mechanical, electrical, and data acquisition subsystems that enable realistic dynamic testing of a railway wheel-set. While the prototype roller rig demonstrated stable actuator motion, smooth rotational behaviour, and vibration responses that aligned with theoretical expectations, the next advancement required accurate measurement of contact forces between the wheel and rollers. The objective of this project is to design, implement, and calibrate a load-cell-based sensing system capable of capturing vertical and tangential forces generated during wheel–roller interaction. The system is integrated into the existing modular structure without altering its core mechanical layout, ensuring compatibility with future upgrades such as metal roller assemblies and closed-loop vibration control. The force-monitoring module includes load-cell selection and placement, signal conditioning circuitry, data acquisition integration, and real-time force visualisation. Testing and validation activities involve applying controlled actuator inputs and comparing measured forces with expected values derived from vibration and contact-mechanics models.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Machine Design A (314) | Mechatronics (424)

Raspberry Pi Laparoscope

Reference	:	KS1
Supervisor	:	Prof Kristiaan Schreve
Can be completed in first semester	:	Yes
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Laparoscopic surgery is a common minimally invasive surgical technique. Typically, a camera and one or two instruments are inserted through small ports in the abdomen. The instruments and camera are typically long, slender components, approximately 10 mm in diameter. The laparoscope consists of a camera head and a telescope which contains a lens assembly. The optical axis is often different from the axial direction of the laparoscope.
The Institute for Biomedical Engineering (IBE) does research using such devices. For the testing phase in many student projects, low-cost prototypes are often needed. This testing in early product development is done in laboratory environments (i.e. on phantom models). Due to the stringent requirements of the theatre environment, surgical laparoscopes are often too expensive for student projects, and they often have proprietary interfaces. The IBE needs a 'look and function like' laparoscope that interfaces effectively with devices frequently used in student projects, e.g. a Raspberry Pi. The imaging properties of the device must be as near as possible to a surgical laparoscope. The device need not be biocompatible or sterilizable since it will only be used on phantom models. In all other respects, it must be as similar to a surgical laparoscope as possible.
Ideally, the end product for this design must be usable during an open day for visitors to get a realistic laparoscopic experience from a surgeon's perspective!

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Mechatronics (424)

3D Printed Stent

Reference	:	KS2
Supervisor	:	Prof Kristiaan Schreve
Can be completed in first semester	:	Yes
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Finite Element Methods (FEM)

Description:
The Institute for Biomedical Engineering will do research into the implant of artificial hearts. Stents are often a key component of these implants. In order to experiment with different concepts, it is necessary to build prototypes that can be tested on a heart phantom or a porcine heart. In this project, various 3D printing technologies, available in the engineering faculty, will be investigated to develop alternative processes to manufacture "look and function like" stent prototypes. This means that the stent prototype is not required to be sterile or made of biocompatible materials. Neither does its strength necessarily need to be exactly as good as a human implantable stent. In terms of shape and size, it must be very similar to the ideal, human implantable, stent. Some of the material properties may have to represent the real stent, e.g. some stents are flexible to a high degree and are expended once in position.
For the heart project, the stents must be implanted in the left ventricle. The manufacture of a phantom of at least the left ventricle is also part of the project. The implant detail in the phantom or porcine heart is outside the scope of this project, but the project must determine the mechanical properties of the stent, experimentally and using finite element methods, and thereby show that it is implantable in the phantom or porcine heart.

Key modules (Scientific and Engineering Knowledge):
Finite Element Methods (414) | Experimental Techniques (414)

Frame Running Tricycle

Reference	:	KS3
Supervisor	:	Prof Kristiaan Schreve
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Finite Element Methods (FEM)

Description:
Frame running is a new Para sport which allows individuals with more severe disabilities, like cerebral palsy, to participate in exercise and sport. These individuals use wheelchairs every day and are given the physiologically important opportunity to ambulate (weight-bear) assisted with the frame runner. Frame running is fun for these individuals and holds great potential for increasing physical activity in this population. It is the fastest growing para sport globally and has recently been added to the paralympics programme. There are strong followings in the UK, Europe and Australia. However, the sport has not yet reached South Africa due to the cost of the frame runner tricycles, despite the need for an activity like this in the country. Stellenbosch University is developing this activity in South Africa through the social impact project Framing Mobility, and want to develop affordable equipment. Therefore, the current project will include designing and manufacturing a low-cost frame runner which can be used by people in South Africa participating in frame running. Website to social impact project: https://www.framingmobility.co.za/. Topic proposed by Prof Pheobe Runciman (Department of Exercise, Sport and Lifestyle Medicine)

Key modules (Scientific and Engineering Knowledge):
Finite Element Methods (414) | Machine Design B (344)

Portable Measurement Arm Joint Design

Reference	: KS4
Supervisor	: Prof Kristiaan Schreve
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Portable measurement arms are versatile tools to take 3D coordinate measurements. They normally offer six degree of freedom movement and the design can be adapted to various measurement envelopes. See https://www.wdhearn.co.za/product/portable-measuring-arms for a selection of typical measurement arms. The arm typically has four revolute joints. The joints are not motorised as a robot arm is. An operator must move the arm manually, typically by grabbing onto the last segment of the arm. The arms offer accuracies in the order to 5 to 50 microns, often depending on the size of the arm.
A key feature of the arms is that their revolute joints provide sufficient holding torque to keep the arm in position after the operator removed their hand from the arm. It would be a very impractical device otherwise!
They joints are also very important in terms of the system accuracy. The run-out tolerances on the joint must be very tight. An encoder is also incorporated in the joint to measure the angular displacement. Finally, there are cables running through all the joints to take the signals to the control unit. Obviously, the joint must be compact and light weight. The holding torque capability must not degrade over time.
A quick internet search did not provide any commercially available joints that meet these requirements. Friction hinges are a low cost, very inaccurate, option that can be used as some inspiration for the design. One website suggests damping grease like Nyogel, but it is doubtful that this will provide enough holding torque.
In this project, a mathematical model of the error propagation of a typical measurement arm must be used to determine the required tolerances for the joint. The model was developed by Leon Hall, a previous skripsie student, in 2009. Using this specification, the joint must be designed and test. Testing is on a single joint only. Building the full arm is outside the scope of this project.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Machine Design A (314)

Mosquito localizer

Reference	:	WJS1
Supervisor	:	Dr Willie Smit
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Build a device that will locate a mosquito in a room. The output of this device should be in such a format that it if fed to a laser, it can shoot the mosquito out of the air.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (315)

Paper plane controller

Reference	:	WJS2
Supervisor	:	Dr Willie Smit
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Design and build a controller for paper planes (similar to https://www.poweruptoys.com/)

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (315)

Hospital based robotic assistant (competition)

Reference	:	WJS3
Supervisor	:	Dr Willie Smit
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The successful candidate will be responsible for the design, build and control implementation of a floor-bound, obstacle-avoiding, pathfinding robot. The robotic assistant should function as a mobile information kiosk, capable of interacting with patients via a custom HMI and leading patients to a designated ward or physician's office. Performance will be measured through various requirements, including but not limited to speed, efficiency, reliability and safety. A full set of customer requirements from the sponsoring industry partner will be provided at the commencement of the project. Successful candidates are expected to conduct themselves in a respectful and professional manner whenever they interact with the sponsoring industry partner, as you will be representing the department and the university.
You will be given the chance to compete against other students with the same topic after the final year projects have been examined. Note that each student will have to adhere to a strict set of rules to ensure the integrity of the competition and fairness towards other participants. Examination of the final year project and competitive performance assessment will be mutually exclusive.
Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Mechatronics (424)

Scale model of heliostat field

Reference	:	WJS4
Supervisor	:	Dr Willie Smit
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
A heliostat field consists of an array of sun-tracking mirrors that continuously reflect sunlight onto a single fixed target, typically a receiver located at the top of a tower. The concentrated heat generated by this setup is used to produce intense heat for electricity generation or various industrial processes.
This project aims to create a functional demonstration model of the heliostat field that visually illustrates the core concept of concentrated solar power (CSP).
The project will integrate mechanical design, electronics, and embedded control. Due to the small scale of the model, nearly all components can be 3D printed.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Mechatronics (424)

Thermal comfort in houses

Reference	:	WJS5
Supervisor	:	Dr Willie Smit
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The student will design and build sensors to be installed in several houses within the same suburb. The objective is to measure the thermal comfort of these houses and investigate how various factors, such as building materials, window sizes, and floor plans, impact this comfort.
The student will conduct thermal calculations or simulations for each house and then compare the theoretical results with actual measurements. The project aims to highlight two key points: 1) the accuracy of the simulations or calculations used by architects, and 2) the variations in thermal comfort resulting from different building materials and house designs.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Mechatronics (424)

Brushless DC motor and propeller database

Reference	: WJS6
Supervisor	: Dr Willie Smit
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
A few years ago, one of our master's students developed a Y4 drone (you can view the video at https://tinyurl.com/y4drone). We have created a model of the drone; however, the model lacks accurate data for the brushless DC motors and propellers. This project aims to compile a database of motors and propellers from online sources and then use this database to design a series of Y4 drones with varying payload weights. This is a purely simulation-based project, and the student will write all the code in MATLAB.

Key modules (Scientific and Engineering Knowledge):
Electrical Drive Systems (324) | Modelling (334)

Development of a Modular Wireless Monitoring and Control Interface for Low-Cost Rural Solar PV Systems

Reference	: CAS1
Supervisor	: Dr Clint Steed
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Low-cost solar PV systems provide energy to meet the modest needs of underserved rural communities. These systems typically consist of established technologies, including solar panels, charge controllers, and batteries. However, while there is extensive historical data on usage and demand profiles for industrial and residential applications, data for rural users remains insufficient. One potential reason is the high cost of measuring and monitoring devices, which significantly exceed the costs of systems without these features. Current systems that measure the battery charge state and PV generation, while also allowing monitoring through access to data logs and device functioning states, are prohibitively expensive for rural applications.
In rural communities, reliable internet and network connections are often unavailable. Therefore, monitoring and measuring devices for these systems must establish a mesh wireless information network to relay data. Additionally, these devices should have the capability to stop power transmission from the system when needed, adding a layer of control to their functionality.
The purpose of this project is to develop a modular, low-cost monitoring and measuring interface for rural solar PV systems. The interface must be cost-effective, capable of measuring and monitoring system states such as the current and voltage of batteries and PV generation, and relay this information through a wireless mesh network. Furthermore, it should include the ability to control and stop power transmission from the system.
The project methodology involves first establishing a comprehensive set of device requirements. Based on these requirements, an electronic device will be developed and subjected to rigorous testing. The device's functionality will be validated by testing its measurement accuracy, control of power relaying, and communication through the wireless network. A cost analysis will ensure the solution remains economically viable. Field testing will evaluate the device's performance in establishing connections, monitoring, and controlling power in practical rural scenarios.
The ideal candidate for this project should possess a strong interest in and capabilities for working with low-voltage power electronics. Additionally, they should have

knowledge and skills related to computer network communication, enabling them to design and implement the wireless mesh network component of the system.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (315)

Developing a Smart Scale System for Hospital Waste Data Capture and Analysis

Reference	:	CAS10
Supervisor	:	Dr Clint Steed
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Background
For hospitals to make better decisions about sustainability, they need accurate data on their waste. However, the current process of collecting waste data—which involves weighing, recording, and classifying waste—is time-consuming and difficult to maintain. This effort prevents hospitals from getting a clear, detailed picture of what waste they produce. This project solves this by adding a mobile smart scale to the existing hospital dirt carts. This technology will automatically record and weigh the waste as it is collected, allowing for a much clearer and more detailed measurement of all waste streams (or "dirt flows") leaving different areas of the hospital. The resulting data can then be used to inform better waste-reduction strategies. Purpose
The purpose of this project is to design, build, and demonstrate a working prototype of a mobile smart scale system that automatically measures and records hospital waste flows.
Methodological Outline
This project is a design, build, and demonstration effort focused on creating a practical data capture tool.
System Design and Integration: We will first design the scale system to be rugged and fit seamlessly onto existing hospital dirt carts. The prototype will be constructed and tested.
User Trial and Validation: The project will conclude with a week-long trial of the cart by hospital employees. The goal of this trial is to prove the system's usefulness and show that it does not disrupt their normal workflow. During this week, we will collect a real, valuable dataset of waste flows. We will then analyze this data and use it to demonstrate how the information can be used to improve the hospital's waste management decisions.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Mechatronics (424)

PoE: A unified energy and information infrastructure for small rural communities

Reference	:	CAS2
Supervisor	:	Dr Clint Steed
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Power over Ethernet (PoE) is a technology standard that offers a unfied electrical-energy and information interface. While recent standards have increased the electrical-energy capacity, it is only suitable for small electronic devices and will not meet the middle class needs. In rural communities however, these offer a viable solution as the users have significantly lower energy. Moreover, this unified interface can form a unified distribution network. This opens up the possibility of automated trading. Purpose The purpose of this project is to create a prototype demonstrating PoE as a distributed energy network for rural solar powered energy and information infrastructure. As such devices should be able to measure and monitor transmission between units. Methodological outline A typical requirements, architecture, specification, and testing methodology should be followed. The following questions are pertinent to the investigation: 1. How well does PoE meet the requirements of households in rural areas? 2. Can it be used to distribute energy between prosumers? 3. What challenges and opportunities emerge from this system? Ideal candidate characteristics Required: Adapt in power electronics

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (315)

Design and Implementation of a Pump Test Workbench for Generating Machine Failure Datasets

Reference	:	CAS3
Supervisor	:	Dr Clint Steed
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Background
Predictive maintenance is a smart technology that uses machine learning to predict when a machine might fail before it actually breaks down, helping to greatly reduce costly machine downtime. These systems rely on data collected during the machine's operation. The core problem is that the available datasets usually contain a large amount of information about normal operation but very little information about the failure state. This lack of failure data makes it difficult to train effective, reliable predictive maintenance algorithms, creating a clear need for high-quality, diverse failure datasets. Purpose
The purpose of this project is to design, build, and test a specialized workbench that can reliably and safely generate a comprehensive dataset for pump failure scenarios. The workbench must be able to collect a variety of important signals, including vibration and electrical data, as well as other relevant signals for diagnostics. Crucially, the workbench needs to allow for changing operating conditions (like speed or load) and must be capable of inducing common bearing failures in a controlled manner to create the necessary failure data for machine learning algorithms. Methodological Outline
This project is a practical engineering exercise focused on the design, build, and test of the physical hardware and software system, with a small demonstration of its intended use.

Key modules (Scientific and Engineering Knowledge):
Electrical Drive Systems (324) | Vibration and Noise (354)

Investigation into the second use of medical waste

Reference	:	CAS4
Supervisor	:	Dr Clint Steed
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Hospitals are under increasing pressure to become more sustainable. Medical waste is expensive to dispose of. Ecosterile machines decontaminate the waste, allowing them to be disposed in conventional landfills. This ester waste is suspected to have a high calorific value. The purpose of this project is to investigate the applications of ester waste in local hospitals. As such the calorific value of the waste should be determined, applications of second use should be investigated, and a feasibility study or prototype should be developed.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Thermofluid Dynamics (344)

Prototyping a mini compactor for soft plastic recycling

Reference	:	CAS5
Supervisor	:	Dr Clint Steed
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Plastics are increasingly creating environmental issues. While alternatives are making their way to marked the adoption rate of alternatives is highly dependant on the view of the younger generation. The EcoChangeMakers program enables high schools students to partake in recycling at schools. A high school initiative is driving high schools to recycle. The design, manufacture, utilisation, and maintenance of hardware for recycling should be local and specific to increase the longevity of such initiatives. Purpose This project aims to develop a proof of concept compactor for soft plastics. This proof of concept will be used regularly by high schools to recycle plastics within the community. Specifications The following specifications are provided for the proof of concept. • The compacted volume target is 400mm x 400mm x 400mm • The material compacted will be HDPE2 and PP5 plastics (maybe No.7 too) • The compactor will be used by high-school students • The compactor will be manufactured by small scale local fabricators and engineering workshops

Key modules (Scientific and Engineering Knowledge):
Machine Design B (344) | Strength of Materials W (334)

Development of an Open-Source, ROS-Compatible Robotic Arm for the Furniture Manufacturing Industry

Reference	:	CAS6
Supervisor	:	Dr Clint Steed
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Background
Robotic arms are a mature and flexible automation technology crucial for enhancing manufacturing competitiveness through the automation of repetitive tasks. However, the adoption of commercial robotic systems is often hindered by their high cost, particularly in regions lacking local manufacturing of these components. This financial barrier restricts small to medium-sized enterprises, such as those in the furniture industry, from leveraging automation. To address this, the development of an open-source robotic arm presents a viable and cost-effective solution, offering the benefits of automation while being more accessible and adaptable to local needs. Purpose
The primary purpose of this project is to develop a functional prototype of an open-source, low-cost robotic arm specifically designed to meet the repetitive task requirements of the furniture manufacturing industry. This includes establishing a cost-effective automation solution for the furniture sector, developing a robot that is compatible with the Robot Operating System (ROS) MoveIt! framework, and creating a simple, intuitive user interface for programming basic operations to maximize accessibility for small-scale manufacturers. Methodological Outline
The project will follow a structured design and development process, beginning with Requirements Elicitation. This will involve conducting a case study or survey to identify and document the specific, repetitive tasks typically performed in the furniture manufacturing industry (e.g., picking, placing, simple assembly) to establish the necessary technical specifications and degrees of freedom (DOF) for the robotic arm prototype. Following this, the Hardware and Software Development phase will focus on the design of the open-source mechanical components and the implementation of control software compatible with ROS MoveIt! for advanced motion planning. This will be followed by the Interface Development phase, where a simple interface (e.g., a graphical user interface or a basic programming-by-demonstration module) will be designed to allow non-expert users to program common tasks. The final phase, Testing and Validation, will involve implementing a selected furniture assembly task on the prototype, testing its functionality and repeatability, and conducting a prelimi-

nary cost analysis to confirm the low-cost and competitive nature of the open-source design.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Mechatronics (424) | Modelling (334)

Using Sterilized Hospital Waste for Energy and Other Applications

Reference	:	CAS7
Supervisor	:	Dr Clint Steed
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Background
Hospitals are under constant pressure to become more environmentally friendly and reduce their running costs. Dealing with medical waste is a major expense. Machines like the Ecosterile system help by cleaning the waste on-site. This makes the material safe, so it can be thrown away in a regular landfill instead of needing costly, specialized disposal. This cleaned material, called "ester waste," is thought to hold a high amount of energy. This project aims to turn this expense into an opportunity by using the energy in this waste for the hospital's own benefit. Purpose
The main goal of this project is to find and prove ways that local hospitals can reuse their clean ester waste to save money and become more sustainable. This will be done by achieving three clear objectives. We must first measure the exact energy content (calorific value) of the waste. Next, we will investigate practical ways to use this waste—for example, turning it into fuel or a reusable material. Finally, we will create either a detailed feasibility study or a small prototype to show how the best reuse idea can be implemented by the hospital. Methodological Outline
The project will follow a three-step method.
First, we will test the waste's energy. We will gather samples of the cleaned ester waste from hospitals and use a device called a bomb calorimeter to precisely measure how much heat or energy it contains. This will confirm its potential as a fuel.
Second, we will research and find the best uses. We will study the most realistic ways to use the energy-rich waste. This research will look at options like making it into a solid fuel (like refuse-derived fuel) or using a process to turn the plastic into oil or other materials. We will focus on simple and cost-effective solutions for a local hospital.
Third, we will create a plan or a working example. We will take the most promising idea and either write a detailed feasibility study (showing the costs, savings, and steps needed for the hospital to adopt it) or build a small prototype that physically shows the waste being converted into a useful resource. Ideal Candidate Characteristics
The best candidate for this project should have a good background in engineering or

energy studies and a strong interest in waste reduction. They must be good at lab work, especially when testing materials for energy content. The candidate also needs strong math and thinking skills to put together the cost analysis and feasibility study. Finally, they should have a practical mindset and understand the real-world needs and cost limits of a working hospital.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Thermofluid Dynamics (344)

Developing a Low-Cost Automated Mobile Chicken Coop for Semi-Rural Farming

Reference	:	CAS8
Supervisor	:	Dr Clint Steed
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Background
In semi-rural areas, such as the surrounding winelands, there are many small areas of land that are not being fully used. These plots are perfect for small-scale farming, which offers several key benefits. It improves food security by producing food locally, simplifies logistics by cutting down on transport needs, and creates jobs for local people. However, the main problem is that these small patches of land are often not easy to access or manage using traditional farming methods. To make this type of farming practical and successful, an efficient, low-cost solution is needed to manage livestock, specifically free-range poultry, in these scattered locations. Purpose
The main purpose of this project is to design and build a cost-effective, mobile chicken coop that can be largely automated. The final product should be a practical tool that allows chickens to safely graze in an area while providing their basic needs, such as shelter, food, and water, for several weeks without constant human attention.
Methodological Outline
The project will focus on the design, development, and testing of the mobile coop.
First, we will establish the design requirements. This involves figuring out the exact size, weight, and materials needed to make the coop mobile and low-cost. We must also determine the needs for automation, such as the mechanisms for moving the coop, dispensing food and water, and securing the birds at night.
Second, we will develop and build the prototype. This phase involves the mechanical design of the movable structure, the selection of low-cost electronic components (like solar panels, batteries, and simple motors), and the programming of the automation capabilities. The system will be designed to manage the coop's movement, protect the birds from predators, and track resource levels (food/water).
Third, we will test and validate the system. The finished prototype will be field-tested on a small, underutilized plot of land for several weeks. The tests will measure its performance on key goals, including the reliability of the automation, the total operating costs, and the time the system can run independently before needing a refill or maintenance. Ideal Candidate Characteristics

The ideal candidate for this project should have a strong background in mechanical or electrical engineering, with a keen interest in farming technology and sustainable systems. They must be skilled in designing and prototyping mechanical structures and have experience with low-cost electronics, sensors, and basic automation or robotics. The candidate should have a practical, hands-on approach and an understanding of the financial and logistical limits of operating in semi-rural communities.

Key modules (Scientific and Engineering Knowledge):
Machine Design B (344) | Mechatronics (424)

Prototyping a clean frugal wood-fired cooking solution for under-served communities

Reference	:	CAS9
Supervisor	:	Dr Clint Steed
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Wood-fire is integral to South African culture. It is used for space heating, cooking, and creating ambience. Yet, in much of the under-served communities, this has become an issue due to inhalation of combustion bi-products leading to repository diseases prevalent in children. While off the shelf fireplaces exist, these are costly. The second challenge that persists is integrating the exhaust system into different housing structure (brick and mortar, wood, or zinc shack) can be challenging. Purpose This project aims to develop a proof of concept providing the woodfire thermal needs including cooking and indoor space heating, while remaining safe. Specifications The following specifications are provided for the proof of concept. • BS EN 1860-1:2024 Appliances, solid fuels and fire-lighters for barbecuing - Barbecues burning solid fuels • ISO 19867-1:2018 Clean cook stoves and clean cooking solutions • ISO 13732-1:2008 Ergonomics of the thermal environment

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Machine Design A (314)

Hospital based robotic assistant (competition)

Reference	:	WSWART1
Supervisor	:	Mr Wayne Swart
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The successful candidate will be responsible for the design, build and control implementation of a floor-bound, obstacle avoiding, pathfinding robot. The robotic assistant should function as a mobile information kiosk, capable of interacting with patients via a custom HMI and leading patients to a designated ward or physician office. Performance will be measured through various requirements, including but not limited to, speed, efficiency, reliability and safety. A full set of customer requirements from the sponsoring industry partner will be provided at the commencement of the project. Successful candidates are expected to conduct themselves in a respectful and professional manner whenever they interact with the sponsoring industry partner as you will be representing the department and the university. You will be given the chance to compete against other students with the same topic after the final year projects have been examined. Note that each student will have to adhere to a strict set of rules to ensure the integrity of the competition and fairness towards other participants. Examination of the final year project and competitive performance assessment will be mutually exclusive. Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Mechatronics (424) | Modelling (334)

Rocket / Drone Gyro

Reference	:	WSWART2
Supervisor	:	Mr Wayne Swart
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The aim of this project is to develop a single package 3-axis gyro system that can orient and stabilize a hobby sized sugar rocket or drone. This project will require a significant application of simulation, modelling and control systems as well as a good grasp on electronic design and applied mathematics. Thorough experimental work will be required to validate the end product.
Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Mechatronics (424) | Modelling (334)

3-axis Magnetic Field Manipulator

Reference	:	WSWART3
Supervisor	:	Mr Wayne Swart
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Many applications of microrobots require magnetic drive systems to actuate the robots, which requires 3-axis magnetic field manipulators for development purposes. Whilst most 3-axis magnetic manipulators are constructed with an arrangement of Helmholtz coil pairs, permanent magnets can provide higher field strengths for the same volume without the need for cooling. The aim of this project is to develop a 3-axis magnetic manipulator utilizing permanent magnet arrays. The project will have an emphasis on mechatronic systems and will require the application of knowledge from electronics design and control systems.
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Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

The Tadpole

Reference	:	WSWART4
Supervisor	:	Mr Wayne Swart
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
In vivo applications of medical microrobots require the robot to operate in a fluid environment. The aim of this project is to design, build and demonstrate a tadpole microrobot that is magnetically driven. The robot should be able to navigate in a fluid environment and deliver a small payload on command. Given the size of microrobots, the sensing and control systems for the robot will not be onboard. Therefore, multiple systems will have to be developed and knowledge of electronic design, sensing and control systems will have to be applied.
Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Modelling (334)

Sumo robot (Competition)

Reference	:	WSWART5
Supervisor	:	Mr Wayne Swart
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project forms part of a competition for which the successful candidates will be required to design and build an autonomous sumo robot that can compete against other participants. The project will require the application of electronic design, modelling and control theory, as well as skills acquired during the E-design module in the first semester. The design needs to satisfy the Robogames (https://robogames.net/rules/all-sumo.php) international sumo robot competition rules for the Mini Sumo robot class. Participants will have the opportunity to compete in a knock-out competition following the examination period. The competition and the examination will be mutually exclusive.
Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (245) | Machine Design A (314)

Micromouse Sphere

Reference	:	WSWART6
Supervisor	:	Mr Wayne Swart
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The aim of this project is to design, build and validate a spherical robot that satisfies the IEEE Micromouse rules (https://ewh.ieee.org/sb/columbus/devry/sacFiles/MicromouseRules.pdf). The robot should be autonomous, i.e. a self-contained platform capable of navigating a maze by sensing obstacles in its surroundings. This project will require an in-depth application of modelling and control systems as well as electronic design. Skills acquired in the E-design module during the first semester will be essential.
Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Livestock Wearable

Reference	:	WSWART7
Supervisor	:	Mr Wayne Swart
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Monitoring livestock bio-signals and gait has great use in precision farming. The aim of this project is to design and build a livestock (specifically sheep) wearable that will be able to record bio-signals over the span of several days without impairing the animal in any way. Although the project leaves room for significant scope expansion, the emphasis will be placed on respiratory rate as function of ambient temperature. The project will require a thorough application of electronic design and rely on the skills acquired in the E-design module that takes place in the first semester. A well-rounded end-product that can be utilised in the field is expected, which will require careful attention toward design for manufacture. This project is in collaboration with the Western Cape Department of Agriculture, and the successful candidate is expected to conduct themselves in a respectful and professional manner whenever they interact with the sponsoring industry partner as you will be representing the department and the university.
Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (245) | Machine Design A (314)

Digital Twin Launchpad: Develop Next-Generation Engineering Tools

Reference	:	NCT1
Supervisor	:	Dr Nicole Taylor
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Are you excited by cutting-edge engineering technologies, such as digital twins, data pipelines, and human-system interaction? Do you want to graduate with skills that are increasingly in demand in modern engineering and research environments? If so, this project may be an ideal fit for you.
This final year project forms part of a larger research initiative to develop the Digital Twin Launchpad - a human-centred, modular platform designed to accelerate the creation of digital twin solutions. The Launchpad will serve as a toolbox of software components that make digital twin development faster, more intuitive, and more accessible for engineers and researchers, particularly those who may not be specialists in data engineering.
The overarching research goals align with the principles of Industry 5.0, emphasising human-centred design and the integration of human factors into digital environments. The platform aims to: • Support the creation of digital twins that not only mirror the physical world but also act as research instruments to automate and advance scientific enquiry. • Provide researchers with ready-to-use tools that minimise the steep learning curve associated with complex data engineering practices. • Enable richer analysis of human factors by embedding human-centred thinking into digital twin development workflows.
Your contribution to the project will require you to select a component of the Digital Twin Launchpad to design and develop. Available project focus areas include: • Data pipeline constructor tools • Data ingestion frameworks • Data model design and hosting tools • Data visualisation modules • User interface for the Digital Twin Launchpad • Research data management and maintenance tools Your developed component will be showcased through a case study implementation involving human digital twins aboard the SA Agulhas II, South Africa's polar supply and research vessel. This case study should support the development of tools to analyse human factors at sea - including motion sickness, discomfort, and sleep disturbance - offering a rare opportunity to contribute to real-world engineering research with meaningful human impact.

By completing this project, you will develop: • Hands-on experience with state-of-the-art digital twin technologies • Skills in data engineering, modelling, visualisation, and developing software architectures • Experience designing human-centred engineering systems • Exposure to real research datasets • A portfolio-ready project demonstrating innovation and technical depth This project is ideal for students eager to work at the frontier of modern engineering and make a tangible contribution to an emerging research platform used in multidisciplinary contexts.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Vibration and Noise (354) | Experimental Techniques (414)

Develop a Modular Mine-Like Testbed for Next-Generation Mining Technology

Reference	: NCT2
Supervisor	: Dr Nicole Taylor
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Are you interested in cutting-edge engineering technologies, such as smart mining systems, human-centred sensing, and Industry 4.0/5.0 automation? Do you want to graduate with hands-on experience in designing reconfigurable, real-world engineering systems? If so, this project may be ideal for you. This final-year project forms part of a broader research initiative to develop a mobile mine testbed - a human-centred, Industry 5.0-aligned environment for testing next-generation mining technologies. The testbed will serve as a mobile simulation of mine-like settings, such as underground mine tunnels, enabling safe, realistic evaluation and showcasing of sensors, wearables, communication networks, and distributed technology systems. It will also complement the existing SIMLAB mining simulation facility (https://sites.google.com/view/mad-research-group/facilities/simlab?authuser=0), which is temporarily unavailable due to ongoing renovations.
The overarching research goals aim to emphasise human, environment and system integration in mine-like settings. The testbed aims to: • Provide a realistic, modular environment for validating smart-mining technologies. • Support the integration of human-centred devices, such as a previously developed wearable human-condition monitoring device for mineworkers. • Enable plug-and-play testing of sensors, communication systems, and distributed computing infrastructures.
Your contribution to the project will require you to design and develop components of the testbed, which include focus areas: • Modular structural design that mimics mine-like settings, such as underground mine tunnels • Integration of environmental and human-centred sensors • Implementation of adaptable communication and networking systems • Development of data acquisition and analysis through computing technologies • Validation and showcasing of plug-and-play operation for a case study
By completing this project, you will develop: • Hands-on experience in modular systems design • Skills in sensor integration, communication networks, and embedded technologies • Experience with human-centred engineering concepts in Industry 4.0/5.0 contexts • Exposure to real mining-research environments and technologies • A portfolio-ready project that showcases innovation and practical engineering ability

This project is ideal for students who want to work at the frontier of smart-mining technologies and contribute to a long-term research platform used across multiple engineering disciplines.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Mechatronics (424) | Experimental Techniques (414)

Rough Seas, Real Humans: Develop a Tool to Analyse Motion Sickness, Discomfort and Sleep Disturbance

Reference	: NCT3
Supervisor	: Dr Nicole Taylor
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Are you curious about how real people experience the harsh conditions of life at sea - particularly the vibration, slamming, and rigid-body motion that can cause discomfort, sleep disturbance, and motion sickness? Do you want to apply your engineering skills to a project with direct human impact and real-world data from a polar research vessel? If so, this project could be the perfect fit. This final-year project focusses on developing a human-centred analysis tool for studying how ship motion affects people aboard the SA Agulhas II, South Africa's polar supply and research vessel. The goal is to transform paper-based and digital datasets into an automated system for analysing human factors related to: • Ship slamming events and comfort effects • Rigid-body motion linked to motion sickness • Sleep disturbance
Your project will support ongoing research into improving human well-being and operational decision-making during scientific voyages in rough sea conditions (https://svrg.sun.ac.za/). Your contribution to the research will include: • Digitise and structure historical paper-bound questionnaires from previous voyages. • Develop algorithms to analyse slamming, motion sickness, and/or sleep disturbance data. • Build an integrated analysis tool that makes these datasets easy to explore and interpret. • Prototype visualisations or dashboards that reveal insights about human experiences at sea.
By completing this project, you will develop: • Experience with data cleaning, digitisation, and structuring • Skills in algorithm development and time-series analysis/signal processing of real-world data collected on the SA Agulhas II during previous voyages • Understanding of human factors engineering in extreme environments • A polished, portfolio-ready tool used in ongoing multidisciplinary research This project is an ideal choice for students who enjoy working with data, solving meaningful engineering problems, and building tools that directly serve people at sea in demanding real-world environments.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Vibration and Noise (354) | Experimental Techniques (414)

Design and Implementation of an Instrumented Test Bench for Performance Analysis of Water Electrolysis System

Reference	: GTH1
Supervisor	: Dr Gerrit Ter Haar
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
This project involves developing a comprehensive data acquisition and control system for an existing hydrogen water electrolysis setup. The student will integrate sensing capabilities including temperature monitoring, differential pressure measurements across the electrolyser stack, flow rate tracking for both water input and hydrogen/oxygen output streams, and real-time power consumption analysis. The project encompasses designing sensor interfaces, implementing a microcontroller-based data acquisition system, developing a graphical user interface for system monitoring, and creating automated safety protocols. Key deliverables include closed-loop control of critical parameters, data logging capabilities for performance analysis, and a modular design allowing future experimental modifications. This hands-on project will provide experience in industrial automation, process control, sensor integration, and renewable energy systems - making it ideal for students interested in green hydrogen technologies or industrial process automation.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (315) | Mechatronics (424)

Upgrade a Membraneless Flow-Through Hydrogen Generator System

Reference	:	GTH2
Supervisor	:	Dr Gerrit Ter Haar
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
This project involves optimizing and scaling a novel membraneless flow-through hydrogen electrolyser that was successfully designed and tested in 2025. The 2025 prototype demonstrated the feasibility of using rectangular electrodes to enable three-dimensional scalability. However, the system was intentionally limited by the use of sodium carbonate electrolyte and stainless steel electrodes to prioritize safety during the initial proof-of-concept phase. Building on 2025's foundation, this 2026 iteration focuses on unlocking the system's full performance potential through three interconnected improvement areas: (1) Advanced electrode optimization: The 2025 testing revealed that electrode configuration significantly impacts both performance and gas separation. This project should explore advanced 3D porous electrode designs using superior materials such as nickel alloys or platinum-coated surfaces, which could provide 2-3x performance improvements. Computational fluid dynamics (CFD) simulation could be used to predict and validate electrode performance. (2) Material system upgrade for aggressive electrolytes: The 2025 system was limited to 80°C with sodium carbonate. To achieve commercial-level performance, upgrade to chemical-resistant materials capable of handling potassium hydroxide (KOH) at elevated temperatures up to 90-100°C. KOH is approximately 4x more conductive than sodium carbonate and will dramatically reduce cell voltage. (3) System scalability and performance validation: The 2025 design demonstrated that rectangular electrodes enable straightforward scaling. This project should add multiple electrode pairs and/or stack cells in series/parallel configurations. Implement quantitative gas purity measurement using laboratory equipment (2025 relied only on visual inspection), measure overall system efficiency, and validate long-term operational stability. This hands-on project offers experience in electrochemical engineering, advanced materials, process optimization, and renewable energy technology, making it ideal for students interested in next-generation green hydrogen production systems.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Numerical Fluid Dynamics (414) | Experimental Tech-

niques (414)

Computational Fluid Dynamic Simulation and Experimental Validation of Two-Phase Flow in PEM Water Electrolysis Cells

Reference	:	GTH3
Supervisor	:	Dr Gerrit Ter Haar
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
Understanding the complex two-phase flow behaviour (liquid water and oxygen bubbles) on the anode side of PEM (Proton Exchange Membrane) electrolysis cells is critical for optimizing hydrogen production efficiency. Oxygen bubbles generated at the anode must be efficiently removed to maintain good electrical contact, prevent voltage losses, and avoid hot spots that can damage the membrane. However, visualizing and accurately predicting this behaviour inside operating cells remains challenging. This project combines advanced computational fluid dynamics (CFD) modelling with experimental flow visualization to develop validated simulation tools for PEM electrolyser optimization.
Students will build a CFD model of two-phase oxygen-water flow in a PEM electrolyser anode, then validate it through direct experimental observation. To enable validation, students will design and integrate a transparent endplate (polycarbonate or optical-grade acrylic) into an operating PEM cell, allowing direct visualization of oxygen evolution and transport through the anode flow field. A high-frame-rate camera system synchronized with the cell's electrical monitoring will capture bubble dynamics under various operating conditions (current density, temperature, water flow rate), enabling direct correlation between flow behaviour and electrochemical performance. Quantitative image analysis of bubble size, coverage, and velocity will then be used to calibrate and refine the CFD model.
The validated CFD model will accurately predict two-phase oxygen-water flow in PEM electrolyser anodes and serve as a tool for accelerating future design optimization by enabling computational exploration of design variations before expensive prototyping. Additional deliverables include documented experimental protocols for flow visualization, quantitative analysis of bubble dynamics under various operating conditions, and design recommendations for improved anode flow field configurations. This project provides hands-on experience in multiphase CFD simulation, experimental fluid dynamics, high-speed imaging and image processing, PEM electrolyser tech-

nology, and data-driven optimization—skills highly valued in the rapidly growing green hydrogen industry.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Numerical Fluid Dynamics (414) | Experimental Techniques (414)

Development and Validation of Digital Twin Models for Hydrogen Microgrid-Refuelling Station Systems

Reference	:	GTH4
Supervisor	:	Dr Gerrit Ter Haar
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Hydrogen microgrid-refuelling station systems represent a cutting-edge solution for South Africa's energy security challenges, enabling long-term storage of renewable energy through hydrogen conversion. These systems store excess solar and wind electricity as hydrogen via electrolysis, providing both grid stabilisation and clean transportation fuel, bridging the gap between intermittent renewable generation and continuous energy demand. However, optimising these complex systems requires accurate predictive models that can simulate component behaviour and operational strategies before deployment. A fully automated commercial hydrogen generator system (HOGEN® S Series 2) acquired second-hand from the University of Cape Town provides a unique opportunity to develop validated digital twin models based on real industrial hardware. Students will first commission the system through installation, sensor diagnostics, and fault-finding to restore full operational capability. Following commissioning, comprehensive performance characterisation will be conducted across the system's operating envelope, collecting detailed experimental data on voltage-current relationships, efficiency curves, gas production rates, and thermal behaviour under various operating conditions. Using this experimental data, students will develop high-fidelity digital twin models in MATLAB Simulink (or similar software) that accurately replicate the electrolyser's behaviour. The modelling approach will combine equivalent circuit parameter identification for electrochemical processes with multi-physics simulations capturing thermal dynamics, mass transport, and system-level interactions. The digital twin will predict system performance, optimise operating strategies, and enable virtual testing of control algorithms without physical hardware operation. Model parameterisation will be rigorously validated against experimental measurements to ensure accuracy across different operating regimes. Students may extend the digital twin framework to encompass additional components of a complete hydrogen microgrid system, such as renewable energy inputs, high-pressure storage dynamics, or grid interface behaviour, enabling simulation of grid-independent renewable hydrogen production scenarios relevant to South Africa's energy landscape. This project provides hands-on experience in industrial sys-

tem commissioning, experimental characterisation, advanced modelling techniques, MATLAB Simulink development, digital twin methodologies, and renewable energy system integration—skills critical for South Africa's emerging hydrogen economy.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Energy Systems M (434) | Experimental Techniques (414)

Degradation Analysis & Lifetime Modelling of H2 Infrastructure Components

Reference	: GTH5
Supervisor	: Dr Gerrit Ter Haar
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical
Required elective	: Maintenance Management

Description:
Comprehensive lifetime assessment for sustainable hydrogen infrastructure Hydrogen microgrid-refuelling systems enable long-term storage of renewable energy by converting excess solar and wind power to hydrogen via electrolysis, providing energy security during grid instability and load shedding events (see https://www.iwu.fraunhofer.de/en/press/2025-Hydrogen-Microgrids-Make-Sun-and-Wind-Storable.html for microgrid functionality). These systems designed for seasonal energy storage face unique operational challenges, particularly when capturing abundant solar energy during summer months and storing it as hydrogen for use during winter periods when solar irradiation is reduced. Components must withstand frequent charge-discharge cycles as renewable energy availability fluctuates, requiring robust infrastructure design for renewable energy integration in regions with seasonal variations and grid instability. Conduct detailed degradation analysis and lifetime modelling of critical components in hydrogen microgrid-refuelling systems that enable seasonal energy storage from renewable sources. Focus includes accelerated ageing studies, degradation mechanism identification, materials characterisation, and predictive lifetime modelling for PEM electrolysers and fuel cells operating under cyclic renewable energy conditions. Develop reliability-centred maintenance strategies for components operating in challenging renewable energy integration scenarios.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Maintenance Management (414)

Hydrogen-Powered Model Train: From Concept to Track Pioneering Fuel Cell Integration in Scale Model Railway Systems

Reference	: GTH6
Supervisor	: Dr Gerrit Ter Haar
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
The H2GEN research team at Stellenbosch University is developing an innovative demonstration project that showcases the practical application of hydrogen technology in transportation: a 1/3-scale hybrid train powered by a fuel cell system. Project Overview This hands-on engineering project integrates a proton exchange membrane (PEM) fuel cell system into a scaled hybrid locomotive to supplement the onboard battery with clean hydrogen power. The project bridges the gap between theoretical hydrogen research and real-world engineering challenges, providing valuable insights for future full-scale rail applications. Key Research Activities System Design & Component Sizing: Optimizing the integration of hydrogen storage, PEM fuel cell system, and DC/DC converter Mechanical Integration: Engineering solutions for component placement within the constrained space of a 1/3-scale locomotive chassis Safety Systems: Implementing comprehensive safety protocols following NFPA 55 standards for hydrogen handling, leak detection, and emergency shutdown procedures Energy Management: Designing control strategies to balance power delivery between fuel cell and battery systems Real-World Testing The prototype will undergo performance testing on a 1-2 km track, evaluating speed, range, fuel efficiency, and system reliability under operational conditions. This provides empirical data on hydrogen consumption, power delivery, and the effectiveness of the hybrid powertrain architecture. Broader Impact This project demonstrates the feasibility of hydrogen fuel cell technology in rail transport, a critical sector for decarbonization in Southern Africa. The knowledge gained will inform future applications in passenger rail, freight transport, and mining operations where electrification infrastructure is limited.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Machine Design B (344)

SCARA Robot PCR Preparation Arm

Reference	:	PEJT1
Supervisor	:	Mr. Piero Trinchero
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Background: Commercially available, multi-station PCR preparation pipetting platforms are of the Cartesian robot design. These machines are rapid, reliable and efficient. However, their large footprint and overall bulkiness can also make them prohibitively expensive.
Requirements: To design a SCARA robot configuration arm with precise motion control for charging 96-well PCR racks with reagent by using a more compact, rotary carousel design. To test a marginally scaled-down prototype to determine if a rotary configuration is equally as rapid, but more compact than a linear one.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Machine Design A (314) | Mechatronics (424)

Speeding Vehicle Barrier

Reference	:	PEJT10
Supervisor	:	Mr. Piero Trinchero
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Background: Gated estates are homes to young children, elderly residents, and common house pets such as dogs and cats. Unfortunately, resident drivers in their vehicles and an ever-increasing number of delivery motorcycle riders break the strict 20 km/h speed limit applied in most of these estates, endangering all of the above-mentioned parties.
Requirements: To design an automatically deployed last resort 'barrier' for use in gated estates' roadways. It would deploy well ahead of the speeding vehicle to serve as a visual warning, not a physical collision barrier. To test a scaled-down prototype of the barrier in terms of how rapidly it deploys and resets after the vehicle has stopped.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Machine Design A (314) | Mechatronics (424)

Elastic Energy Storage Mechanism for a Crossbow

Reference	:	PEJT2
Supervisor	:	Mr. Piero Trinchero
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Finite Element Methods (FEM)

Description:
Background: Crossbow limbs made from laminated wood or composites are bulky, sensitive to moisture and temperature, and require regular tuning that may decrease accuracy. An alternative elastic energy storage system could reduce size and simplify maintenance, while reproducing the power and precision archery sport shooters expect.
Requirements: To design a different method of elastic energy storage for an archery crossbow that reproduces the performance and precision of a traditional crossbow's limbs, without any of the above-mentioned disadvantages. To perform outdoor target shooting tests with a full size version of the modified crossbow.

Key modules (Scientific and Engineering Knowledge):
Finite Element Methods (414) | Machine Design B (344) | Strength of Materials W (334)

Collapsable Footbridge for Emergency Services

Reference	:	PEJT3
Supervisor	:	Mr. Piero Trinchero
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Finite Element Methods (FEM)

Description:
Background: In disaster scenarios such as flooding rivers or earthquakes, access to affected areas is often hindered by damaged or impassable infrastructure. Paramedics and other emergency service personnel require a portable, quickly deployable footbridge to safely evacuate people and transport essential supplies.
Requirements: To design a stable, load-bearing footbridge that can be collapsed for ease of transport and deployment. To perform outdoor tests on a scaled-down prototype of the bridge using static weights and human test subjects.

Key modules (Scientific and Engineering Knowledge):
Finite Element Methods (414) | Machine Design B (344) | Strength of Materials W (334)

Motorized Mount for an Outdoor Camera

Reference	:	PEJT4
Supervisor	:	Mr. Piero Trinchero
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Background: Motorized outdoor camera mounts are essential tools for wildlife observation, environmental monitoring, and time-lapse filming. For example, researchers and enthusiasts may want to remotely monitor an eagle hatchling's growth and developmental progress in its cliffside nest.
Requirements: To create a remotely controlled, motorized outdoor camera mount that provides smooth, responsive movements and high resistance to small vibrations. To test the mount and camera assembly outdoors in windy Cape conditions, or indoors using a suitable fan to simulate the same.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Machine Design A (314) | Mechatronics (424)

Mobile Terrain Mapping Platform

Reference	:	PEJT5
Supervisor	:	Mr. Piero Trinchero
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Background: Good 3D terrain mapping may facilitate some small applications such as constructing on uneven building sites, or larger ones like creating a new golf course. Understanding the landscape can optimize operational costs and ensure overall safety. However, traditional mapping systems can be both complex and expensive.
Requirements: To design a mobile terrain mapping platform that produces a 3D map of the terrain it covers by using simple sensors while maintaining gyroscopic stability. To test a proof-of-concept prototype outdoors on uneven terrain.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Machine Design A (314) | Mechatronics (424)

Heavy Duty Mover's Hand Truck

Reference	:	PEJT6
Supervisor	:	Mr. Piero Trinchero
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Finite Element Methods (FEM)

Description:
Background: Moving bulky, heavy items is a common task in warehouses, some door-to-door deliveries, and urban relocations, often requiring significant physical effort and posing risks of operator injury. Traditional hand trucks can be unstable on uneven terrain, leading to inefficiencies and safety concerns.
Requirements: To design a hand truck that reduces operator strain and improves the general stability of bulky, heavy loads. To test a full-scale overloaded prototype on difficult to negotiate, uneven terrain.

Key modules (Scientific and Engineering Knowledge):
Finite Element Methods (414) | Machine Design B (344) | Strength of Materials W (334)

Commercial Coil Spring Tester

Reference	: PEJT7
Supervisor	: Mr. Piero Trinchero
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Background: Springs are one of the most common mechanical components used in a wide range of products and machinery. Spring manufacturers depend on reliable testers to determine the performance, accuracy and consistency of large batches of their mass-produced springs.
Requirements: To design a coil spring tester for small to medium-sized tension and compression springs; the design must include a calibration function. To test a full-scale prototype of the tester on a suitable range of mass-produced springs.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Machine Design A (314) | Mechatronics (424)

Alternate Bicycle Brake Design

Reference	:	PEJT8
Supervisor	:	Mr. Piero Trinchero
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Finite Element Methods (FEM)

Description:
Background: Bicycles disk brakes provide superior stopping power to the much older calliper design. However, all disk brakes suffer from brake fade as the friction elements heat up during hard and prolonged use.
Requirements: To explore an alternate bicycle brake design that is as efficient, or better than current disk brakes at reducing brake fade. To test the efficacy of this design on an off-road trail bicycle. NOTE: The student will need to own such a bicycle and be willing to test their prototype on it.

Key modules (Scientific and Engineering Knowledge):
Finite Element Methods (414) | Machine Design B (344) | Strength of Materials W (334)

Self-Balancing Wheel Hub

Reference	:	PEJT9
Supervisor	:	Mr. Piero Trinchero
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Background: Washing machine drums rely on floating mounts and static counterweights to deal with their unbalanced rotating loads. This system rarely works correctly, particularly with laundry items that soak up a lot of water such as bath towels. Requirements: To design a self-balancing wheel hub for the drum of front-loading washing machine. The hub must dynamically dampen an unbalanced load rotating at typical spin-cycle speeds. To test a scaled-down prototype of the dampened machine drum.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Machine Design A (314) | Mechatronics (424)

Automatic Wheel Choker Design for Industrial Vehicles

Reference	:	LZ1
Supervisor	:	Prof Lijun Zhang
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
• Structural design and simulation of a wheel choker. • Adaptable to different terrain conditions. • Adaptable to different vehicle dimensions. • Structural decoupling from the rigid body of the vehicle. • Mechatronics students only.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (245) | Electronics (315)

Structural Design of A Defect Detection and Classification System for Small Agricultural Produce

Reference	:	LZ2
Supervisor	:	Prof Lijun Zhang
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
• Structural design of an agricultural produce classification platform. • Automation of the structure for automatic feeding, rotation, and possible flipping of the produce • Develop an automated produce sorting mechanism to sort produce into different grades across multiple output streams.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (245) | Electronics (315)

Defect Detection and Classification of Small Agricultural Produce Based on Computer Vision

Reference	:	LZ3
Supervisor	:	Prof Lijun Zhang
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Development of computer vision–based defect detection algorithms. Camera mounting mechanism design Online monitoring software design for continuous monitoring of the detection results. Automatic sorting of the produce according to the level of defects detected.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Design (E) (314) | Mechatronics (424)

Speed synchronisation of a robotic arm with a conveyor belt

Reference	:	LZ4
Supervisor	:	Prof Lijun Zhang
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Design a small-scale conveyor belt Design a speed measurement device for the belt Design a robotic arm that can synchronise with the speed of the belt Develop an algorithm to control the robotic arm to pick up objects from the conveyer belt while the speed of the conveyor belt varies.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314)

Development of a nozzle for a selective powder deposition 3D printer

Reference	:	JVDM1
Supervisor	:	Dr Johan van der Merwe
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The purpose of this project is to develop a nozzle for a low cost, multi-material selective powder deposition 3D printer. The printer works by depositing both model and support materials in a removable crucible, one line at a time. This requires path planning and control of two powder deposition nozzles relative to a three-axis printing bed. Afterwards, the crucible is placed in an oven and the model material is fused at high temperatures, without melting the support material. Once cooled, the part is then removed from the crucible, and the support material cleaned off.
The scope of this project is limited to the deposition of powder materials on a flat surface with the aid of an appropriately designed test bench (no z-axis, crucible or oven required). Safety protocols for handling powdered materials must be investigated and strictly adhered to.
The use of generative AI for any purpose will be actively discouraged in this project due to the documented harm that it causes to learning. Otherwise, the usual rules regarding responsible use of AI will apply throughout.
Recommended reading:
Chianrabutra, S., Mellor, B.G. and Yang, S., 2014. A dry powder material delivery device for multiple material additive manufacturing. In 2014 International Solid Freeform Fabrication Symposium. University of Texas at Austin.
Magalhaes, S., Sardinha, M., Vicente, C., Leite, M., Ribeiro, R., Vaz, M. and Reis, L., 2021. Validation of a low-cost selective powder deposition process through the characterization of tin bronze specimens. Proceedings of the Institution of Mechanical Engineers, Part L: Journal of Materials: Design and Applications, 235(12), pp.2681-2691.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (315) | Mechatronics (424)

Continued development of an automated mandible reconstruction plate bending machine

Reference	:	JVDM2
Supervisor	:	Dr Johan van der Merwe
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Plates are often used during the reconstruction of large mandible defects. These plates are traditionally bent to shape intra-operatively. This process is time-critical, strenuous, weakens the plate due to plastic deformation, and ultimately relies on surgeon experience for success. To expedite reconstructive surgery, patient-specific solutions focus on pre-operatively shaping reconstruction plates, usually either by producing 3D printed models of the patient's anatomy against which a plate may be bent by hand, or additively manufacturing custom implants entirely. Custom plates delay treatment and potentially increase cost due to the requirements for one-of-a-kind design, manufacture, and transport, after the patient first presents for treatment. Pre-bending plates have many of the same disadvantages of intra-operative shaping, although are less time-critical, and may be less costly due to the economies of scale and logistics.
Therefore, to combine the advantages of custom plates with off-the-shelf solutions, this project is concerned with the development of an automated mandible reconstruction plate bending machine. The idea is to automatically pre-bend stock mandible reconstruction plates according to a patient-specific bending profile. This could potentially afford more precise control over the final shape, avoid plate fracture via larger bending radii and stress-relief, as well as provide opportunities for cost- and time-effective production. The bending machine must focus on achieving continuously varying out-of-plane bending radii and twist for reconstructing defects that do not involve the mandibular angle. Spring-back must be modelled and compensated for to achieve accurate control over the bend.
The scope of the project will be limited to the bending of a flat bar designed to mimic mandible reconstruction plates. Furthermore, in-plane bending is not part of the project scope. Testing will be conducted using a bending profile that reflects a typical mandible. No testing will be performed on human subjects. A concept demonstrating single radius plate bending has been developed. The candidate will be expected to analyse and refine the design of the previous concept demonstrator, and extend its functionality to continuous out of plane bending and twist.

The use of generative AI for any purpose will be actively discouraged in this project due to the documented harm that it causes to learning. Otherwise, the usual rules regarding responsible use of AI will apply throughout.
Recommended reading:
Ellis, E., Schubert, W., 2021. Plate bending. Available at: https://surgeryreference.aofoundation.org/cmf/basic-technique/plate-bending (Accessed: 14 November 2023).
Hua, M., Baines, K. and Cole, I.M., 1999. Continuous four-roll plate bending: a production process for the manufacture of single seamed tubes of large and medium diameters. International Journal of Machine Tools and Manufacture, 39(6), pp.905-935.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Sumo robot (competition)

Reference	: JVDM3
Supervisor	: Dr Johan van der Merwe
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
This project forms part of a competition for which the successful candidates will be required to design and build an autonomous sumo robot that can compete against other participants. The project will require the application of electronic design, modelling and control theory, as well as skills acquired during the E-design module in the first semester. The design needs to satisfy the Robogames international sumo robot competition rules for the Mini Sumo robot class.
Participants will have the opportunity to compete in a knock-out competition following the examination period. However, the competition and the examination will be mutually exclusive.
The use of generative AI for any purpose will be actively discouraged in this project due to the documented harm that it causes to learning. Otherwise, the usual rules regarding responsible use of AI will apply throughout.
Recommended reading:
Unified Sumo Robot Rules, 2023. Available from: <https://robogames.net/rules/all-sumo.php>. [28 November 2025].
Antunes, B.G., Bezerra, L.F., Canella, A.L. and Pinto, M.F., 2024, November. Development of an Autonomous Robot for Competitive and Educational Purposes in the Sumo Category. In 2024 Brazilian Symposium on Robotics (SBR), and 2024 Workshop on Robotics in Education (WRE) (pp. 209-214). IEEE.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Modelling (334)

Development of a reconfigurable cable driven robot to simulate mandibular loading conditions

Reference	: JVDM4
Supervisor	: Dr Johan van der Merwe
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
The temporomandibular joint (TMJ) connects the mandible (lower jaw) to the skull. In case of injury or disease, mandibular reconstruction is required to restore normal function to the mandible. Understanding the forces acting on the mandible is crucial during implant design and verification, such as when performing Finite Element Analyses (FEA). However, it is difficult to measure the mechanics of the mandible during activities of daily living such as speaking or chewing.
Therefore, the purpose of this project is to build a reconfigurable cable driven robot that will reproduce the various static and dynamic loading conditions experienced by the mandible by simulating the actuation of aggregate muscle groups. The student will be required to investigate anatomical muscle attachment points, force vectors and magnitudes, and effectively reproduce published FEA results (strain measurement) in a laboratory environment. Feedback control of at least one muscle actuator will be required.
The scope of the project is limited to demonstrating the device with the aid of an anatomical model. Human subjects will not be involved. Review of relevant FEA papers will be required, but performing an FEA analysis is out of scope.
The use of generative AI for any purpose will be actively discouraged in this project due to the documented harm that it causes to learning. Otherwise, the usual rules regarding responsible use of AI will apply throughout.
Recommended reading:
Celebi, N., Rohner, E.C., Gateno, J., Noble, P.C., Ismaily, S.K., Teichgraeber, J.F. and Xia, J.J., 2011. Development of a mandibular motion simulator for total joint replacement. Journal of Oral and Maxillofacial Surgery, 69(1), pp.66-79.
Cheng, K.J., Zhang, Q.Q., Zhang, F., Wang, R. and Liu, Y.F., 2024. Biomechanical behavior of temporomandibular joint movements driven by mastication muscles. International Journal for Numerical Methods in Biomedical Engineering, 40(10), p.e3862.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Mechatronics (424) | Modelling (334)

Development of a reconfigurable cable driven robot to simulate loading conditions experienced by a reverse shoulder implant

Reference	: JVDM5
Supervisor	: Dr Johan van der Merwe
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
The glenohumeral joint (GHJ) connects the humerus to the scapula. In case of injury or disease, joint reconstruction is required to restore normal function to the shoulder, via implantation of a reverse shoulder replacement. Understanding the forces acting on the shoulder is crucial during implant design and verification, such as when performing Finite Element Analyses (FEA). However, it is difficult to measure the mechanics of the shoulder during activities of daily living.
Therefore, the purpose of this project is to build a reconfigurable cable driven robot that will reproduce the various static and dynamic loading conditions experienced by the shoulder by simulating the actuation of aggregate muscle groups. The student will be required to investigate anatomical muscle attachment points, force vectors and magnitudes, and effectively reproduce published FEA results (strain measurement on the implant) in a laboratory environment. Feedback control of at least one muscle actuator will be required.
The scope of the project is limited to demonstrating the device with the aid of an anatomical model. Human subjects will not be involved. Review of relevant FEA papers will be required, but performing an FEA analysis is out of scope.
The use of generative AI for any purpose will be actively discouraged in this project due to the documented harm that it causes to learning. Otherwise, the usual rules regarding responsible use of AI will apply throughout.
Recommended reading:
Guo, R., Ferle, M., Nebel, D. and Hurschler, C., 2023. The development and evaluation of an in-vitro shoulder simulator with active muscle simulation. Scientific Reports, 13(1), p.4423.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Mechatronics (424) | Modelling (334)

Design of a passive, low-profile, disposable sensor system for measuring mandible motion

Reference	:	JVDM6
Supervisor	:	Dr Johan van der Merwe
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The temporomandibular joint (TMJ) connects the mandible (lower jaw) to the skull. In case of injury or disease, temporomandibular replacements are surgically implanted to restore normal function to the joint. Understanding the motions of the TMJ is crucial during implant design and verification.
The purpose of this project is to design a low profile and disposable sensor for measuring the three-dimensional motions of the mandible during activities of daily living. The motions of a 3D model of the mandible must be measured and described in terms of an anatomical coordinate system during activities such as speaking or yawning. The measurement system may not rely on optical tracking, and must instead consider passive sensing elements, mounted directly against the teeth via dental cement or 'invisible braces', to avoid interfering with normal behavior during measurement and ensure patient safety.
While the system is intended to be used within a dentist's practice, by the dentist themselves, the scope of the project is limited to demonstrating the system in a laboratory with the aid of an anatomical model. Human subjects will not be involved. Miniaturization must be accounted for in the design, but will not be required.
The use of generative AI for any purpose will be actively discouraged in this project due to the documented harm that it causes to learning. Otherwise, the usual rules regarding responsible use of AI will apply throughout.
Recommended reading:
Woodford, S.C., Robinson, D.L., Abduo, J., Lee, P.V. and Ackland, D.C., 2023. Kinematics of the jaw following total temporomandibular joint replacement surgery. Journal of Biomechanics, 159, p.111741.
Zambrana, N., Sesma, N., Fomenko, I., Dakir, E.I. and Pieralli, S., 2022. Jaw tracking integration to the virtual patient: A 4D dynamic approach. The Journal of Prosthetic Dentistry.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Mechatronics (424) | Experimental Techniques (414)

Design of a passive, low-profile, disposable sensor system for measuring mandible loads

Reference	:	JVDM7
Supervisor	:	Dr Johan van der Merwe
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The temporomandibular joint (TMJ) connects the mandible (lower jaw) to the skull. In case of injury or disease, temporomandibular replacements are surgically implanted to restore normal function to the joint. Understanding the loads of the TMJ is crucial during implant design and verification.
The purpose of this project is to design a low profile and disposable sensor for measuring the three-dimensional loads of the mandible during activities of daily living. The biomechanics of a 3D model of the mandible must be measured and described in terms of an anatomical coordinate system during activities such as chewing or biting. The measurement system must consider passive sensing elements, mounted directly against the teeth via dental cement or 'invisible braces', to avoid interfering with normal behaviour during measurement and ensure patient safety.
While the system is intended to be used within a dentist's practice, by the dentist themselves, the scope of the project is limited to demonstrating the system in a laboratory with the aid of an anatomical model. Human subjects will not be involved. Miniaturization must be accounted for in the design, but will not be required.
The use of generative AI for any purpose will be actively discouraged in this project due to the documented harm that it causes to learning. Otherwise, the usual rules regarding responsible use of AI will apply throughout.
Recommended reading:
Ledet, E.H., Liddle, B., Kradinova, K. and Harper, S., 2018. Smart implants in orthopedic surgery, improving patient outcomes: a review. Innovation and entrepreneurship in health, pp.41-51.
Merema, B.B., Sieswerda, J.J., Spijkervet, F.K., Kraeima, J. and Witjes, M.J., 2022. A contemporary approach to non-invasive 3D determination of individual masticatory muscle forces: a proof of concept. Journal of personalized medicine, 12(8), p.1273.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Mechatronics (424) | Experimental Techniques (414)

Design of a Stewart robot concept demonstrator for cervical spinal disc wear simulation

Reference	:	JVDM8
Supervisor	:	Dr Johan van der Merwe
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Cervical spinal discs connect the vertebrae in the neck, facilitating motion and load bearing while protecting the spinal cord. Injury or disease may require surgical replacement with an implant. Development of new implants is challenging and requires multiple stages of verification and validation prior to testing on human subjects. Simulated wear testing is one such process, whereby cyclic loads and motions that are representative of the in-vivo situation are applied to physical implant prototypes to simulate conditions for an accelerated life cycle.
The aim of this project is to investigate the use of a Stewart platform for the purposes of cervical spinal disc wear simulation. A Stewart platform (6UPS) is a type of parallel robot, consisting of six radially arranged linear actuators that act on a mobile platform. The advantages of a parallel robot compared to the more traditional serial kinematic actuator chain (e.g., a robot arm), include greater stability, precision, and load bearing capacity per weight; all potentially desirable characteristics for wear simulation. However, the dynamic analyses of parallel robots are challenging, making their design and control more complex.
This is a challenging project. The candidate will be expected to perform independent self-study of third-year level Stewart platform modelling and control theory, model the system in MATLAB and explore the workspace through inverse kinematic simulation, in order to perform detailed design including actuator and sensor selection. It is recommended that the candidate purchases a copy of "Parallel Robots" by Taghirad. Load, stress, and failure analyses must be conducted on the machine components under simplified, worst-case assumptions.
The initial concept demonstrator should focus on reproducing the kinematics described by the relevant ISO 18192-1:2011 standard, although the full requirements of the standard, such as environmental control, lubrication, and intermittent decanting of wear debris, must be taken into consideration throughout the design. Possible extended scope includes dynamic force modelling and control. Actual wear simulation will not be required.
The use of generative AI for any purpose will be actively discouraged in this project

due to the documented harm that it causes to learning. Otherwise, the usual rules regarding responsible use of AI will apply throughout.
Recommended reading:
ISO 18192-1, 2011. Implants for surgery - Wear of total intervertebral spinal disc prostheses. Part 1: Loading and displacement parameters for wear testing and corresponding environmental conditions for test
Taghirad, H.D., 2025. Parallel robots: mechanics and control. CRC press.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Mechatronics (424) | Modelling (334)

Design of a pin-on-plate friction test rig for biomedical engineering material pair characterisation

Reference	:	JVDM9
Supervisor	:	Dr Johan van der Merwe
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Articulating joint implants are made from engineering material pairs, typically a metal alloy and a polymer or ceramic component. Various material pair combinations and surface treatments interact differently within the complex physiological environment of a synovial joint. Therefore, thorough characterisation of material pair combinations is crucial to implant design.
The aim of this project is to design a pin-on-plate friction test rig for biomedical engineering material pair characterisation. The test rig must conduct reciprocal motion of a polymer or ceramic plug against a metal counterface under axial loading. The test apparatus must maintain appropriate environmental conditions, such as serum lubricant and temperature control. Finally, the apparatus must facilitate determination of the dynamic friction coefficient within a laboratory environment according to relevant engineering standards.
The use of generative AI for any purpose will be actively discouraged in this project due to the documented harm that it causes to learning. Otherwise, the usual rules regarding responsible use of AI will apply throughout.
Recommended reading:
Cowie, Raelene M., Adam Briscoe, John Fisher, and Louise M. Jennings. "Wear and Friction of UHMWPE-on-PEEK OPTIMA™." Journal of the Mechanical Behavior of Biomedical Materials 89 (2019): 65-71.
Forster, H. and Fisher, J., 1999. The influence of continuous sliding and subsequent surface wear on the friction of articular cartilage. Proceedings of the Institution of Mechanical Engineers, Part H: Journal of Engineering in Medicine, 213(4), pp.329-345.

Key modules (Scientific and Engineering Knowledge):
Mechatronics (424) | Modelling (334) | Experimental Techniques (414)

1.2 Mechanics Division

Ship motion and fatigue estimation on a model-scale ship structure

Reference	:	ABEK1
Supervisor	:	Prof Anriette Bekker
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Finite Element Methods (FEM)

Description:
This work entails the scaling and construction of a model-scale ship hull structure for investigation in the Stellenbosch Towing Tank. The structure will be characterized dynamically through modal analysis in a laboratory environment. Understanding of the dynamics of the structure will lead to insight into the mode shapes and possible stress hot-spots. The structure will then be exposed to excitation through water interaction in the Stellenbosch Towing Tank or CSIR basin test facility. Analytical frameworks are to be developed for motion and fatigue estimation of the structure, based on the measurements selected. This work serves to inform the current debate on the usefulness of structural digital twins for the shipping industry. The project is somewhat open in scope and lends itself to several potential areas of technical depth including finite element investigation, computational fluid dynamics or mechatronic design. The potential practical implication of this work includes a contribution to vessel management in South African deep sea mining operations. Furthermore, the structure is poised to contribute to future short courses for the maritime industry, particularly the vibrant activity in catamaran hulls in the Western Cape.

Key modules (Scientific and Engineering Knowledge):
Strength of Materials W (334) | Vibration and Noise (354) | Experimental Techniques (414)

The investigation of wheel forces and track deflection for South Africa's new fleet of passenger trains

Reference	: ABEK2
Supervisor	: Prof Anriette Bekker
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: Finite Element Methods (FEM)

Description:
An instrumented track section could serve as a cost-effective investment to evaluate the condition of train wheels. The aim of the project is to enable the extraction of wheel loads from track measurements on a passing train. The approach will start with the investigation and of vertical loads. The idea is to monitor the shear strain and vertical track deflection on a section of rail such that the force can be measured for a full revolution of a rail wheel. First, the project can be tackled by using a McCauly beam structure, where a force is progressively moved to simulate the passage of a rail wheel. From here, the instrumentation of a laboratory track section can be planned and executed. Measurement configurations including strain, displacement and acceleration will be designed and practically investigated using loading by a single wheelset. Artificial defects could be induced in a controlled manner to evaluate the effect on the laboratory load signals and how these map to the rotational domain of the wheel. With sufficient progress, operational installation and measurement can be considered.
Journal article with background: https://doi.org/10.1016/j.measurement.2022.111297
The Gibela Engineering Research Chair (at the Department of Mechanical and Mechatronic Engineering) will support the project, arrange access to Gibela depots and trains, and provide funding for prototyping. Students will enjoy access to industry, additional development opportunities and colloquia.
Read more at: https://www.gerc.sun.ac.za

Key modules (Scientific and Engineering Knowledge):
Finite Element Methods (414) | Strength of Materials W (334) | Vibration and Noise (354)

Subsea drill signal analysis to inform real-time material composition

Reference	:	ABEK3
Supervisor	:	Prof Anriette Bekker
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
A seabed sampling drill is used to collect samples for a near-shore marine diamond mining operation. The drill forms part of a larger sampling system, consisting of a seabed sampling tool which supports the drill, its power source and instrumentation as well as a surface vessel that processes the samples. While the drill is well instrumented with a number of mechanical and acoustic sensors, the insight generated from these sensors is currently limited in scope. An opportunity has been identified to use the acoustic sensors on the seabed sampling tool to provide real time insight into the composition of the material being drilled. A large set of acoustic data is available for analysis as well as other mechanical data such as drill torque and observations of various samples from geologists deployed on the sampling vessel. The student will be required to apply signal processing techniques to these data sets to determine whether material composition and volume can be indicated in a quantitative manner. Scale testing may be required to provide control datasets against which in-field data can be correlated.
This work offers an opportunity for a student with high initiative and a tendency towards innovative practical work.

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Vibration and Noise (354) | Experimental Techniques (414)

LiDAR and complimentary measurements to digitize the rail environment.

Reference	: ABEK4
Supervisor	: Prof Anriette Bekker
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:
Rail networks comprise expansive distances and geographic areas. This challenges the ability of rail operators to inspect and monitor this asset on a continuous basis. Train-mounted technologies such as LiDAR, cameras and accelerometers, coupled with data-driven algorithms augment human ability to keep track of evolving track environments. Literature points to applications such as the automated extraction of vegetation encroachment, rail position and catenary geometry from LiDAR. Additionally, LiDAR technology has been harnessed to capture the geometrical context of rail accident scenes as well as measurements related to permanent sensor installations on rail tracks. It is proposed to evaluate the use of LiDAR in the rail context, possibly in combination with other onboard sensing equipment, to initiate onboard track monitoring for South Africa's passenger trains, built by the Gibela Rail Transport Consortium.
The Gibela Engineering Research Chair (at the Department of Mechanical and Mechatronic Engineering) will support the project, arrange access to Gibela depots and trains, and provide funding for prototyping. Students will enjoy access to industry, additional development opportunities and colloquia.
Read more about our work on : https://www.gerc.sun.ac.za

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Design (E) (314) | Experimental Techniques (414)

Improved measurement of powder spreadability for additive manufacturing

Reference	: DB1
Supervisor	: Prof Deborah Blaine
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
Laser powder bed fusion (LPBF) is an additive manufacturing technique that uses a high power laser to fuse thinly spread layers of metal powder together, to progressively build the required part shape, layer by layer. Metal powders consist of a range of powder particle shapes and sizes which causes them to pack, flow and spread in different ways. Powder characterisation techniques attempt to provide a set of parameters that will help engineers in designing powder manufacturing processes that produce high quality, defect-free parts. For LPBF, powder spreadability is an important parameter in ensuring consistent powder layers and final material quality. SU's Materials laboratory has a custom-built powder spreadability test rig that was designed to help characterise powders in terms of their suitability for LPBF processing. A previous project designed a wedge-system to allow for fine adjustment (10 um) of the layer height for spreadability testing; however, the accuracy of this adjustment is negatively affected by the stacking errors of tolerance in the system and requires refinement. The goal of this project is to implement design adjustments to allow for more accurate and repeatable spreadablity testing using the wedge height adjustment mechanism. As such, it will entail a design review, new design implementation, testing of the design changes through evaluation of the spreadability of commonly used AM powders in the laboratory.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Experimental Techniques (414)

Evaluation of a flash apparatus for measuring thermal conductivity of metal powder structures

Reference	:	DB2
Supervisor	:	Prof Deborah Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The laser flash method is an advanced application of the original flash method, developed in 1961 by Parker et al, to measure the thermal conductivity of materials. Commercially available laser flash equipment is expensive and has to be imported into South Africa. A low-cost version of this apparatus, using an LED source for heating, was designed and built in 2025, and initial experiments showed that it yields promising results. The aim of this project is to review the design, implement design changes to improve the measurements, and use the apparatus to measure the thermal conductivity of loose metal powder and sintered structures.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Heat Transfer A (414) | Machine Design A (314)

Metal powder screen printing apparatus: design improvement and evaluation

Reference	:	DB3
Supervisor	:	Prof Deborah Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
A laboratory scale 3D screen printing apparatus has been designed and built for manufacturing thin metal structures from metal powder slurries. While the screen printing mechanism works well, the equipment has yet to produce a cohesive thin metal film as required. The goal of this project is to investigate different slurry formulations and screen printing parameters with the purpose of printing a cohesive thin metal film. Additionally, the process requires a heating stage for drying the slurry after deposition. The heating mechanism for the build plate must be designed and integrated into the existing system. The design must be evaluated for defect-free drying of the thin films after slurry deposition.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Heat Transfer A (414) | Experimental Techniques (414)

Improved ambulation mechanism for a 3D printed, robotic toy unicorn

Reference	:	DB4
Supervisor	:	Prof Deborah Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
3D printing using plastic filaments opens up a world of possibility for rapid prototyping of designs. However, very often initial design run into issue with the surface roughness, tolerances and orthogonal mechanical properties that are inherent to 3D printed parts. In a previous project, a solar-powered toy unicorn was designed and built using 3D printing to be used as an educational toy to illustrate a variety of engineering concepts. The unicorn is able to walk, with the ambulation mechanism based on a crank and shaft design that is also 3D printed. However, due to surface roughness and tolerance challenges with the printed parts, the mechanism does not operate very smoothly and the small connector pins are prone to fracture. For this project, the student is required to refine the design of the crank and shaft with gearbox mechanism such that it can still be integrated into the overall design of the current unicorn, but so that the unicorn is able to walk more smoothly.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314)

Design of a graphic user interface (GUI) for an educational experimental stress analysis apparatus

Reference	:	DB5
Supervisor	:	Prof Deborah Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
An experimental stress analysis apparatus that consists of a series of connected beams structures that have been instrumented with strain gauges has been designed as a teaching and learning tool for the undergraduate module, Strength of Materials W244. At present, the apparatus can capture strain measurements using portable strain gauge measurement equipment. However, it requires a learning assistant to fully explain the connection between the output readings and different loading scenarios. The goal of this project is to design and programme a simple graphic user interface (GUI) that clearly guides students in demonstrating the stress states that arise for different loading scenarios using this apparatus, and allows students to test out and troubleshoot their solutions. The system can potentially incorporate interfacing with generative AI platforms for "tutorbot" chats as students explore the different loading scenarios.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Experimental Techniques (414)

Sacrificial anode for water heater with embedded wear indicator

Reference	:	DB6
Supervisor	:	Prof Deborah Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	22933042
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
US Patent US6932891B2 expired in 2024 which means it is now freely available for general use. The patent describes the design of a radio frequency identification (RFID) wear indicator that is embedded in a sacrificial anode for a water heater (https://patents.google.com/patent/US6932891B2/en). Sacrificial anodes are instrumental in ensuring extended life of water heaters (geysers) but visual checking of them is very poorly monitored in most households as it requires removing and visually inspecting the anode on a regular basis. With this design, a radio signal can be monitored, allowing for remote monitoring of the anode. The goal of this project is to design a process for retrofitting locally available sacrificial anodes with the embedded sensor, experiment with and evaluate different setups, and identify a solution that would be viable for commercialisation in South Africa.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Experimental Techniques (414)

Sacrificial anode depletion detection for water heater

Reference	: DB7
Supervisor	: Prof Deborah Blaine
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:
US Patent US20140218005A1 was abandoned in 2016 which means it is now freely available for general use. The patent describes a hardware circuit for measuring the current between a sacrificial anode and a water heater shell for monitoring anode depletion (https://patents.google.com/patent/US20140218005A1/en). Sacrificial anodes are instrumental in ensuring extended life of water heaters (geysers) but visual checking of them is very poorly monitored in most households as it requires removing and visually inspecting the anode on a regular basis. With this design, monitoring the anode rod current allows for remote monitoring of the system and can provide a means of indicating when the anode needs to be replaced. The goal of this project is to design a process for retrofitting locally available sacrificial anodes with the anode current detection device, evaluating the setup, and identifying a solution that would be viable for commercialisation in South Africa.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Experimental Techniques (414)

Corrosion solutions for 18650 lithium-ion rechargeable batteries

Reference	:	DB8
Supervisor	:	Prof Deborah Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
18650 Lithium-ion batteries are a standardised and widely-used type of cylindrical rechargeable battery, utilised over a wide range of consumer products such as torches, laptops, electric vehicles and power tools. A common issue that arises in apparatus that use these batteries is corrosion of the positive terminal. The goal of this project is to investigate potential causes for this corrosion, and to evaluate different approaches to counteracting this corrosion. Currently, aluminium foil tape is used as a sacrificial anode, or contact grease as a protective coating. A full investigation into the mechanisms that govern these interventions, and a description of how each solution acts to prevent corrosion is required, along with recommendations for best use case scenarios.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Mechatronics (424) | Experimental Techniques (414)

Sustainable, temperature-sensitive beehive ventilation system

Reference	:	DB9
Supervisor	:	Prof Deborah Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Man-made beehives suffering from overheating in the hot summer months when temperatures inside the hive exceed 36 deg Celcius. As many beehives in South Africa are located in remote rural spaces, it is not always possible to reach all the beehives in hot weather to open ventilation systems. Additionally, permanently open ventilation systems are often glued closed by the bees during cooler months to maintain optimal temperatures for the hive. The goal of this project is to design a ventilation system that will automatically open when temperatures inside the hive exceed 35 deg Celcius. The ventilation system must not rely on electronic control, i.e. this must be a mechanically controlled system, and must be protected from the bees natural instinct to close openings to the hive.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Machine Design A (314) | Machine Design B (344)

Vibration-Assisted Powder Flow Optimization For Multi-Material recoater System

Reference	:	RB1
Supervisor	:	Mr Raymond Botete
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Laser Powder Bed Fusion (LPBF) is a leading additive manufacturing process for producing high-precision metal components. Despite its advantages, incorporating multiple materials within a single build remains a significant challenge due to difficulties in accurate and selective powder deposition. A previous project developed an adaptive powder recoater capable of spreading multiple powders on the same build platform, but testing revealed critical limitations in powder flow, layer accuracy, and material integrity. To address these issues, this project focuses on enhancing the nozzle design by integrating a piezoelectric vibration system to improve powder flow and reduce clumping. The student will experimentally determine optimal vibration frequencies and amplitudes for different metal powders and validate their effect on achieving uniform deposition across varying layer heights and spreading patterns. Deliverables include a functional vibration-enabled nozzle assembly, a dataset correlating vibration parameters with deposition quality, and recommendations for operating conditions to ensure consistent multi-material spreading.

Key modules (Scientific and Engineering Knowledge):
Vibration and Noise (354) | Experimental Techniques (414)

Automated Powder Deposition Control For Multi-Material Recoater System

Reference	:	RB2
Supervisor	:	Mr Raymond Botete
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Accurate powder deposition in multi-material LPBF requires precise coordination of nozzle actuation, valve timing, and powder flow. The previous adaptive powder recoater prototype experienced start/stop deposition artifacts and unreliable valve performance due to limited electronics and firmware capabilities. This project aims to upgrade the control system, implementing advanced timing logic to synchronize solenoid valve actuation with vibration and nozzle motion, reducing deposition errors and improving repeatability.
The student will develop, integrate, and test the upgraded control system on the existing prototype, evaluating improvements in deposition consistency and reliability. Deliverables include a fully functional automated control system, test results demonstrating reduced start/stop artifacts, and operational guidelines for reliable multi-material deposition.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Mechatronics (424)

Contamination Control & Purge Strategies For Multi-Material Recoater System

Reference	:	RB3
Supervisor	:	Mr Raymond Botete
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Laser Powder Bed Fusion (LPBF) enables high-precision metal fabrication, but using multiple powders in a single build introduces the risk of cross-contamination, compromising layer quality and material integrity. The existing adaptive powder recoater prototype demonstrated the ability to deposit multiple powders, yet residual material on the nozzle and build surface led to cross-contamination between successive powders. This project focuses on designing, developing, and integrating an effective cleanup mechanism into the prototype to remove residual powders between material changes.
The student will be responsible for conceptualising a cleanup solution, implementing it on the prototype, and experimentally validating its effectiveness. Deliverables include a fully integrated, functional cleanup system, experimental data demonstrating reduced cross-contamination, and operational guidelines for reliable multi-material deposition in LPBF processes.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Experimental Techniques (414)

Tabletop dynamic linear fatigue tester

Reference	:	NMD1
Supervisor	:	Dr Nur Mohamed Dhansay
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
This project requires the design, build and testing of a small scale (tabletop) fatigue tester. Ideally, it needs to house a compact tension specimen and conform to the ASTM E647 standard. The fatigue tester will have to control and monitor the load and frequency of the dynamic cycles in various ways. The interest lies in that the student will have to incorporate a (or multiple) suitable crack detection method(s) which can automatically adjust the load level accordingly(variable amplitude loading as opposed to constant amplitude loading). This means that the fatigue tester should also be capable of being controlled via the stress intensity factor. Given that the system will have linear motion, suitable alignment technique needs to be in place to assure that specimen alignment is accurate and does not cause any significant/unwanted bending. This tester should also incorporate a camera attachment for both remote viewing and control.
It is expected that the student follows an established design methodology in this project. Typically, it is an iterative process and involves design requirements to be established, showing how the concept is developed, a detailed design and analysis of the selected design, and an evaluation of the design.
For mechanical students, there will be aspects of mechatronics which you will be required to learn on your own. For mechatronics student, there will be aspects of mechanical which you will be required to learn on your own.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Vibration and Noise (354)

Small scale rolling contact fatigue tester for railway track

Reference	:	NMD2
Supervisor	:	Dr Nur Mohamed Dhansay
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Rolling contact fatigue (RCF), as the name suggests, is a near surface fatigue process which occurs between surfaces which are in contact with each other. This phenomenon if typically found in bearings, gears, railway lines etc. The issue lies in that testing RCF for railways, typically, occur at full scale. This of course requires a large space and resources to move the test specimens (railway line and wheelset) to the relevant location.
The student is required to design, build and test a small scale RCF tester specific to railway tests. This means that the equipment needs to consider straight and curved rolling contact regimes, inner and outer styled wheel flanges, mapping of the lateral force and angle of attack, amongst others, to analyse the wear effects on the wheelset and railway track.
It is expected that the student follows an established design methodology in this project. Typically, it is an iterative process and involves design requirements to be established, showing how the concept is developed, a detailed design and analysis of the selected design, and an evaluation of the design.
For mechanical students, there will be aspects of mechatronics which you will be required to learn on your own. For mechatronics student, there will be aspects of mechanical which you will be required to learn on your own.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Electrical Drive Systems (324) | Vibration and Noise (354)

Rotating bending fatigue machine

Reference	:	NMD3
Supervisor	:	Dr Nur Mohamed Dhansay
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
This project requires the design, build and testing of a rotating bending fatigue tester. Ideally, it needs to conform to the ASTM E2948 and/or ISO1143. The fatigue tester will have to control and monitor the load(s) and frequency of the dynamic cycles in various ways. The system should also be capable of being controlled and monitored remotely. This type of testing allows for fully reversed, uniaxial type, fatigue loading using rotation rather than linear actuation. It is highly desired for the student to add torsional fatigue to the system, resulting in a multiaxial type fatigue testing machine. It is expected that the student follows an established design methodology in this project. Typically, it is an iterative process and involves design requirements to be established, showing how the concept is developed, a detailed design and analysis of the selected design, and an evaluation of the design.
For mechanical students, there will be aspects of mechatronics which you will be required to learn on your own. For mechatronics student, there will be aspects of mechanical which you will be required to learn on your own.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Electrical Drive Systems (324) | Vibration and Noise (354)

Basic introduction to finite element method practical kit for 3rd year students

Reference	:	NMD4
Supervisor	:	Dr Nur Mohamed Dhansay
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Finite Element Methods (FEM)

Description:
This project requires the design, build, and test of an experimental setup to be used which will introduce 3rd year students to finite element methods. Typically, engineering problems can be solved by analytical, experimental and numerical approaches. The student is expected to build an experimental setup, such as but not limited a truss system, and verify its accuracy through analytical (what you've learnt through the strength of materials chain) and numerical approach (FEM). The experimental setup will require a variety of instrumentation such that the relevant stresses/loads and strains/displacements can be measured, monitored and controlled.
It is expected that the student follows an established design methodology in this project. Typically, it is an iterative process and involves design requirements to be established, showing how the concept is developed, a detailed design and analysis of the selected design, and an evaluation of the the design.

Key modules (Scientific and Engineering Knowledge):
Finite Element Methods (414) | Machine Design A (314) | Strength of Materials W (334)

Bicycle frame fatigue testing machine

Reference	:	NMD5
Supervisor	:	Dr Nur Mohamed Dhansay
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
This project requires the design, build and testing of a bicycle frame fatigue testing machine. While it is common practice to obtain fatigue data for a material by using standardised geometries, such as compact tension specimens. it is also common practice to then test the actual application geometry (in this case, a bicycle frame) to determine its integrity. It is desired that a bicycle frame fatigue tester is built which can, ideally, test various components of the frame, i.e. pedals, forks, handlebars, saddle etc. Realistically, the design will have to be modular such that it will test one component at a time and then adjusted to test a different frame component when desired. This machine should be able to replicate the expected real-world riding forces. Of particular interest is out-of-phase and in-phase loading conditions. This is when under certain riding conditions, the load shifts from one side (of pedal and/or handlebar) to the other (out-of-phase). The are other riding conditions where load is applied simultaneously to both pedals and/or both sides of the handlebar (in-phase). The fatigue tester will have to control and monitor the load and frequency of the dynamic cycles in various ways.
It is expected that the student follows an established design methodology in this project. Typically, it is an iterative process and involves design requirements to be established, showing how the concept is developed, a detailed design and analysis of the selected design, and an evaluation of the design.
For mechanical students, there will be aspects of mechatronics which you will be required to learn on your own. For mechatronics student, there will be aspects of mechanical which you will be required to learn on your own.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Electrical Drive Systems (324) | Vibration and Noise (354)

Improvment on the design of a rotating drum mechanism for measuring dynamic angle of repose

Reference	:	GOV1
Supervisor	:	Mr Preyin Govender
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The dynamic angle of repose refers to the maximum angle at which a material can be piled or stacked without the particles sliding or cascading off. Unlike the static angle of repose, which is determined when the material is at rest, the dynamic angle takes into account the effects of motion or flow. It is commonly observed in powders and is influenced by factors such as particle shape, size distribution, and moisture content. The dynamic angle of repose is important in industries like material handling, where the behavior of bulk materials during transportation or storage must be understood and managed. The student will be tasked with improving the design and manufacturing of a cost friendly rotating drum that was used to determine the dynamic angle of repose for various metal powders.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

The Effect of Powder Size Distribution (PSD) of WC-Co powders on Build Quality through laser powder bed fusion

Reference	:	GOV2
Supervisor	:	Mr Preyin Govender
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
The purpose of this investigation analysises how the powder size distribution (PSD) of WC-Co hardmetal powders influences powder spreading, porosity formation, and microcracking during Laser Powder Bed Fusion (LPBF). WC-Co is a challenging material to process through additive manufacturing due to its high melting point, low ductility, and susceptibility to thermal cracking. The PSD strongly affects powder flowability, packing density, and energy absorption, making it a key parameter in achieving high-quality LPBF builds.
In this study, the student will be tasked with preparing various powder sizes by sieving existing WC-Co powder. Each batch will be characterised for particle morphology and flowability. Small LPBF test cubes will be printed using identical process parameters to isolate the effect of PSD. Microstructural analysis will quantify porosity, crack density, and final hardness. The results will help determine which PSD provides the best balance of spreadability and thermal behaviour for printing dense, crack-minimised WC-Co components. This work supports the optimisation of hardmetals processing for wear applications in mining and tooling.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Strength of Materials W (334)

Laser Re-melting Effect on WC-Co Manufactured via Laser Powder Bed Fusion

Reference	: GOV3
Supervisor	: Mr Preyin Govender
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
The aim of this investigation is to explore how in-situ laser re-melting influences the surface quality, microstructure, and crack formation of WC-Co hardmetals produced by Laser Powder Bed Fusion (LPBF). WC-Co commonly suffers from high thermal gradients during printing, leading to microcracking, binder (Co) depletion, and poor surface finishes. Laser re-melting, where the laser scans the top layer again after deposition, may help reduce surface roughness, heal shallow defects, and improve microstructural homogeneity.
In this study, small WC-Co samples will be manufactured using identical base parameters, but with varying degrees of re-melting. Each condition will be evaluated for changes in surface roughness, porosity, and crack density. Microhardness testing will assess whether re-melting alters carbide distribution or binder pooling. The outcomes will help determine whether re-melting is an effective, low-cost strategy to enhance build quality and reduce defect formation in LPBF-produced WC–Co components, supporting improved performance for wear-resistant applications.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Strength of Materials W (334)

Microcracking Analysis of WC-Co Using Varias Printing Parameters in Laser Powder Bed Fusion

Reference	:	GOV4
Supervisor	:	Mr Preyin Govender
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
WC-Co is a hard-metal composite where tungsten carbide particles provide extreme hardness and wear resistance, while the cobalt binder gives toughness. It is widely used in cutting tools, drilling bits, wear-resistant components, mining equipment, and additive-manufactured hard-metal parts. This project examines how various printing parameters influence microcrack formation in WC-Co parts manufactured using Laser Powder Bed Fusion (LPBF). WC-Co hardmetals are prone to severe thermal stresses during printing, often resulting in microcracks that reduce strength and reliability. Understanding how process parameters control these defects is essential for improving printability and material performance.
The study will print a series of small LPBF samples using combinations of different printing parameter sets with varying energy density. Each sample will be analysed for crack density, porosity, and microhardness, providing a comprehensive view of how thermal gradients and melt pool behaviour contribute to cracking. The results will identify parameter combinations that minimize microcracking and optimise density, offering valuable insight for producing high-quality WC-Co components through LPBF.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Strength of Materials W (334)

Extracting external fixation device configurations from X-ray images using computer vision

Reference	:	PFL1
Supervisor	:	Mr Philip Ligthart
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Tygerberg hospital has wealth of X-ray imaging data of patients with fractured long bones and external fixation devices. This project is a pilot study to build a framework to extract the geometric configuration of the external fixation device, like the Ilizarov frame, from the X-rays. The student will create a synthetic dataset of pairs of 2D images where and then classical computer vision techniques using Matlab or Python, not machine learning, will be used to extract the 3D device configuration. Parameters such as distances between components, wires or pins orientation, etc. can then be determined. There will be many variables that have an effect on the framework and the effect of these on the quality of extracted 3D configuration should be explored. Finally, a proof of concept may be tested using X-ray images.
This project will require a lot of programming and learning classical computer vision topics. Mathematics and geometry play a big part in computer vision.

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Modelling (334) | Experimental Techniques (414)

Design of a parallel manipulator for tibia gait simulation

Reference	:	PFL2
Supervisor	:	Mr Philip Ligthart
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
In this project, the student will design, build, and test a robotic test rig capable of replicating the complex three-dimensional kinematics of the human tibia during a gait cycle. Physiological knee motion involves coupled rotation and translation, to capture this, a 3D parallel manipulator to control flexion, abduction, and internal rotation is required. This device can then be used to replicate healthy or injured gait motion captured in a gait laboratory.
The primary objective is high-fidelity trajectory control to ensure the device accurately follows physiological motion data. A secondary goal is the integration of force control to simulate ground reaction loads. Affordable loadcells and camera based tracking may be necessary. This is a multidisciplinary project requiring mechanical design, kinematic modeling, programming, and experimentation.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Modelling (334) | Experimental Techniques (414)

Development and testing of a low-cost stereo digital image correlation system

Reference	: MNS1
Supervisor	: Dr Melody Neaves
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:
The Materials Engineering Research Group (MATENG - https://www0.sun.ac.za/mateng/) is routinely involved in the mechanical testing of many materials. One type of material characterisation involves determining mechanical properties of materials by measuring deformation through digital image correlation (or DIC for short). DIC is a powerful non-contact, camera vision-based technology that measures deformation of the surface of mechanically stressed specimens. During testing, software tracks the motion of a random speckle pattern applied to the specimen surface across multiple images.
Commercial DIC setups are extremely expensive given the use of specialised camera and lens equipment. The hardware costs coupled with the licencing fees of commercialised DIC software makes the setup difficult to introduce into small laboratory environments for material property testing. For instance, the cost of the current LaVision DIC setup available in the MATENG laboratories is approximately R 1 500 000. The possibility of using easily accessibly low-cost cameras and/or lenses for DIC has largely been left unexplored. MATENG has previously explored the use of low-cost optics in planar DIC wherein one camera is used. Low-cost stereo systems (where more than one camera is used), however, are not widely available.
This project aims to develop a stereo DIC system using low-cost optics and camera combinations for DIC applications. By evaluating the impact of these cost-effective options on the accuracy of stereo DIC measurements, this research seeks to make DIC technology more accessible to a wider range of researchers and industries.
Project objectives include: 1. Identify suitable low-cost optics and camera combinations: Explore various affordable lens and camera options compatible with stereo DIC systems. Examples include borescopes, Raspberry Pi camera modules, and/or cell phone cameras. 2. Establish the calibration needs of the low-cost stereo setup: Calibration should allow for camera triangulation, distortion removal and image to global coordinate conversion. 3. Develop a standardised testing methodology: Establish a robust experimental framework to assess the performance of the stereo DIC setup under controlled conditions such as rigid body displacement and out-of-plane

testing. 4. Quantify the impact of low-cost optics on stereo DIC accuracy: Evaluate the influence of factors such as calibration efficacy, resolution, distortion, and noise on the precision and reliability of stereo DIC measurements. To this end, the commercial software or open-source software can be used. 5. Provide recommendations for low-cost stereo DIC systems: Offer practical guidelines for selecting and configuring affordable components to achieve acceptable levels of accuracy and precision.
The candidate need not develop new DIC software from scratch but should motivate the selection of existing software and calibration methods. They should demonstrate mastery of this software/methods and should ideally be able to integrate the chosen software into the DIC system by potentially modifying existing code in the case of open-source options.
Being comfortable or being willing to learn Python will be to your advantage.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Experimental Techniques (414)

Design of a “Pawsitive Playmate”: An automated multi-format dog training system

Reference	:	MNS10
Supervisor	:	Dr Melody Neaves
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Canine enrichment is essential for a dog’s mental well-being, preventing boredom and destructive behaviour. While human interaction is the gold standard for training, owners cannot always be present to provide consistent engagement. There is a growing need for automated "smart" devices that can autonomously guide a dog through positive reinforcement exercises. Existing market solutions are often limited to simple ball launchers or timed feeders or focus on corrective, punitive measures such as shock collars. This project aims to develop a comprehensive, all-in-one system that uses sensors and cues to gamify training, providing both mental stimulation and reward-based learning.
The student is required to design and build a low-cost, safe, standalone mechatronic device that functions as a positive reinforcement training hub. The system must feature multiple "games" or interaction modes (e.g., memory games, agility prompts, or "wait" commands). It must include sensors to detect proximity and specific actions (like touching a target), as well as a dispensing mechanism for delivering treats. The design should integrate an app/software that allows a dog-owner to adjust settings. Note: The final system will be validated through bench testing and simulated inputs; no testing on live animals is required or permitted for this project.
The overall goal of this project is to design, build, and test an automated, sensor-based training device (named the “Pawsitive Playmate”) that engages a user through multi-sensory cues and dispenses rewards upon the successful completion of various interactive tasks. The objectives include:
1. Research and select suitable positive reinforcement exercises for canine pets. 2. Design a robust, pet-safe enclosure theoretically capable of withstanding interaction with an enthusiastic dog. 3. Develop a reliable mechanical system for storing and dispensing treats of various sizes without jamming. 4. Integrate sensors to detect presence and validate task completion. 5. Program the device to run multiple distinct "game modes" using visual and/or audible cues. 6. Verify the system’s performance by simulating canine behaviour (e.g., manual activation of sensors) to demonstrate reliable state-switching and reward dispensing logic.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Machine Design B (344) | Mechatronics (424)

Design, construction, and commission of a miniature tension/compression testing rig

Reference	: MNS2
Supervisor	: Dr Melody Neaves
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:
The Materials Engineering Research Group (MATENG - https://www0.sun.ac.za/mateng/) is routinely involved in the mechanical testing of many materials. Some of these materials are either very rare or very expensive to produce. Conventional tension/compression testing machines often require specimens that are too large and that are too expensive to fabricate for these rarer materials. In many aerospace industries, for instance, the development of novel materials is largely slowed due to the expensive material requirement involved in thoroughly qualifying their properties. Miniaturisation of mechanical test specimens has therefore become increasingly important in these cases to use materials efficiently for testing.
Combined with specimen miniaturisation of specimens, powerful non-contact methods, such as Digital Image Correlation (DIC), allow deformation/strain to be measured in a full-field and data-rich manner. In other words, getting more information out of less material poses significant advantages to the efficiency of modern materials testing methods.
This project aims to provide a solution by designing, constructing, and commissioning a miniature tension/compression testing rig capable of accurately measuring the mechanical properties of small-scale specimens. To remain within the budget, the rig will be designed to accommodate a variety of low-strength materials, including 3D printed polymers and/or low-grade aluminium alloys. The rig should be capable of operating in tension and in compression and be capable of measuring load and displacement. Closed-loop control should also be applied to control and measure displacement. Compatibility with non-contact/optical deformation measurement techniques (such as DIC) should also be considered in the design specifications of the rig and should be ultimately tested in the final commissioning stage.
Project objectives include: 1. Formulation of specifications for the miniature tension/-compression testing rig 2. Concept development and selection 3. Construction and assembly of final design 4. Development of user-interface and displacement control system 5. Verification of the setup through tensile testing suitable materials (such as low strength 3D printed polymers and/or low-grade aluminium) using DIC and

comparing to literature
Design and programming skills will come in handy.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Machine Design B (344)

Development and testing of an adaptive illumination system for digital image correlation

Reference	:	MNS3
Supervisor	:	Dr Melody Neaves
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The Materials Engineering Research Group (MATENG - https://www0.sun.ac.za/mateng/) is routinely involved in the mechanical testing of many materials. One type of material characterisation involves determining mechanical properties of materials by measuring deformation through digital image correlation (or DIC for short). DIC is a powerful non-contact, camera vision-based technology that measures deformation of the surface of mechanically stressed specimens. During testing, software tracks the motion of a random speckle pattern applied to the specimen surface across multiple images.
The accuracy of DIC is critically dependent on the quality of the captured images, which require a high-contrast, well-defined speckle pattern. However, achieving optimal illumination is challenging; images that are under-lit (poor contrast) or over-lit (saturated) will cause the correlation algorithm to perform poorly or even fail. This project addresses the need for a robust, low-cost system that can intelligently manage illumination to ensure high-quality data acquisition, removing the "guesswork" from experimental setups.
The system must be built using low-cost components, such as a microcontroller (e.g., Raspberry Pi or Arduino) and an associated camera module. The final system should operate as a closed-loop, stand-alone device that can be integrated into a typical mechanical testing setup. A user interface should be developed that can give fast or live feedback to a user about the quality of the lighting on their specimens. The system should be automated as far as possible and should also be tested under different conditions and using different light sources.
The project goal is to design, build, and validate a closed-loop system that automatically analyses images of a speckle pattern and adjusts the test environment lighting to consistently maintain optimal image quality for DIC analysis. The project objectives include:
1. Select and/or develop the necessary hardware and software for the adaptive illumination system. Examples include microcontrollers, LEDs, drivers, etc. 2. Research and define real-time image processing "quality metrics" that can quantify the suitabil-

ity of a speckle pattern. Examples include those based on its histogram, contrast, or saturation levels or even other speckle pattern quality metrics such as MIG, subset entropy, SSSIG, etc. 3. Design a control algorithm that uses the quality metrics as feedback to actively manage the lighting environment. 4. Develop a simple, user-friendly interface for the system. 5. Implement the control system to demonstrate its ability to automatically achieve and maintain an optimal speckle pattern under different environmental conditions. 6. Validate the system's performance by comparing the quality of images and subsequent DIC analysis results from the "smart" system versus manually operated lighting.
Helpful Skills: Programming (Python/C++/MATLAB), microcontroller interfacing (e.g., Arduino, Raspberry Pi), basic electronics and circuit design, interest in image processing and machine vision

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Electronics (245) | Experimental Techniques (414)

Evaluating speckle pattern transfer methods for digital image correlation

Reference	: MNS4
Supervisor	: Dr Melody Neaves
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:
Have you ever wanted to be a tattoo artist but instead of skin, you work on engineering materials? The technique of Digital Image Correlation (DIC) relies heavily on the quality of the speckle pattern (or a unique "tattoo") applied to the material (eg. metal/polymer) specimen surface. DIC tracks the deformation of the specimen by correlating the deformation of the unique features of this speckle pattern. While traditional spray-painting is the industry standard, it is often inconsistent, user-dependent, and does not offer a wide variety in terms of speckle pattern types (such as lines, geometric shapes and even patterns that resemble animal fur patterns). Oftentimes it is preferred to be able to artificially generate optimal and consistent speckle patterns. However, the challenge that remains is the transfer and application of this synthetic speckle pattern from software to a physical specimen surface. Emerging techniques, such as transfer printing, temporary tattoos, or direct stamping, offer potential alternatives for creating high-fidelity, repeatable patterns. For example, this paper (https://iopscience.iop.org/article/10.1088/0957-0233/26/9/095201) uses a water transfer printing method (similar technology as a fake tattoo application). This project seeks to investigate and benchmark these pattern application methods against the traditional spray-paint approach to determine their viability for DIC measurement in materials testing.
The project requires the student to select and master at least two alternative speckle pattern application methods (e.g., water transfer printing, laser toner transfer, stamping etc.) and compare them against a control group of spray-painted specimens. The comparison must be quantitative, using standard DIC performance metrics as well as general pattern performance metrics (such as adhesion to specimen, ductility, representative of underlying specimen deformation during a tensile test, high contrast, etc.).
The project goal is to systematically evaluate and validate alternative speckle pattern transfer techniques to establish a robust, repeatable, and user-independent method for preparing DIC specimens. The objectives include: 1. Review literature to identify promising low-cost pattern transfer technologies suitable for various material surfaces

(metallic, polymeric, etc.). 2. Develop a procedure for applying patterns using at least two selected alternative methods and the standard spray-paint method. 3. Design and conduct mechanical tests (e.g., rigid body motion or hand-crank tensile tests) to capture images of the patterns under displacement/deformation. 4. Quantify the performance of each method using suitable metrics. 5. Produce a comparative study recommending the most effective application method based on (amongst others) the ease of use, cost, and measurement accuracy.

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Strength of Materials W (334) | Experimental Techniques (414)

Design, build and testing of a 3D hologram-like RGB LED fan display

Reference	:	MNS5
Supervisor	:	Dr Melody Neaves
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Some claim that in the Venn diagram of art and engineering, there is little to no overlap. However, several artistic inventions have been the result of the synergy between the technical prowess of engineering and the sensory enjoyment associated with the arts. One example of this is the development of 3D "holographic" RGB LED fan displays. These fan displays are not truly holographic but do give a holographic effect and have captivated audiences for decades, offering a mesmerising visual experience by allowing the 3D rendering of objects in a physical space. However, traditional holographic displays are often complex and expensive to produce. This project aims to design, build, and test a 3D hologram-like RGB LED fan display that is both affordable and versatile. By leveraging the power of RGB LEDs and the principles of optical illusion, this project will create a stunning visual display that can be used for various applications, such as university open days, entertainment, and education.
The LED fan display should be safe, portable and relatively low-cost. Furthermore, an app or software with a user-friendly interface should be developed to allow for the operation of the fan as well as for uploading custom designs for display. Existing software may not be used "as is" in this project – the custom software development should be integrated within the design. Consideration should be given to multi-facet design aspects such as the integration of multiple fans to allow for a larger, coordinated display unit. The design should incorporate additive manufacturing parts and should be visually appealing. Project objectives include:
1. Formulation of specifications for a 3D hologram-like RGB LED fan display unit 2. Generation and evaluation of concepts 3. Selection of suitable materials and components for manufacturing of the final design 4. User-interface/software development and integration 5. Testing of the unit using several visual display inputs of varying complexity (including static and dynamic scenes).
Electronic design, soldering and programming skills will be very handy in this project.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (245) | Electronics (315)

Design, build and testing of a "Dot-Bot" 3D speckle pattern printer

Reference	: MNS6
Supervisor	: Dr Melody Neaves
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:
Accurate Digital Image Correlation (DIC) relies heavily on the quality of the speckle pattern applied to a material. Traditional methods like spray painting are often challenging to control, leading to inconsistent pattern feature sizes and difficulties when applying patterns to non-planar surfaces. Recent research has demonstrated that "smart" DIC patterns can be created using modified 3D printing technology (for example, see this paper: https://doi.org/10.1007/s11340-021-00720-x). This project aims to design, manufacture and validate a dedicated, low-cost desktop 3D speckle pattern printer specifically designed to automate this patterning process in a repeatable manner.
The printer should work on a variety of planar and non-planar surfaces. A user-friendly app should be developed from scratch for this purpose. There should be an option to select custom speckle pattern designs and/or the ability to upload and process existing images of speckle patterns. The speckle patterns should be validated using speckle pattern quality metrics available in literature and/or through DIC performance (such as rigid body displacement).
The project objectives are to: 1. Formulate specifications for the hardware and software of the Dot-Bot system. 2. Develop and evaluate different concepts to achieve automated 3D speckle pattern printing. 3. Manufacture and/or assemble the final design. 4. Development and integration of the user-interface/software/firmware. 5. Validate the system's performance by generating repeatable speckle patterns on both planar and non-planar surfaces (e.g., inclined or curved) using various materials. 6. Evaluate the patterns for suitability for use in DIC using existing metrics.
Knowledge of 3D printing and some G-code generation basics would be very helpful.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Machine Design B (344) | Experimental Techniques (414)

Design, build and testing of a multi-colour 3D printer

Reference	: MNS7
Supervisor	: Dr Melody Neaves
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical and/or Mechatronic
Required elective	: None

Description:
Standard Fused Deposition Modelling (FDM) is typically monochromatic, limiting the visual communicative power of printed parts. Multi-colour printing allows for more interesting colourful prints. See this commercial printer here as an example: https://www.youtube.com/watch?v=rluJj3NEdQA. There are several applications for such a device. One interesting engineering visualisation application for such a printer involves printing Finite Element Analysis (FEA) stress maps directly onto a part, creating topographic maps with elevation colour-coding, or embedding clear text and labels into components. However, achieving clean colour changes without "bleeding," jamming, or excessive waste requires precise mechatronic control over filament retraction and purging. Although the Department of Mechanical and Mechatronic Engineering currently own a commercial system, low-cost options for multi-colour capable printers are currently limited.
The student is required to design and build a user-friendly FDM 3D printer equipped with a specialised extrusion system for multi-colour output. The system must be capable of printing a single benchmarking object with at least two (preferably more) distinct colours automatically. Existing code/software can be used, provided some modification is demonstrated for the multi-colour functionality.
The following project objectives are applicable: 1. Development and evaluation of several concepts for the mutli-colour FDM 3D printer. 2. Manufacture and assembly of the final design. 3. Development and integration of a user-interface/software. 4. Commissioning of the device using two or more colour filaments. 5. Validate the system by printing a multi-coloured test artifact and evaluating the final printed part in comparison to an existing commercial system.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Machine Design B (344) | Mechatronics (424)

Design, build, and testing of a CNC engraver for non-planar geometries

Reference	:	MNS8
Supervisor	:	Dr Melody Neaves
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Computer Numerical Control (CNC) technology is ubiquitous in modern manufacturing, typically utilized for cutting flat sheet materials into 2D shapes. However, many functional engineering components require fine detail machining on non-planar (curved or irregular) surfaces. While commercial desktop engravers exist (e.g., Creality CR-Laser Falcon Engraver), they are often costly and are often limited to flat surfaces This project addresses the need for an accessible, low-cost desktop machine capable of executing sophisticated engraving toolpaths on both flat and uneven topographies, enabling students to prototype or customise designs without relying on expensive workshop facilities.
The student is required to design and construct a low-cost, 3D CNC engraving system from first principles or modular components. The system must feature a rigid motion stage and an engraving mechanism capable of engraving soft-to-medium density materials (such as wood, plastics, or tooling board) in both planar and non-planar design geometries. The machine must be safe, user-friendly and a user interface should be developed to enable control of the machine.
Project objectives include the following:
1. Formulate design requirements for the CNC engraver system. 2. Generate and evaluate concepts for the system. 3. Design and manufacture the final selected concept. 4. Develop and/or integrate the software and user-interface for the system. 5. Commission the CNC engraver system. 6. Evaluate the performance of the system by engraving sufficiently complex planar and non-planar geometries and comparing to engravings produced through commercial systems.
Design skills and knowledge of G-code or CNC machines will be very useful in this project.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Machine Design B (344)

Design, build and testing of a low-cost rotary CNC system

Reference	:	MNS9
Supervisor	:	Dr Melody Neaves
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Standard three-axis CNC routers are limited to machining flat, prismatic stock. To machine cylindrical geometries—such as chess pieces, camshaft prototypes, or decorative table legs—a rotary axis (fourth axis) is required. Industrial "turn-mill" centres are prohibitively expensive and complex for small-scale prototyping. There is a significant need for a dedicated, low-cost desktop machine that combines the flexibility of a rotary axis with the precision of a router spindle to machine complex cylindrical forms.
The student is required to design and build a CNC machine that features a rotary axis (A-axis) as a primary motion component. The machine must be capable of holding cylindrical stock of various sizes and suitable low-strength material types (eg. word, plastic, etc.) and executing continuous rotary toolpaths to create 3D features around the circumference of the part. The system should be safe to use and user-friendly. A user interface should be developed to enable control of the machine.
Project objectives include the following: 1. Formulate design requirements for the rotary CNC router system. 2. Generate and evaluate concepts for the system. 3. Design and manufacture the final selected concept. 4. Develop and/or integrate the software and user-interface for the system. 5. Commission the rotary CNC router system. 6. Validate the machine by carving a suitably complex test piece to demonstrate synchronisation between the rotary and linear axes. This includes comparing the machine's capabilities with commercial options.
Design skills and knowledge of G-code or CNC machines will be very useful in this project.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Machine Design B (344)

Automatic guitar tuner

Reference	:	BMN1
Supervisor	:	Dr Brendon Nickerson
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Guitars require that their strings be tuned in order to play them. This can either be done manually, or using some sort of aid such as a tuning fork, a battery powered tuner, or a cell phone app. Recent develops include an automatic battery powered tuner that not only measures the string pitch, but also automatically rotates the tuning head to achieve a desired pitch. These devices can be great for beginners to easily tune their guitar, however they can be expensive. The aim of this project is to design, manufacture, and test a low-cost automatic guitar tuner. The device should be able to measure the current frequency a guitar string is tuned to, and adjust it to a given frequency. The device could be designed to be standalone, or attached permanently to a guitar. The device must make use of feedback control, and allow a user to select between multiple tuning configurations. The responsible student will need to show that they have followed an established design methodology in the execution of this project.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Vibration and Noise (354) | Experimental Techniques (414)

Low cost vibration measurement device

Reference	:	BMN2
Supervisor	:	Dr Brendon Nickerson
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Vibration measurement is conducted in a variety of industries for various applications. High accuracy measurement systems are typically expensive. The aim of this project is to design, manufacture and test a low-cost vibration measurement system. The device can be designed for a specific application, or for general use. The device can be compared to a commercial vibration measurement system, and other low cost solutions such as a cell phone app. The responsible student will need to show that they have followed an established design methodology in the execution of this project.

Key modules (Scientific and Engineering Knowledge):
Vibration and Noise (354) | Experimental Techniques (414)

The Impact of Load Carrying on a Bicycle

Reference	:	BMN3
Supervisor	:	Dr Brendon Nickerson
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	25945939
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
This proposal was identified and recommended by Tailfin, a leading manufacturer of high-performance bicycle luggage systems, highlighting its relevance to real-world load-carrying efficiency and rider performance. This study will investigate how different methods of attaching loads to a bicycle influence vibrational losses, ride efficiency, comfort and handling.
The current understandings of the performance of bicycles under load are primarily focused on aerodynamics and weight penalties. Some studies have shown that using panniers can reduce the speed of the bike in comparison to compact bikepacking bags. However, the effect of vibrational induced losses due to loads on a bicycle is often overlooked. When a bicycle experiences vibrations due to the road or terrain it is moving on, it can cause energy to dissipate through the frame, its components and the loads and load interfaces that are connected to it. The way in which the load is connected to the bicycle, whether it be a rigid rack, semi-flexible mount or soft seat pack, can affect the bicycles vibrational dynamics which in turn can affect the efficiency and comfort of the bicycle.
Thus, the aim of this project is to investigate how load carrying, mounting stiffness and compliance can affect vibrational losses and overall bicycle efficiency under real-world load carrying conditions.
The study will therefore encompass the following: • Measuring and comparing vibrational transmissions through the bicycles frame for different load-mounting configurations. • Measuring and comparing vibrational transmissions through the load and mounting set-up based on mounting stiffnesses and compliance. • Measuring and comparing the bicycles efficiencies in each scenario under real-world conditions. • Measuring and comparing the bicycles efficiencies in each scenario under a controlled testing scenario by designing and integrating a testing station. • Making recommendations for stiffness, compliance and load-mounting configurations based off of analysed results.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Vibration and Noise (354) | Experimental Techniques (414)

Onboard Track Stiffness Measurements

Reference	:	BMN4
Supervisor	:	Dr Brendon Nickerson
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Track stiffness plays a crucial role in the safety and performance of train operations. The track structure consists of several layers, namely: rail clips, rail pads, sleepers, ballast, sub-ballast and subgrade. The overall stiffness of the track is used to characterize the contributions from all the track layers. Understanding track stiffness can assist rail operators with maintenance planning and asset management. The traditional hydraulic jack-loading method is accurate, but is restricted to local measurements, hence, this topic proposes to design, build and test an onboard (contactless) track stiffness measurement device.
The Gibela Engineering Research Chair (at the Department of Mechanical and Mechatronic Engineering) will support the project, arrange access to Gibela depots and trains, and provide funding for prototyping. Students will enjoy being part of a team, having access to industry, additional development opportunities and colloquia.

Key modules (Scientific and Engineering Knowledge):
Strength of Materials W (334) | Vibration and Noise (354) | Experimental Techniques (414)

Sumo robot (Competition)

Reference	:	BMN5
Supervisor	:	Dr Brendon Nickerson
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project forms part of a competition for which the successful candidates will be required to design and build an autonomous sumo robot that can compete against other participants. The project will require the application of electronic design, modelling and control theory, as well as skills acquired during the E-design module in the first semester. The design needs to satisfy the Robogames international sumo robot competition rules for the Mini Sumo robot class (https://robogames.net/rules/all-sumo.php). Participants will have the opportunity to compete in a knock-out competition following the examination period. The competition and the examination will be mutually exclusive.
Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Mechatronics (424)

Design, build, and test a low-cost insulin delivery system

Reference	:	VS1
Supervisor	:	Dr Vinay Shekhar
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Globally, there are over 400 million diabetics of which over 8 million are type 1 diabetes. Type 1 diabetes can be characterized by the partial or complete loss of insulin production in the patient. As a result, a patient must typically inject insulin into their body subcutaneously 3-4 times per day. This system of insulin delivery while cost effective, comes with limited control for the patient. Insulin pumps provide an effective alternative in terms of insulin intake control, but are cost-prohibitive for most people, particularly in the developing world. A typical pump can cost over R100000 for the initial device, with annual consumable costs of over R50000.
The goal of this project is to design a low-cost insulin pump that is capable of accurately delivering the required dosage of insulin and can be controlled via bluetooth by the user. Security should be considered to ensure that only correctly paired devices can operate the pump.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Modelling (334)

Design, build, and test a classroom demonstration tool for the planar mechanisms presented in Introductory Machine Design 254

Reference	:	VS2
Supervisor	:	Dr Vinay Shekhar
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
A key concept in the second-year module, Introductory Machine Design 254, is the understanding of the kinematics of planar mechanisms. The aim of this project is to develop a series of class demonstration tools to help students with the visualisation of the different planar mechanisms studied in the course. These tools should be adjustable to allow for different motions to be obtained. Real-time imaging analysis should be used to determine the position, velocity, and acceleration of the mechanisms and code should be developed to compare the analytical solutions to the demonstration solutions.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Modelling (334) | Vibration and Noise (354)

Design, build, and test a balancing rig for classroom practical demonstration in Introductory Machine Design 254

Reference	:	VS3
Supervisor	:	Dr Vinay Shekhar
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Rotating unbalance and shaft balancing is a topic presented in the second-year module, Introductory Machine Design 254. Stellenbosch University does not currently have a set up to demonstrate the effects of shaft unbalance in a classroom/practical environment. While the effect of an unbalanced rotating shaft on the bearing loads can be shown via video examples, the built device will allow students to develop an analytical solution for balancing the rotating shaft and then implement their solution and observe the results on the loads transferred to the bearings and the structure attached to the rotating shaft.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Modelling (334) | Vibration and Noise (354)

Numerically determine the rupture threshold for an armor steel to projectile impact

Reference	:	VS4
Supervisor	:	Dr Vinay Shekhar
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Finite Element Methods (FEM)

Description:
The goal of this project is to use Finite Element Analysis (FEA) techniques to simulate the dynamic interaction between a projectile and an armor steel target plate to determine the rupture threshold of plate. The study aims to provide insights into the material's dynamic structural response and determine the key failure mechanisms. The project will involve the selection of a suitable armor steel and the characterization of the material selected. A 3D numerical model will need to be developed including the selection of appropriate element formulations and material models. The numerical model results must be validated against appropriate published data.
In this project, you will be required to learn a new FEA tool, namely, LS-DYNA. This software will be used for the dynamic simulations on the steel specimens. You will also need to choose your material and design your material tests during the first semester as this data is essential for characterising the material models in the FEA software. Thus, you will need to learn about the existing material models and the material tests required to characterise these models.

Key modules (Scientific and Engineering Knowledge):
Finite Element Methods (414) | Strength of Materials W (334) | Experimental Techniques (414)

Redesign of an extendable threshold to close the gap on platforms

Reference	:	VS5
Supervisor	:	Dr Vinay Shekhar
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
To improve the accessibility of train carriages, there is a proposal to redesign the thresholds of the trains to close the gap between the train and the platform edge. This will allow people with mobility issues to more easily board and exit the train carriages, and also allow items such as heavy suitcases to be loaded. The design must be capable of withstanding repeated impact loads from passengers and luggage, while also being able to operate in a range of environmental conditions such as rain and high temperatures. It needs to incorporate sensors to locate the edge of the platform and motors to extend the platform until the gap is closed. This project is being run by the Gibela research group.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Vibration and Noise (354)

Design, build, and test a handheld tool to aid the visually impaired with watching sport

Reference	:	VS6
Supervisor	:	Dr Vinay Shekhar
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Engaging with live sport can be challenging for individuals with visual impairments. While one has access to commentary when watching sport at home, this facility is not readily available in the stadium. Additionally, much of the commentary relies on the listener being able to also watch the action on the field. Most modern sports, such as cricket, football, and rugby already employ sensors and cameras to track the action in real-time. This project aims to use this existing technology to develop a low-cost, light, handheld device that will allow a visually impaired person to follow the action of a game. Audio feedback via speakers or earphones can be used to relay key information such as fouls, injuries, and goals, while vibration at different places on the device can be used to indicate when the ball has gone out of play. The device should be impact and weather-resistant to allow for use both indoors and outdoors.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Vibration and Noise (354) | Experimental Techniques (414)

Investigation of the effects of out-of-autoclave curing on the performance of composite specimens.

Reference	:	VS7
Supervisor	:	Dr Vinay Shekhar
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Composite materials are extensively used in aerospace, automotive, and defence applications due to their excellent properties, such as corrosion resistance and high strength-to-weight ratio and their ability to be used to manufacture components with complex shapes and optimised stiffness and deformation characteristics. Out-of-autoclave curing serves as an alternative and cheaper option for the manufacture of composite specimens due to the lower comparative costs of large volume temperature-controlled ovens.
In this project, at least two fibre-reinforced matrix materials should be considered in combination with Nomex honeycomb core moulded in a sandwich format. The primary aim is to investigate the strength properties of the cured composite specimens manufactured under different curing conditions (autoclave vs oven). The specimens would need to be manufactured at AAT Composites in 2026 due to the unavailability of the oven. This project requires direct participation with an industrial partner at their facilities and will require careful time management and early engagement with AAT Composites. This project has been proposed by AAT Composites, who will be able to assist with raw materials and provide technical assistance on layup configurations to investigate.

Key modules (Scientific and Engineering Knowledge):
Strength of Materials W (334) | Experimental Techniques (414)

Composite specimen material testing and material model characterisation

Reference	:	VS8
Supervisor	:	Dr Vinay Shekhar
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Finite element (FE) models are useful for the rapid design iteration and optimisation of composite parts for under different loading conditions that can be difficult to replicate easily and quickly in a laboratory testing environment. The accuracy of the FE simulations is highly dependent on the accuracy of the material data for the composite materials used. Thus, there is a need to characterise these materials in accordance with testing standards. This project requires direct participation with an industrial partner at their facilities and will require careful time management and early engagement with AAT Composites.
This project aims to characterise the mechanical behaviour of woven fibre-reinforced polymer (FRP) composite specimens manufactured and supplied by AAT Composites. Material testing will be performed based on appropriate testing standards and utilising non-contact displacement measurement techniques such as Digital Image Correlation (DIC). The data collected will need to be fitted to suitable material models and validated against simple FE models. The data generated will aid the numerical analysis team at AAT Composites in the development of FE models for the components being developed at their facilities.

Key modules (Scientific and Engineering Knowledge):
Finite Element Methods (414) | Strength of Materials W (334) | Experimental Techniques (414)

Design, Build, and Test a Next-Generation Soft Robot

Reference	:	MPV1
Supervisor	:	Prof Martin Venter
Can be completed in first semester	:	Yes
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Dive into the fascinating world of Soft Robotics, a revolutionary new field that trades rigid joints for flexible, damage-resistant soft materials. Imagine robots that can squeeze through tight spaces, safely interact with the human body, or survive harsh environments! This project is your chance to design and build a robot for a real-world application in areas like advanced medicine (e.g., surgical tools) or challenging exploration (e.g., disaster relief, space).
You'll select a compelling application, then take your design from concept to a fully tested prototype. This hands-on project will build your skills in advanced materials, actuation, and control systems.
Mechatronics Students: Design a robust soft robot integrated with a feedback control system to achieve precise, controlled movement.
Mechanical Students: Focus on passive embedded control, where the robot's material properties and clever physical design inherently manage its motion and interaction.
This is a high-impact capstone where you'll contribute to an emerging technology with huge commercial and research potential. Ready to build the future of robotics?

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Machine Design A (314) | Machine Design B (344)

Engineering the Ultimate Everyday Carry (EDC) Device

Reference	:	MPV2
Supervisor	:	Prof Martin Venter
Can be completed in first semester	:	Yes
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Tired of bulky tools? This is your chance to design and build a compact, multifunctional, and durable device that solves a real-world problem. Everyday Carry (EDC) is a booming field, and we challenge you to create a tool that users won't leave home without.
Your Mission This project is a complete product development cycle. You'll move beyond theory to create a tangible, marketable device:
Define the Need: Research user pain points (daily tasks, emergencies, convenience).
Design & Material Selection: Engineer a device that balances strength, portability, and ergonomics, with a focus on aesthetics and ease of use.
Prototype & Test: Bring your CAD model to life through rapid prototyping, then conduct rigorous testing to prove its reliability and functionality.
If you are passionate about product design, practical engineering, and creating something you can hold (and potentially commercialize!), this project is for you. Finish your degree with a polished, portfolio-ready physical product!

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Machine Design A (314) | Machine Design B (344)

Design, Build, and Test Smart Home Gym Equipment

Reference	:	MPV3
Supervisor	:	Prof Martin Venter
Can be completed in first semester	:	Yes
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Join the future of fitness! The Smart Home Gym market is exploding, and this is your chance to engineer a device with massive commercial appeal. We challenge you to design, build, and test next-generation equipment that makes at-home strength training more efficient and engaging.
Choose Your Focus You could develop an electronically controlled weight system, a device that gives real-time feedback on form, or focus on a compact, innovative design like an adjustable dumbbell or barbell set.
Mechanical Students: Focus on materials selection, manufacturing, durability, and mechanical innovation. Design the clever, efficient form factor and the robust, user-friendly mechanisms for weight adjustment.
Mechatronics Students: Focus on electromechanical systems and feedback control. Design the electronic actuation, integrate sensors, and create a smart control system for seamless, automated operation.
This capstone is perfect if you want to create a tangible, portfolio-ready product that sits at the intersection of engineering, human factors, and a high-growth consumer market.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Machine Design A (314) | Machine Design B (344)

Biomicry of Bee-Wing Mechanics Using Compliant Mechanisms and Actuation

Reference	:	MPV4
Supervisor	:	Prof Martin Venter
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	22927530
Direction	:	Mechatronic
Required elective	:	None

Description:
Background: Bees and other insects generate complex wing trajectories that are difficult to reproduce with traditional rigid mechanisms. By focusing on soft robotics and pneumatic actuation, this project explores how compliant systems can be used to mimic natural motion without relying on rigid hinges or high-speed motors. Beyond biomimetics, this work contributes to the broader development of pneumatic logic and soft actuation strategies that could be applied to other robotic systems requiring oscillatory or cyclic motion.
Aim: The aim of this project is to design, build, and experimentally validate a soft robotic system that mimics the characteristic figure-eight wingbeat motion of a bee using pneumatic actuation and logic-based control. The work seeks to demonstrate that biologically relevant flapping patterns can be achieved without conventional mechanical joints or electronic control, instead relying on compliant structures and air-driven circuits.
Objective: The objective is to develop a bio-inspired robotic wing system that reproduces the bee's figure-eight flapping pattern using pneumatically driven actuators. This will involve: 1) Analyzing the biomechanics of bee wing motion to extract the kinematic requirements 2) Designing and fabricating a compliant wing mechanism coupled to pneumatic actuators, and developing pneumatic logic circuits capable of generating the oscillatory control signals required for flapping 3) Validating the system experimentally by comparing the achieved wing trajectories to biological reference data.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Machine Design A (314) | Modelling (334)

Design, build and test an image segmentation and classification system using Convolutional Neural Networks

Reference	:	MPV5
Supervisor	:	Prof Martin Venter
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	25137891
Direction	:	Mechatronic
Required elective	:	None

Description:
In this project, Convolutional Neural Networks are used to develop and implement an image segmentation and classification system. The objective is to design a model capable of accurately detecting and categorising objects within digital images. The system will use a Raspberry Pi camera module to capture live image data, enabling real-time object identification and analysis. The project will include deep learning architecture design and training, data analytics integration for model evaluation, and hardware implementation for real-time testing and validation.

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Design (E) (314)

1.3 ThermoFluids Division

Design, manufacture and prove a Real Time mobile track condition monitoring system

Reference	:	PGB1
Supervisor	:	Eng Peter Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The project will determine the capability of the designed equipment, sensors and cameras to determine the condition of the rail infrastructure. As an example, a camera can be used to monitor track defects (rail gauge, missing rail clamps, worn rail, etc), monitor the staggering of the overhead catenary wire, etc.
The mechanical student would be responsible for designing, manufacture and proving of the instrument carriage, while the mechatronic student would be responsible for the instrumentation and data collection and analysis
The Skripsie will be carried out in cooperation with Ceres Rail, Headquarters at the Cape Town harbour near the Royal Yacht Sailing Club

Key modules (Scientific and Engineering Knowledge):
Machine Design B (344) | Mechatronics (424) | Strength of Materials W (334)

Investigate, analyse and recommend on the properties of various bearing materials

Reference	:	PGB10
Supervisor	:	Eng Peter Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
These are to be used on steam locomotives and the investigation would recommend each bearing material for each particular application. The focus is on steam locomotives and rail stock.
This project will be conducted with the cooperation of Ceres Rail, near to the Royal Yacht Club in Cape Town Harbour

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Machine Design B (344)

Design and Manufacture of a High Speed Model Aircraft for a National Competition

Reference	:	PGB2
Supervisor	:	Eng Peter Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The Aeronautical Society of South Africa (AESSA) annually holds a competition for the design and demonstration of a high speed model aircraft. The University of Stellenbosch has never entered this competton. The project will entail the design from concept to assembly and flight, of a powered aircraft hopefully to compete in the competition if suitable sponsorship can be found. The major specifications include Maximum Wingspan 1200mm Motor an unmodified electric E-flight Park 450 or equivalent Control Transmitter transmitting on the 2.4GHz frequency Power Source 3 cell Lithium Polymer (LiPo) battery or equivalent The full rules and regulations for the competition are available on request.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Mechatronics (424)

Design, develop, manufacture and prove a Camera Trap For Monitoring Wild Life and General Security

Reference	:	PGB3
Supervisor	:	Eng Peter Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	22103945
Direction	:	Mechatronic
Required elective	:	None

Description:
The present cameras are bulky and expensive. They are battery driven and subject to being seriously damaged by vandals and baboons. They are stand alone and need to be inspected on a regular basis to collect captured footage and replace batteries.
If a new design of a camera trap using batteries rechargeable from a photovoltaic panel and using smart phone technology, the camera could report changes in its environment including capturing photographs of vandals. The data could be transmitted in real time via an sms, WhatsAp or similar messaging system to a central control centre. This would make regular on-site monitoring where cameras have been located in very difficult areas in the mountains.
Possibly use an external taser to protect the equipment..
The designed camera traps would be tested and improved in the field

Key modules (Scientific and Engineering Knowledge):
Computer Systems (214) | Electronics (245) | Mechatronics (424)

Design, develop, manufacture and prove a Coach Window Lowering System upgrade

Reference	:	PGB4
Supervisor	:	Eng Peter Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Investigate the old coach window lowering mechanism and design, develop and test to prove the new mechanism will replicate the way the original rail system worked. The Skripsie will be carried out in cooperation with Ceres Rail, Headquarters at the Cape Town harbour near the Royal Yacht Sailing Club

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Machine Design B (344) | Maintenance Management (414)

Design, develop and prove a machine to clean surface garbage off rivers and lakes

Reference	:	PGB5
Supervisor	:	Eng Peter Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Especially in urban areas garbage ends up being thrown into rivers and lakes. Such garbage is unsightly and can be the source of disease and bad smells which are deleterious and a danger to local communities. It is proposed design and test a machine to clear this rubbish so the water and environment can be kept sanitary and acceptable

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electrical Drive Systems (324) | Mechatronics (424)

Design, develop and prove a powered trolly suitable as a test bed for train tracks and their environment

Reference	:	PGB6
Supervisor	:	Eng Peter Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The test trolley will be powered by an electrical motor and will be used to inspect train tracks and the adjoining environment such as vegetation which invades and can hinder the motion of the train. It would also provide a suitable instrumentation platform as well as being able to provide computer facilities for data collection and analysis. The trolley will be supplied and will be converted to a stable test bed.
The skripsie will be performed in conjunction with Ceres Rail based in the Cape Town Harbour opposite the Royal Cape Town Sailing Club

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Mechatronics (424)

Bulk Water Supply Scheme – Design of New Water Borehole Field

Reference	:	PGB7
Supervisor	:	Eng Peter Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	20299028
Direction	:	Mechanical
Required elective	:	None

Description:
The aim of the project is to explore, develop and equip a new water borehole field outside the existing Sutherland water supply area. The number of boreholes is to be determined, based on the water demand of the district. The Mechanical scope of works for the new supply field is broken down as follows: • Each borehole site, numbers dependent on water yield tests, will be equipped with: o submersible bore pump/motor, sized to pump the safe yield, within the designed pipe network o All supporting pipework, valves, enclosure, ancillary equipment, etc • Possible pump station, as local control and higher efficiency pumping point, depending upon final well-field design and hole locations

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Numerical Fluid Dynamics (414)

Design, develop, manufacture and prove a Diver Alert System

Reference	:	PGB8
Supervisor	:	Eng Peter Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
For safety reasons, it is necessary to check that the Driver of a locomotive is still capable. For this purpose it is required that s/he is capable by requiring that s/he signal, using a button for instance. If the check is not activated, the brakes of the train would be activated and the train brought to a stop. The system would incorporate a variable time device which would need activation at different intervals based on the specific area in which the tyrain is operating
This project will be conducted with the cooperation of Ceres Rail, near to the Royal Yacht Club in Cape Town Harbour

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Maintenance Management (414)

Design, develop, manufacture and prove a new oil burner "atomiser" head for a steam locomotive

Reference	:	PGB9
Supervisor	:	Eng Peter Blaine
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
The Steam Locomotives use atomised oil as fuel. The project would analyse suitable oil atomiser systems so that it would be possible to increase the efficiency of the burn. Samples of the present atomisers would be made available.
The design would be manufactured and tested. If successful, it would then make it possible for them to be manufactured in South Africa
This project will be conducted with the cooperation of Ceres Rail, near to the Royal Yacht Club in Cape Town Harbour

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Thermofluid Dynamics (344)

Evaluating the effect of swirl number on the turbulent flow field in a Micro Gas Turbine's combustor

Reference	:	FENN1
Supervisor	:	Mr Chaz Fenner
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
During the combustor design process, engineers have several variables to consider to maximise combustion efficiency and reduce heat transfer and pressure losses. These outcomes often exhibit complex relationships that necessitate the use of computational fluid dynamics and usually require engineering compromises.
In the combustion of gaseous fuel, reaction progress is limited by the chemical reaction rate and the mixing rate of the fine-scale structures. The fuel choice determines the former; however, the latter is controlled by mechanical subcomponents that must be designed for the desired fuel and operating conditions.
This project will involve the design and 3D numerical analysis of a MGT combustor. The aim is to determine the effect of swirl number and dilution jet design on the flow field in the primary and secondary zones of the flametube.

Key modules (Scientific and Engineering Knowledge):
Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344)

Optimisation of a Micro Gas Turbine Combustor through System-Level simulation in Flownex SE

Reference	:	FENN2
Supervisor	:	Mr Chaz Fenner
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
Gas turbine systems are widely used for power generation and propulsion due to their high power density and reliability. The combustor is a critical component that determines the overall efficiency, emissions, and thermal performance of the system. Traditional high-fidelity simulations of combustion and heat transfer processes, such as those performed in CFD, provide detailed insights but are computationally expensive and time-consuming.
Flownex SE, a one-dimensional thermofluid network simulation software, offers a powerful alternative for system-level analysis and optimisation. By coupling simplified physical models of combustion, heat transfer, and fluid flow, Flownex enables rapid evaluation of multiple design configurations and operating conditions, making it highly suitable for preliminary design and optimisation studies.
In this project, a gas turbine system model will be developed in Flownex SE with a particular focus on the combustor subsystem. Key operating and geometric parameters will be varied to study their influence on overall system performance, including efficiency, pressure losses, and outlet temperature profiles. The goal is to identify optimal operating conditions and design adjustments that improve system performance.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344)

Analysing the effect of different fuels on a peaking gas turbine.

Reference	:	FENN3
Supervisor	:	Mr Chaz Fenner
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
South Africa's peaking power generation relies heavily on peaking gas turbines, which currently operate on expensive diesel fuel. Recent studies have demonstrated that converting these systems to LPG could reduce fuel costs by up to 50%, while LNG offers additional environmental benefits and potential for hydrogen blending in future decarbonization efforts.
This project investigates how different fuels affect turbine performance, efficiency, and emissions using thermodynamic modeling and reduced-order combustion analysis. Students will develop engineering tools (e.g. Python and Flownex) to evaluate fuel alternatives and understand the technical trade-offs between cost, performance, and environmental impact. The work directly supports South Africa's energy security goals and provides insight into fuel flexibility strategies for the national grid.

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Energy Systems M (434) | Thermofluid Dynamics (344)

Computational fluid dynamics modelling of Cylinder head Swirl with in an IC engine cylinder

Reference	:	GILA1
Supervisor	:	Dr Andrew Gill
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
Cylinder head swirl ratio is a dimensionless number and is defined as the ratio between the circumferential velocity and the axial velocity of the air inside an IC engine cylinder bore at a specified inlet valve lift. Typically, cylinder heads with low swirl ratios result in high exhaust smoke emissions and poor fuel consumption at lower engine speeds and the swirl ratio can be increased by modifying the cylinder head, valve and seat ring geometry.
A previous student commissioned a physical test bench in order to enable the experimental investigation of cylinder head swirl using particle image velocimetry and hot wire anemometry, and discovered complex flow structures within the cylinder, notably multiple vortices. The present project requires that flow within the test bench be modelled using both basic fluid mechanics methods and CFD in an attempt to reproduce the experimental data previously gathered, as well as to add insight into the flow structures within the cylinder head of a diesel engine.
A similar study can also be conducted by a second student for single cylinder spark-ignition engine (the CFR engine).

Key modules (Scientific and Engineering Knowledge):
Modelling (334) | Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344)

Propeller/rotor design for quadcopter drones

Reference	:	GILA2
Supervisor	:	Dr Andrew Gill
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
The purpose of this project is to do proper evaluation of a range of propellor designs for quadcopter drones using engineering methods.
Propeller design is well established for manned aircraft. However, quadcopter drone propellors or rotors operate in a very different Reynolds number range, and have, for the most part, only been investigated by hobbyists without a technical background. This project will require the student to systematically investigate the various novel propellor or rotor design options for quadcopter drones such as ducted fans, toroidal propellers, swept blades, and alternative blade aerofoil profiles, and make recommendations on improving propeller design for efficiency. Improvements in quietness, stall resistance or other relevant aspects of drone operation are also desirable. A previous student attempted to build a test-bench for propellors, but was only partly successful, and it may require significant re-design for this project.
This project will involve classical fluid mechanics/turbomachinery modelling and design techniques such as blade element design, as well as fabricating and testing propellers.
CFD is not strictly required, but is recommended (for simple supplementary aerodynamic simulations).

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Thermofluid Dynamics (344) | Vibration and Noise (354)

Modelling and Design of a Small Stirling Engine

Reference	:	GILA3
Supervisor	:	Dr Andrew Gill
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
The student will be required to develop a predictive numerical model for performance of a Stirling-cycle engine. The student will then use the model to design, construct and test a small portable Stirling engine. Stirling engines potentially have applications in solar-thermal power generation, waste heat energy recovery, as well as many other fields. The ideal Stirling cycle also offers very high theoretical efficiency. However, their relative simplicity makes them popular as hobby projects.
A previous student performed a simple simulation and constructed a stirling engine, but was able to get it to run continuously under its own power. His project can be used as a starting point, if desired.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Modelling (334) | Thermofluid Dynamics (344)

Design and construction of a small portable calibration wind tunnel

Reference	:	GILA4
Supervisor	:	Dr Andrew Gill
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The student will design and construct a portable incompressible flow "desktop wind-tunnel" for the purposes of calibrating flow measurement devices such as hot wire probes and 3-hole cobra-probes in situ, as well as for demonstrations. With the M&M laboratories unavailable for the next few years, such a piece of equipment would be very useful, and will continue to be used afterwards whenever its portability and low power requirements are required.
The project scope will be adjusted, depending on whether the student assigned this topic is a mechanical or mechatronics student.
If a mechatronics student is assigned the topic, the project would be greatly improved by means of some sort of closed-loop control-system to maintain constant flow rate through the tunnel.
For a Mechanical student, a detailed study of the internal flow within the test section, using advanced measurement techniques (and possibly CFD) would be desired.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Experimental Techniques (414)

Tensile strength and impact-resistance testing of resin 3D-printed components

Reference	:	GILA5
Supervisor	:	Dr Andrew Gill
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
The aim of this project is to systematically determine how the strength and other mechanical properties of 3D printed resin parts vary for different printer settings and setup (exposure time per layer, layer thickness, orientation, post-printing curing time, etc.), for at least six readily available consumer-grade resins (three water-washable and three non-water washable resins).
SLA or resin 3D printers use a UV light source to selectively cure UV-sensitive resins. However, the way in which the strength and mechanical properties of 3D-printed resin parts are affected by varying the printing parameters and the orientation of the printed part are not well known at present. Nor is it clear how much the properties vary between resin manufacturers or different types of resin.
A previous student has conducted a set of exploratory tensile tests of 3D-printed "dog-bone" test-sections, which yielded interesting results. The aim is to expand the range of these tests to include more resins, and more importantly, to test impact-resistance in addition to tensile testing. To do the latter, the student will be required to design and build a small impact-testing machine.
Previous experience of 3d printing, particularly SLA printing, would be advantageous (but is not required)

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Strength of Materials W (334) | Experimental Techniques (414)

Hospital based robotic assistant (Competition)

Reference	:	GILA6
Supervisor	:	Dr Andrew Gill
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The successful candidate will be responsible for the design, build and control implementation of a floor-bound, obstacle avoiding, pathfinding robot. The robotic assistant should function as a mobile information kiosk, capable of interacting with patients via a custom HMI and leading patients to a designated ward or physician office. Performance will be measured through various requirements, including but not limited to, speed, efficiency, reliability and safety. A full set of customer requirements from the sponsoring industry partner will be provided at the commencement of the project. Successful candidates are expected to conduct themselves in a respectful and professional manner whenever they interact with the sponsoring industry partner as you will be representing the department and the university. You will be given the chance to compete against other students with the same topic after the final year projects have been examined. Note that each student will have to adhere to a strict set of rules to ensure the integrity of the competition and fairness towards other participants. Examination of the final year project and competitive performance assessment will be mutually exclusive. Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Mechatronics (424)

Sumo Robot (Competition)

Reference	:	GILA7
Supervisor	:	Dr Andrew Gill
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project forms part of a competition for which the successful candidates will be required to design and build an autonomous sumo robot that can compete against other participants. The project will require the application of electronic design, modelling and control theory, as well as skills acquired during the E-design module in the first semester. The design needs to satisfy the Robogames international sumo robot competition rules for the Mini Sumo robot class (see https://robogames.net/rules/all-sumo.php) . Participants will have the opportunity to compete in a knock-out competition following the examination period. The competition and the examination will be mutually exclusive.
Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Mechatronics (424)

The Effect of Arrow head geometry and mass on archery arrow performance

Reference	:	GILA8
Supervisor	:	Dr Andrew Gill
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
The aim of this project is to investigate the aerodynamic and dynamic behavior of arrows with broadhead and field points (with the option to include additional arrow head designs), in order to attempt to arrive at a better understanding of arrow dynamics that can be used to generate an improved arrow design for a given purpose. A number of different arrow head geometries are used in modern archery and bow hunting (broadheads, field points, blunts, etc) and during the middle ages, there was even more variation in design. The aerodynamic stability of a finned projectile, such as an arrow, can be determined by determining how far behind the center of gravity the aerodynamic center of pressure is located. Therefore, some arrow designs with a large surface area, such as broadhead or historical swalllowtail head will move the aerodynamic center of pressure forward, which can negatively affect stability. However, the mass of the arrow head affects the center of gravity of the arrow, moving it forward if the head and thereby improving stability. However, the mass of the arrow also has an effect on how fast it can be propelled by a bow of given draw-weight, due to the conservation of momentum and energy.
This project is primarily a numerical modelling project, but with the option of manufacturing and testing prototype arrows to confirm numerical findings.

Key modules (Scientific and Engineering Knowledge):
Modelling (334) | Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344)

Development of a 1D model of a residential-scale air-to-water transcritical CO2 heat pump with an internal heat exchanger

Reference	: RAH1
Supervisor	: Dr Rashid Haffejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
Heat pump systems have seen significant adoption rates as energy efficiency becomes increasingly important in a push to reduce emissions. CO2 (R744) is a safe, cheap, and natural refrigerant with low global warming potential. Due to the low critical temperature of CO2 (31 °C), a transcritical vapour compression cycle is used to achieve high water heating temperatures (> 90°C), leveraging high CO2 temperatures in the supercritical region. The integration of an internal heat exchanger may improve system efficiency due to the higher gas cooler outlet temperatures.
The aim of this project is to develop two simplified component-level (1D) models of a residential-scale transcritical CO2 heat pump, (i) with an internal heat exchanger, and (ii) without an internal heat exchanger. The model must be able to sufficiently characterize each heat pump component, including the compressor, water-heating gas cooler, expansion valve, (dry) air-source evaporator, and internal heat exchanger. The internal heat exchanger will be sized and modelled in detail. The two developed models will be verified using existing results from either literature or from an existing model. Once developed and verified, the two models will be compared to each other to evaluate the effects of an internal heat exchanger, and to quantify the performance of the heat pump system at typical ambient conditions observed in Cape Town.
Python is the selected programming environment for this project. The candidate should have a interest in programming/simulation and thermofluids.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Development of a 1D model of an industrial-scale air-to-water transcritical CO2 heat pump with an internal heat exchanger

Reference	: RAH2
Supervisor	: Dr Rashid Haffejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
Heat pump systems have seen significant adoption rates as energy efficiency becomes increasingly important in a push to reduce emissions. CO2 (R744) is a safe, cheap, and natural refrigerant with low global warming potential. Due to the low critical temperature of CO2 (31 °C), a transcritical vapour compression cycle is used to achieve high water heating temperatures (> 90°C), leveraging high CO2 temperatures in the supercritical region. The integration of an internal heat exchanger may improve system efficiency due to the higher gas cooler outlet temperatures.
The aim of this project is to develop two simplified component-level (1D) models of an industrial-scale transcritical CO2 heat pump, one with and another without an internal heat exchanger. The model must be able to sufficiently characterize each heat pump component, including the compressor, water-heating gas cooler, expansion valve, (dry) air-source evaporator, and internal heat exchanger. The internal heat exchanger will be sized and modelled in detail. The two developed models will be verified using existing results from either literature or from an existing model. Once developed and verified, the two models will be compared to each other to evaluate the effects of an internal heat exchanger, and to quantify the performance of the heat pump system at typical ambient conditions observed in Joburg.
Python is the selected programming environment for this project. The candidate should have a interest in programming/simulation and thermofluids.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Performance comparison of a residential-scale air-to-water conventional heat pump and transcritical CO2 heat pump through 1D simulation

Reference	: RAH3
Supervisor	: Dr Rashid Haffejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
Heat pump systems have seen significant adoption rates as energy efficiency becomes increasingly important in a push to reduce emissions. CO2 (R744) is a safe, cheap, and natural refrigerant with low global warming potential. Due to the low critical temperature of CO2 (31 °C), a transcritical vapour compression cycle is used to achieve high water heating temperatures (> 90°C), leveraging high CO2 temperatures in the supercritical region. However, this requires high operating pressures which can result in large losses across the expansion valve compared to conventional heat pump systems.
The aim of the project is to develop two simplified component-level (1D) heat pump models: (i) of a conventional heat pump system, and (ii) of a transcritical CO2 heat pump system, for residential-scale applications. The models must be able to sufficiently characterize each heat pump component, including the compressor, water-heating gas cooler, expansion valve, and (dry) air-source evaporator. The models will be verified using existing results from either literature or from an existing model. Once developed and verified, the two models will be compared to each other to quantify performance differences and feasibility of the two systems with respect to ambient conditions in Joburg.
Python is the selected programming environment for this project. The candidate should have a interest in programming/simulation and thermofluids.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Performance comparison of an industrial-scale air-to-water conventional heat pump and transcritical CO2 heat pump through 1D simulation

Reference	: RAH4
Supervisor	: Dr Rashid Haffejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
Heat pump systems have seen significant adoption rates as energy efficiency becomes increasingly important in a push to reduce emissions. CO2 (R744) is a safe, cheap, and natural refrigerant with low global warming potential. Due to the low critical temperature of CO2 (31 °C), a transcritical vapour compression cycle is used to achieve high water heating temperatures (> 90°C), leveraging high CO2 temperatures in the supercritical region. However, this requires high operating pressures which can result in large losses across the expansion valve compared to conventional heat pump systems.
The aim of the project is to develop two simplified component-level (1D) heat pump models: (i) of a conventional heat pump system, and (ii) of a transcritical CO2 heat pump system, for industrial-scale applications. The models must be able to sufficiently characterize each heat pump component, including the compressor, water-heating gas cooler, expansion valve, and (dry) air-source evaporator. The models will be verified using existing results from either literature or from an existing model. Once developed and verified, the two models will be compared to each other to quantify performance differences and feasibility of the two systems with respect to ambient conditions in Cape Town.
Python is the selected programming environment for this project. The candidate should have a interest in programming/simulation and thermofluids.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Development of a 1D model of a transcritical CO2 heat pump in Flownex SE for performance characterization in typical South African ambient conditions

Reference	: RAH5
Supervisor	: Dr Rashid Haffejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
Heat pump systems have seen significant adoption rates as energy efficiency becomes increasingly important in a push to reduce emissions. CO2 (R744) is a safe, cheap, and natural refrigerant with low global warming potential. Due to the low critical temperature of CO2 (31 °C), a transcritical vapour compression cycle is used to achieve high water heating temperatures (> 90°C), leveraging high CO2 temperatures in the supercritical region, but requiring high operating pressures.
The aim of this project is to develop a 1-D model of a transcritical CO2 heat pump in Flownex SE. The model must be able to sufficiently characterize each heat pump component, including the compressor, water-heating gas cooler, expansion valve, and (dry) air-source evaporator. The developed model will be verified using existing results from either literature or from an existing model. Once developed and verified, the model will be used to characterize performance of the heat pump system for average South African ambient conditions observed on a yearly basis. Transient modelling on a daily-basis may also be explored.
The candidate should have a strong interest in thermofluids and simulation. Knowledge of C# scripting may be beneficial to develop custom scripts in Flownex SE, but is not required.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Development of a 1D model of a simple recuperated supercritical CO2 Brayton cycle in Flownex SE to investigate nominal and part-load operation

Reference	: RAH6
Supervisor	: Dr Rashid Haffejee
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
Supercritical CO2 (sCO2) Brayton power cycles have garnered significant research attention for its high thermal efficiency and compact physical footprint compared to traditional steam Rankine cycles. This makes sCO2 Brayton cycles highly suitable to use as a topping/bottoming cycle to retrofit existing power cycles, thereby increasing power generation and energy efficiency.
While there are several sCO2 Brayton cycle layouts, this project focuses on a simple recuperated Brayton cycle layout, applied to retrofit an existing Rankine cycle. The candidate will develop a 1D model of the cycle in Flownex SE, capturing cycle component characteristics at a high level. The recuperator can have a significant impact on cycle performance. Thus, detailed recuperator modelling will be integrated into the model. The developed model will be verified at nominal load. Thereafter, the model will be transiently simulated to investigate system performance and response across load changes.
The candidate should have a strong interest in thermofluids and simulation. Knowledge of C# scripting may be beneficial to develop custom scripts in Flownex SE, but is not required.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Investigation on the operation of a simple recuperated sCO2 Brayton power cycle near the critical point using 1D simulation

Reference	:	RAH7
Supervisor	:	Dr Rashid Haffejee
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Supercritical CO2 (sCO2) Brayton power cycles have garnered significant research attention for high thermal efficiencies and compact physical footprints compared to traditional steam Rankine cycles. This makes sCO2 Brayton cycles highly suitable to use as a topping/bottoming cycle to retrofit existing power cycles, thereby increasing power generation and energy efficiency. The critical point poses several challenges with cycle operation. Rapid fluctuations in thermo-physical properties of sCO2 near the critical point can significantly affect both compressor behaviour (due to density changes) as well as pre-cooler behaviour (due to specific heat changes). These variations are further exacerbated with load fluctuations influencing heat loads and cycle operation.
This project focuses on a simple recuperated Brayton cycle layout, applied to retrofit an existing Rankine cycle. The candidate will develop a 1D model of the cycle in Python, capturing cycle component characteristics at a high level. The developed model will be verified using a previously designed set of nominal cycle parameters. The model will then be used to explore two sets of cycle operating parameters: (i) low-side pressure and temperature away from the pseudo-critical point (as previously designed), and (ii) low-side pressure and temperature closer to the critical point. This may require re-design of each cycle component to suit the developed operating parameter specifications. The consequent designed cycles will be investigated for various load conditions, which will characterise cycle performance across the operating range.
Python is the selected programming environment for this project. The candidate should have a interest in programming/simulation and thermofluids.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Hospital based robotic assistant (competition)

Reference	:	RAH8
Supervisor	:	Dr Rashid Haffejee
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The successful candidate will be responsible for the design, build and control implementation of a floor-bound, obstacle avoiding, pathfinding robot. The robotic assistant should function as a mobile information kiosk, capable of interacting with patients via a custom HMI and leading patients to a designated ward or physician office. Performance will be measured through various requirements, including but not limited to, speed, efficiency, reliability and safety. A full set of customer requirements from the sponsoring industry partner will be provided at the commencement of the project. Successful candidates are expected to conduct themselves in a respectful and professional manner whenever they interact with the sponsoring industry partner as you will be representing the department and the university.
You will be given the chance to compete against other students with the same topic after the final year projects have been examined. Note that each student will have to adhere to a strict set of rules to ensure the integrity of the competition and fairness towards other participants. Examination of the final year project and competitive performance assessment will be mutually exclusive.
Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (315) | Mechatronics (424)

Sumo robot (Competition)

Reference	:	RAH9
Supervisor	:	Dr Rashid Haffejee
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project forms part of a competition for which the successful candidates will be required to design and build an autonomous sumo robot that can compete against other participants. The project will require the application of electronic design, modelling and control theory, as well as skills acquired during the E-design module in the first semester. The design needs to satisfy the Robogames international sumo robot competition rules for the Mini Sumo robot class, see: https://robogames.net/rules/all-sumo.php. Participants will have the opportunity to compete in a knock-out competition following the examination period. The competition and the examination will be mutually exclusive.
Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Modelling (334)

RWH1: Heat Release Modelling and Knock Intensity of Fuels in a Single Cylinder Engine

Reference	: RWH1
Supervisor	: Mr Richard Haines
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
The objective of this project is to develop a single-zone diagnostic heat release model including a knock model to analyse the burn rate of a fuel and determine knock intensity when knock is observed during combustion process. These models will be used in the future for engine practicals and biofuels research projects.
The Department has two single cylinder spark ignitions engines, namely, a Cooperative Fuel Research (CFR) engine, which is universally used to measure fuel octane, and a Ricardo Hydra research engine for combustion analysis. Test data from previous test campaigns using these engines will be used for the heat release modelling and knock intensity calculations.
Lotus Engine Simulation (LES) software, a user-friendly package, will be used to simulate the performance of the two engines to provide additional information.
Work to be done/objectives to be attained by the student in this project: 1. Literature review: Internal combustion engines, knock/abnormal combustion in spark ignition engines, alternative fuels, fuel properties, thermodynamic and heat-transfer modelling of internal combustion engines. 2. Develop a lumped system, single-zone diagnostic model for the CFR engine and Ricardo Hydra engine for motoring (no combustion) and fired conditions (regular operation). Investigate and select a method for calculating knock intensity. 3. Model the CFR and Ricardo Hydra engines using Lotus Engine Simulation (LES) software. Validate the LES model results using previous engine test data. 4. Compare the in-cylinder pressure data, modelled rate of heat release. mass fraction burned and the knock intensity results obtained for the CFR engine, the Ricardo Hydra engine and the LES simulation results.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

RWH2: Single-zone and Two-zone Heat Release Model Comparison for a Twin Cylinder, Air Cooled Petrol Engine

Reference	: RWH2
Supervisor	: Mr Richard Haines
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
The objective of this project is to compare the heat release results obtained for a lumped system, single-zone diagnostic model with those obtained for a two-zone diagnostic model. Heat release modelling is the application of the first law of thermodynamics to the IC engine fuel combustion process.
The Department has a modified Honda twin cylinder engine which is used for an engine practical and for fuel-to-fuel testing projects and these models will be used in the future for research and teaching. Test data from previous test campaigns using this engine will be used for the heat release modelling.
Lotus Engine Simulation (LES) software, a user-friendly package using a single-zone model, will be used to simulate the twin cylinder engine performance. Results from the LES simulation will be compared to actual test data and the lumped system, single-zone model.
Work to be done/objectives to be attained by the student in this project: 1. Literature review: Internal combustion spark ignition engines, combustion processes, alternative fuels, fuel properties, thermodynamic and heat-transfer modelling of internal combustion engines. 2. Develop a lumped system, single-zone heat release model and a two-zone heat release model for the twin cylinder engine. Use test data captured during previous test campaigns to validate the models and compare the results. 3. Using Lotus Engine Simulation (LES) software, set up an engine simulation based on the Honda engine specifications. Validate the LES model using previous test data. 4. Compare the results from the two diagnostic models (single-zone and two-zone) to those simulated results from the LES model.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

RWH3: Development of an IC Engine Control Module for Simulating the Operation of Fuel Injection and Ignition Systems

Reference	: RWH3
Supervisor	: Mr Richard Haines
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
The objective of this project is to design, build and test a control module to simulate the operation of the fuel injection and ignition systems of an internal combustion (IC) engine, on an electronic test bench. The module must simultaneously simulate the fuel injection and spark discharge events as required for different engine operating conditions.
The Department has a single cylinder carburetted engine with magneto, which will be upgraded in the future with fuel injection and electronic ignition, using this prototype module. The specifications for this engine will be used to identify the functional requirements and configuration of the ECU (module), hardware and sensors.
Lotus Engine Simulation (LES) software will be used to simulate the operation of this single cylinder engine and can provide additional performance information, if required.
Work to be done/objectives to be attained by the student in this project: 1. Literature review: Internal combustion spark ignition engines, combustion processes, abnormal combustion and spark knock, engine control units/modules, fuel injection systems, ignitions systems, automotive sensors and instrumentation, microcontrollers, etc. 2. Using Lotus Engine Simulation (LES) software, set up an engine simulation for the single cylinder carburetted engine. 3. Design, build and test a ECU (module) including associated sensors/hardware to control the fuel injection timing and duration, and spark discharge (ignition) timing as required for different engine operating conditions, with the goal of using it on the engine. Specify and procure the microcontroller, sensors and hardware to be used for controlling the fuel injection event and the ignition timing.

Key modules (Scientific and Engineering Knowledge):
Electronics (245) | Electronics (315) | Mechatronics (424)

RWH4: Development of a Compact Mass Air Flow Rate Measurement Device for a Single Cylinder Engine

Reference	:	RWH4
Supervisor	:	Mr Richard Haines
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The objective of this project is to design, build and test a compact mass air flow rate measurement device to measure the mass air flow rate of clean air drawn into a single cylinder engine during operation, in real time.
The mass air flow rate of clean air drawn into an engine is used to determine the volumetric efficiency of the engine. The device must filter out the pulsations in the air flow due to the engine operation, and determine the average flow rate.
The Department has a number of single cylinder engines mounted on engine test beds and the device will be used in the future to measure the mass flow rate of the clean air drawn into these engines.
Lotus Engine Simulation (LES) software, a user-friendly package, will be used to simulate the operation of a Department's single cylinder engine which will provide information on typical mass air flow rates for different operating conditions.
Work to be done/objectives to be attained by the student in this project: 1. Literature review: Internal combustion spark ignition engines, combustion processes, volumetric efficiency, air-fuel ratio, exhaust gas emissions, automotive sensors and instrumentation, microcontrollers, etc. 2. Using Lotus Engine Simulation (LES) software, set up a simulation for a representative single cylinder engine, to obtain the air flow rates for different operating conditions. 3. Design, build and test the device on an electronic test bench to meet the requirements. Specify and procure the microcontroller, sensors and hardware to be used for the device. The device must include a digital readout to display the air flow rate. The device should provide an output (BNC connector) to capture the air flow rate using a data acquisition system.

Key modules (Scientific and Engineering Knowledge):
Electronics (245) | Electronics (315) | Mechatronics (424)

RWH5: Development of an Electronic Throttle Body Controller for an Internal Combustion Engine

Reference	:	RWH5
Supervisor	:	Mr Richard Haines
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The objective of this project is to design, build and test a controller for an electronic throttle body on an electronic test bench. The controller must control the position of the throttle plate based on air flow requirements and provide real time information on the throttle plate position (angle or percentage opening).
Prior to the use of fuel injection systems, a carburettor was used, which includes a butterfly valve (throttle plate) to control the flow rate of the air drawn into the engine. On modern engine designs with fuel injection, an electronic throttle body incorporating a throttle plate and position sensor is used to control the mass air flow (MAF) rate which is measured by a MAF sensor.
The Department has a Ricardo Hydra single cylinder, spark ignition engine, the specifications for which will be used for developing the electronic throttle body controller and the user interface requirements.
Lotus Engine Simulation (LES) software, a user-friendly package, will be used to simulate the engine performance and provide information on throttle (air flow rate) requirements at different operating conditions.
Work to be done/objectives to be attained by the student in this project: 1. Literature review: Internal combustion spark ignition engines, combustion processes, engine control units/modules, throttle bodies, automotive sensors and instrumentation, microcontrollers, etc. 2. Using Lotus Engine Simulation (LES) software, set up an engine simulation for the single cylinder engine, to obtain typical air flow rates for different operating conditions. 3. Design, build and test a microcontroller-based controller on an electronic test bench to control the position of the throttle plate for different operating conditions. Specify and procure the microcontroller, sensors and hardware to be used for controlling the electronic throttle body and user interface. The controller must provide real time digital readout of the throttle position. The controller must also be compatible with outputs from a test bed PLC that is used to control engine speed and load.

Key modules (Scientific and Engineering Knowledge):

Electronics (245) | Electronics (315) | Mechatronics (424)

Design and performance of a rocket stove

Reference	:	TMH1
Supervisor	:	Mrs Taneha Hans
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Indoor cooking and heating with traditional biomass produces harmful emissions that cause adverse health effects, especially in informal housing that is not well-ventilated. The inefficient burning of biofuels is an unsustainable use of natural resources, motivating the investigation into improved cooking stoves. A rocket stove is to be investigated due to more complete combustion and low emissions.
This project entails: - Design and manufacture a rocket stove with a focus on: - enabling secondary combustion. - turbulent mixing. - ignition and re-fuelling. - Testing to - determine heat release rates. - analyse the flue gas.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Machine Design B (344)

Design, build and test a solar food dehydrator

Reference	:	TMH2
Supervisor	:	Mrs Taneha Hans
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Most of South Africa's dried fruit is dried on drying trays in the sun after they have been picked, due to the ease of production and affordability. Humid conditions may cause fungal growth, and direct sun exposure causes vitamin degradation and can affect the colour and taste of food.
The biggest drawback with this type of drying is that producers have little control over climatic conditions and, as such, the final product quality. A farmer would like to change to the use of a drying tunnel to reduce exposure to pests and adverse weather while ensuring high quality final products. The farmer wants to use a dehydrator that is not reliant on electricity to preserve harvest by using solar resource. This project requires the following steps:
1. Design a solar food dehydrator 2. Build and test a prototype 3. Develop a mathematical model for the adiabatic saturation process based on the ambient conditions 4. Discuss the feasibility for a farmer in the Western Cape

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Machine Design B (344)

Solar Cell Performance as applied to a UAV

Reference	:	RH1
Supervisor	:	Dr René Heise
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
The performance predictions for solar cells on a real UAV is required. This involves the calculation of the view factors, solar radiation strength and associated environmental factors for a vehicle operating at ground level to stratospheric altitudes. Design a user-friendly tool that predicts the performance of the solar cells for a series of supplied aircraft geometries. Note that this project requires a significant portion of self-study to familiarize the student with the applicable engineering methods, such as aircraft performance, solar models and geometry handling of files such as STL and IGES surface models.

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Modelling (334) | Thermofluid Dynamics (344)

Fuel Cell Bay climatic prediction for a UAV

Reference	:	RH2
Supervisor	:	Dr René Heise
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
H2 fuel cells are very sensitive to the environment they operate in. This includes dust and temperature to mention a few. Conceptualise and develop a tool that can be used to define the design of the fuel cell bay for a stratospheric UAV. Note that this project requires a significant portion of self-study to familiarize the student with the applicable engineering methods, such as aircraft performance, functioning of fuel cells and applicable thermodynamic models.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Modelling (334) | Thermofluid Dynamics (344)

Sterile Moth dispenser for a Fixed wing UAV

Reference	:	RH3
Supervisor	:	Dr René Heise
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The use of sterile moths to control orchard pests is an effective method of using environmentally friendly pest controls. The challenge is effectively dispersing the insects. This project is a follow on project that aims to implement the findings of a previous skripsie, amongst miniaturising the electronic controls and implementing a robust GUI to control the device. Note that this project requires a significant portion of self-study and has an interesting interface to multi-body / bulk material dynamics systems and even aerodynamics that have an influence on the design decisions.

Key modules (Scientific and Engineering Knowledge):
Electronics (245) | Mechatronics (424)

Design a Blower for a Blown-Flap system for a UAV

Reference	:	RH4
Supervisor	:	Dr René Heise
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
The take-off and landing performance of a fixed wing UAV is to be improved by making use of a blown flap system. The system requires a bower that supplies a specified volume of air at a specified pressure. Being applicable to a flying vehicle, weight, installed volume and power requirements are constraints that need to be carefully adhered to. This project is in support of a Master's level project, so coordination with other team members is required.

Key modules (Scientific and Engineering Knowledge):
Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344)

Ablative Liner Modelling for a Rocket Motor

Reference	: RH5
Supervisor	: Dr René Heise
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
Solid Rocket motors operate at high temperatures, but for a limited time. Ablative materials are used to protect the structural elements – the ablative liner firstly functioning as a thermal barrier and actively cooling the material by the ablating of the resin matrix that absorbs energy and carried the heat downstream. Develop a numerical tool that can be used to design and analyse the performance of the ablative material. Note that this project requires a significant portion of self-study to familiarize the student with the applicable properties of ablative materials and applicable thermodynamic models.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Modelling (334) | Thermofluid Dynamics (344)

Design and implement a Hardware-In-the-Loop system

Reference	:	RH6
Supervisor	:	Dr René Heise
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	26872102
Direction	:	Mechatronic
Required elective	:	None

Description:
A guidance kit is being developed as part of a larger industrial project that aims to establish a guidance and navigation design capability. Design a HIL system that interfaces with the systems being established in a commercial environment.

Key modules (Scientific and Engineering Knowledge):
Electronics (245) | Mechatronics (424) | Modelling (334)

Calibration of material properties of dolerite rock for discrete element modelling (DEM)

Reference	:	JEH1
Supervisor	:	Prof Jaap Hoffmann
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Calibrating material properties (Young's modulus, Poisson ratio, static and dynamic friction coefficients, etc.) based on a single validation case may result in a whole range of non-unique property combinations.
You must devise multiple small test cases for validation purposes. Your test must be designed for enough particles to give you a reliable result, but the time required for DEM simulations also increase with time. That, combined with a method to systematically work towards a unique set of properties, might require development of optimization skills.
Students are not exposed to DEM on undergraduate level, so 1) you must develop a basic understanding of the theory to grasp the implications of material properties on the modelling results, and 2) learn to use a commercial DEM code
Crushed dolerite rock particles are available for testing.

Key modules (Scientific and Engineering Knowledge):
Modelling (334) | Experimental Techniques (414)

Thermal radiation extinction coefficient in a packed bed

Reference	:	JEH2
Supervisor	:	Prof Jaap Hoffmann
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
A test facility to determine the extinction coefficient in a packed bed was developed in 2025. The power supply was insufficient to heat the bed to temperatures where thermal radiation dominates. This project is aimed at revising the heating and measurement system. A packed bed can be approximated as a participating medium in radiation calculations. This allows easy implementation of radiation heat transfer in a computational fluid dynamics model. Such a model requires representative values of the absorption and scattering coefficients in the bed. The candidate must develop a mathematical model to capture the radiation phenomena, and then extract the relevant radiation properties of the bed experimentally. It is conceivable that the radiation effect can be measured in a bed with a heated top, cooled bottom and insulated sides.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Experimental Techniques (414)

Particle image velocimetry in a randomly packed bed of ellipsoidal particles

Reference	:	JEH3
Supervisor	:	Prof Jaap Hoffmann
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
Thermal energy storage is necessary to offset the diurnal cycle in solar energy applications. For solar thermal energy, sensible heat storage in packed beds of crushed rock is an inexpensive, practical, and flexible solution. For the design of packed beds, heat transfer and pressure drop data for packed beds are required. To date, most work has been done with spherical particles.
Transition to turbulence in packed beds happen at Reynolds numbers of about 350 (depending of the definition), and most of the available work has been done with beds of regular spaced glass or Perspex rods or direct numerical (CFD) simulation. The former is not representative of a randomly packed bed, whilst the latter requires huge computational resources.
An intern from Eindhoven University of Technology developed a PIV system for a structured bed of ellipsoidal particles. Techniques and materials in manufacturing transparent particles have been developed during the internship. This study aims to proceed with the PIV in a moving fluid.
A small wind tunnel with transparent walls that can be filled with transparent particles immersed in a liquid with a matching refractive index will make the flow in the voids visible. Some discrete element modelling might be useful to determine particle positions and orientations. Other techniques may be used to obtain the position and orientation of the particles.
Specialist skills are required to machine and polish the particles; the technology is available in the Western Cape.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Numerical Fluid Dynamics (414)

Determine the heat transfer coefficient for an irregular particle

Reference	:	JEH4
Supervisor	:	Prof Jaap Hoffmann
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Thermal energy storage is necessary to offset the diurnal cycle in solar energy applications. For solar thermal energy, sensible heat storage in packed beds of crushed rock is an inexpensive, practical, and flexible solution. For the design of packed beds, heat transfer and pressure drop data for packed beds are required. To date, most work has been done with spherical particles.
One way of describing the heat transfer coefficient in a packed bed, if to use data for a single particle and correct it for the presence of neighbouring particles. The objective here is to determine the heat transfer coefficient for a single particle at different Reynolds numbers and different flow angles.
Challenges are heating the particle and measuring surface temperatures with minimum disruption of the flow.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Experimental Techniques (414)

Redesign the air distribution system for a mini packed bed thermal energy storage system

Reference	:	JEH5
Supervisor	:	Prof Jaap Hoffmann
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
Thermal energy storage is necessary to offset the diurnal cycle in solar energy applications. For solar thermal energy, sensible heat storage in packed beds of crushed rock is an inexpensive, practical, and flexible solution. For the design of packed beds, heat transfer and pressure drop data for packed beds are required. To date, most work has been done with spherical particles.
The air flow in an experimental thermal energy storage system at Stellenbosch is highly biased, leading to local hot and/or cold spots that reduce storage efficiency.
Redesign the air distribution system of the facility to achieve a more uniform flow distribution through the bed. Use CFD to quantify the improvement.
Computational fluid dynamics might proof to be valuable (but not essential) in this project.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Numerical Fluid Dynamics (414) | Experimental Techniques (414)

Validate Misdorp's 1D model of heat transfer in a packed bed

Reference	:	JEH6
Supervisor	:	Prof Jaap Hoffmann
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Thermal energy storage is necessary to offset the diurnal cycle in solar energy applications. For solar thermal energy, sensible heat storage in packed beds of crushed rock is an inexpensive, practical, and flexible solution. For the design of packed beds, heat transfer and pressure drop data for packed beds are required. To date, most work has been done with spherical particles.
For air/crushed rock thermal energy storage systems, the thermal capacity of the air is so small compared to that of the rocks, that it can safely be ignored. Some researchers suggest that crushed rock is used as a cheap filler material in molten salt systems. Misdorp developed a theoretical model to describe 1D heat transfer in such systems, but his model needs validation.
Develop a test facility, say by using crushed rocks and a liquid, that can provide the necessary data to validate Misdorp's model.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Experimental Techniques (414)

Simulation of a gravitational water vortex turbine

Reference	:	JEH7
Supervisor	:	Prof Jaap Hoffmann
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
A previous skripsie tested a scale model of a gravitational water vortex turbine experimentally. The torque generated was so small that it could not be measured reliably. Upscaling looks promising, but the facilities at M&M is insufficient to test at large scale. It is expected of the student to simulate the turbine at a larger scale, and suggest further improvement.

Key modules (Scientific and Engineering Knowledge):
Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344)

Falling droplet heat exchanger

Reference	:	JEH8
Supervisor	:	Prof Jaap Hoffmann
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
One way of generating hydrogen is the copper-chlorine cycle. The cycle is a good match for molten salt solar thermal systems, as the maximum temperature required (530 °C) corresponds to the hot salt temperature (560 °C) in a central receiver system. Ontario Tech in Canada build a laboratory scale demonstration model of such a plant. They identify the particle heat exchanger as an area that require further development. In this heat exchanger, moisture is removed from the CuCl2 (aq) solution. The preferred way is to atomize the CuCl2, and use the large surface area of the droplets. The task is to develop an atomizer that is capable of controlling droplet size, and measure heat (or mass) transfer rates from the droplets.

Key modules (Scientific and Engineering Knowledge):
Thermofluid Dynamics (344) | Experimental Techniques (414)

Determine the Ergun coefficients for crushed rock particles of different sizes

Reference	:	JEH9
Supervisor	:	Prof Jaap Hoffmann
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Solar thermal energy is a promising clean alternative for fossil fuels. The intermittent nature of solar energy calls for thermal energy storage if this energy is to be available on demand. Packed beds of crushed rocks is a cheap and environmentally benign way of storing thermal energy for several hours.
To design storage facilities, correlations are required that relate rock size and shape, and packing density to the pressure drop across a packed bed at different velocities. The Ergun equation is such a tool, but it is mostly used for spherical particles and performs poorly for non-spherical particles. In this project, the student is expected to expand the Ergun equation to crushed rock particles.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Thermofluid Dynamics (344) | Experimental Techniques (414)

Computational modelling and analysis of the steady-flow haemodynamic environment of bicuspid aortic valves

Reference	:	LGK1
Supervisor	:	Mrs. Lindi Grobler Kock
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
Accurate estimation of the transvalvular pressure gradient (TPG) is essential when diagnosing the severity of aortic stenosis (AS). The simplified Bernoulli equation used in clinical practice to estimate the TPG relies on simplifying assumptions which have been shown to both under- and overestimate the TPG depending on the cardiac output (CO) and the type and severity of AS.
In this project, steady-state CFD simulations of blood flowing through bicuspid aortic valve (BAV) pathologies at peak systole will be developed. This project will aim to evaluate the flow environment and pressure gradients across different BAV phenotypes and for a range of COs to determine the performance of the simplified Bernoulli equation in each case.
This project uses StarCCM+ to develop and run the CFD simulations, image processing to calculate the opening area ratio and determine the severity of the stenosis, and fundamental fluid mechanic principles to derive a theoretical secondary loss TPG model.
The student will be expected to largely self-learn the use of StarCCM+ using available documentation, tutorials, and examples, with guidance provided as needed.

Key modules (Scientific and Engineering Knowledge):
Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344)

Experimental design and analysis of the steady-flow pressure gradients across bicuspid aortic valves

Reference	:	LGK2
Supervisor	:	Mrs. Lindi Grobler Kock
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Accurate estimation of the transvalvular pressure gradient (TPG) is essential when diagnosing the severity of aortic stenosis (AS). The simplified Bernoulli equation used in clinical practice to estimate the TPG relies on simplifying assumptions which have been shown to both under- and overestimate the TPG depending on the cardiac output (CO) and the type and severity of AS.
In this project, the steady-flow pressure gradient across 3D-printed bicuspid aortic valve (BAV) models opened at peak systole will be determined experimentally using a glycerin-based blood analog. This project aims to evaluate the pressure gradients across different BAV phenotypes and for a range of COs, and determine the performance of the simplified Bernoulli equation in each case.
BAV models opened at peak systole will be used. A steady-flow hydraulic test bench will be designed and built to measure the pressure gradient across the BAV models. The experimental results will be compared to an analytical TPG estimation and the standard clinical simplified Bernoulli TPG estimation. This project uses image processing to calculate the opening area ratio and determine the severity of the stenosis, and fundamental fluid mechanic principles to derive a theoretical secondary loss TPG model.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Experimental Techniques (414)

The effect of aortic root diameter on the pressure recovery in aortic stenosis

Reference	:	LGK3
Supervisor	:	Mrs. Lindi Grobler Kock
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
Accurate estimation of the transvalvular pressure gradient (TPG) is essential when diagnosing the severity of aortic stenosis (AS). The effect that the aortic root diameter has on pressure recovery, and therefore the net TPG, is important to consider when evaluating the accuracy of the clinical simplified Bernoulli TPG estimation across different types and severities of AS.
In this project, steady-state CFD simulations will be performed to determine the effect that various aortic root diameters have on the downstream energy dissipation and static pressure recovery for various degrees of calcific and rheumatic AS. Using StarCCM+, CFD models will be developed for various degrees of calcific and rheumatic AS and various aortic root geometries for a range of physiological flow rates. The TPG will be estimated through post-processing of the simulated results. The simulated results will also be compared to the clinical simplified Bernoulli TPG estimation for each geometric configuration.
The student will be expected to largely self-learn the use of StarCCM+ using available documentation, tutorials, and examples, with guidance provided as needed.

Key modules (Scientific and Engineering Knowledge):
Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344)

Preliminary design and performance analysis of an axial compressor for a 250 N microjet engine

Reference	:	RL1
Supervisor	:	Prof. Ryno Laubscher
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
This study has three primary objectives. First, a Flownex SE simulation model of the CAT 250N microjet engine will be developed using the provided turbomachinery maps to establish benchmark results. Second, a mean line simulation code will be developed to design a small axial compressor as a replacement for the current radial compressor. Third, once the compressor design is complete, the mean line simulation will be used to generate a new compressor map. This map will then be integrated into the Flownex model to evaluate the performance differences between the engine configurations using the newly designed axial compressor versus the original radial compressor at different operating speeds. The design output should include a drawing pack for the high-level modifications to be made to the engine.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Thermofluid Dynamics (344)

Forecasting of the power plant boiler thermal performance using deep learning

Reference	:	RL2
Supervisor	:	Prof. Ryno Laubscher
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Develop a time-series deep learning model to predict heat exchanger heat loads in a power plant boiler using one year of operational data. Inputs include air flow rates, fuel flow rates, water flow rates, and temperatures. Students should select one deep learning architecture from LSTM, GRU, or transformer models, and compare its performance against multiple traditional machine learning approaches including linear regression, random forest, XGBoost or LightGBM, and support vector regression. Students must engineer temporal features such as lagged variables, rolling statistics, and rate of change for non-sequential models to ensure fair comparison. The final deliverable is a comparative analysis addressing which approach performs best, whether temporal dependencies justify deep learning complexity, and the accuracy versus computational efficiency trade-offs, with recommendations for practical deployment.

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Modelling (334)

Design and harware-in-the-loop testing of a microjet engine controller

Reference	:	RL3
Supervisor	:	Prof. Ryno Laubscher
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project aims to develop and validate a hardware-in-the-loop (HIL) control system for a 250 N microjet engine. A system model of the engine will be created using either Flownex SE or MATLAB Simulink to capture key thermodynamic and dynamic behaviors. A robust control system will then be designed in Simulink to manage three critical operating modes: engine start-up sequencing, steady-state load following, and controlled shutdown. The control algorithms will be implemented on a microcontroller board (e.g., Arduino) and interfaced with the simulation model for real-time HIL testing. The project will demonstrate stable operation across all three modes with comprehensive safety interlocks including overspeed protection, temperature limits, and flame-out detection. This HIL methodology enables safe, repeatable testing of control strategies and validates controller performance without risking physical hardware or incurring operational costs.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Modelling (334)

Investigation of 3D-printed centrifugal compressor rotors

Reference	:	MRM1
Supervisor	:	Mr Matthew Meas
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	26876949
Direction	:	Mechanical
Required elective	:	None

Description:
Additive manufacturing using reinforced polymer filaments enables the production of centrifugal compressor rotors suited for less demanding applications. This project investigates the aerodynamic performance, material behaviour, and failure mechanisms of 3D-printed rotors under testing. It will run in conjunction with a Mechatronic Engineering final year project that will focus on the design an development of a test bench for centrifugal compressors. Initial experiments will consider truncated cone rotors lacking blades to establish baseline structural and aerodynamic characteristics. The subsequent tests will include rotors with blade profiles that are to be determined using mean line calculations or computational design tools. The performance predictions from the mean line calculations will be used in the aerodynamic performance assessment of the rotors. A sensor arrangement and simple signal processing code is to be developed, as a modular addition to the test bench system of the companion project, to facilitate the vibration analysis for material monitoring of the rotor during testing. Material properties, production constraints, and expected operational stresses will be examined to interpret the performance limits of the various rotor designs and failure risks. The project requires an understanding of fluid mechanics, turbomachinery fundamentals, and vibration analysis techniques.

Key modules (Scientific and Engineering Knowledge):
Thermofluid Dynamics (344) | Vibration and Noise (354)

Development of a test bench for small centrifugal compressors

Reference	:	MRM2
Supervisor	:	Mr Matthew Meas
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Centrifugal compressors require a controlled environment for operational testing. This project concerns the design, construction, and commissioning of a test bench to facilitate the operation of small (sub-100 mm rotor diameter) compressors. It will run in conjunction with a complementary final year engineering project that will investigate the feasibility of 3D-printed centrifugal compressor rotors.
A functional decomposition is to be undertaken and applied to the scope of work to be done, which shall entail: - Mechanical design of the hardware for mounting, aligning, and containing the compressor, drive system, and instruments. - Selection and integration of drive- and control electronics. - Development and deployment of embedded software to regulate the operating conditions of the test bench. - Integration and calibration of suitable instrumentation for measuring key operating parameters and communications to allow recording of data for subsequent analysis. - Incorporation appropriate interlocks and emergency shutdown mechanisms to ensure safe functioning of the test bed. Regular, facilitated consultation among all parties will ensure alignment of functional requirements and compatibility of the testing infrastructure, and support a comprehensive validation process for each project.

Key modules (Scientific and Engineering Knowledge):
Electrical Drive Systems (324) | Machine Design A (314) | Vibration and Noise (354)

Annealing chamber for 3D printed objects

Reference	:	MRM3
Supervisor	:	Mr Matthew Meas
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
Annealing is a heat treatment process that may be used to reduce internal stresses and improve the strength and thermal resistance of 3D-printed parts. This project entails the design and construction of an enclosure capable of uniformly heating multiple objects at once. Heat delivery may be accomplished using a liquid medium or other means that ensures an even temperature distribution. The enclosure should be configurable to accommodate different sizes and quantities of objects. To augment the functionality of the device, the chamber should also feature a ventilation system allowing it to serve as a filament and component dryer. The scope of the project shall include: - Thermal and mechanical design and modeling of the system to ensure consistent temperatures in various physical configurations. - Implementation of suitable heating and ventilation control features. - Integration of sensors for monitoring and data recording. - Incorporation of appropriate safety features. The project will combine principles of heat transfer, mechanical design, and control systems to create a versatile post-processing solution that enhances the durability and quality of 3D-printed components.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Modelling (334)

Integrated process modeling of small-scale linear Fresnel organic rankine cycle concentrating power systems

Reference	:	MRM4
Supervisor	:	Mr Matthew Meas
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Concentrating Solar Power (CSP) technology harnesses sunlight by focusing it to generate heat, which can then be converted to electricity. Linear Fresnel reflector (LFR) systems use multiple flat or slightly curved mirror strips arranged in parallel rows to concentrate sunlight onto a fixed, linear receiver tube. This configuration offers cost and land-use advantages compared to curved mirror systems and efficiently generates high-temperature heat for power production. The Organic Rankine Cycle (ORC) is a thermodynamic cycle using organic refrigerants as working fluids, enabling efficient electricity generation from moderate-temperature heat sources such as LFR systems. Hydrofluoroolefin refrigerants are well regarded for these applications due to their favourable environmental and thermal properties.
The purpose of this project is to develop a validated system model for a small-scale LFR-ORC CSP plant. Analytical models of the solar collector—validated with ray tracing—and key components including the pump, turbine, condenser, and recuperator will be created based on literature data. These component models will be integrated within FlowNex Simulation Environment to analyze system performance under varying configurations, including simple and recuperated cycles.
Additionally, component sizing will be estimated to assess economic feasibility. The levelized cost of energy (LCOE) of the modeled CSP system will be compared to an equivalent photovoltaic and battery storage system using the NREL System Advisor Model.
This integrated modeling effort will generate detailed technical and economic insights to support design development and feasibility evaluation of ORC CSP systems.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Outlet Kinetic Energy Recovery in Axial Flow Fans

Reference	:	CJM1
Supervisor	:	Prof Chris Meyer
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
A fair amount of kinetic energy is present at the outlet of axial flow fans which is, as a rule, lost in induced draught applications. It is however possible to recover some of this kinetic energy through the use of outlet guide vanes and/or diffusers. Recovery increases the fan static pressure and would thus allow the fan to reach higher volume flow rates for a given aerodynamic load. Investigate methods of kinetic energy recovery and subsequently design a number of devices that would enable recovery for a given axial flow fan. The devices are to be constructed and experimentally tested in the small BS848-facility. It is required to not only identify which devices perform better but also offer insights into why they work better.

Key modules (Scientific and Engineering Knowledge):
Machine Design B (344) | Thermofluid Dynamics (344) | Experimental Techniques (414)

Design, Build and Test a High Speed Boat

Reference	:	CJM2
Supervisor	:	Prof Chris Meyer
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The department is in the unique position to operate a large towing tank that allows for the testing of model scale, high speed watercraft. For a given set of full-scale vessel specifications, the student is tasked with the following: 1) Design the vessel using an appropriate methodology. The implementation would most likely be in the form of a computer program. 2) Scale the design to a model size that would be appropriate for testing in the department's towing tank. 3) Construct a model making provision for the adjustment of centre of gravity and total displacement. 4) Test the model in the towing tank and compare the results to the original specifications.
As multiple students can choose this topic, unique vessel specifications will be provided to each student.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Thermofluid Dynamics (344) | Experimental Techniques (414)

Characterization of Brushless DC Motor and Propeller Performance

Reference	:	CJM3
Supervisor	:	Prof Chris Meyer
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The efficiency of a brushless direct current motor (BLDC) is a function of the rotational speed of and the load or torque experienced by the motor. In drone or other aerial vehicle applications, the matching of BLDC performance characteristics and that of the primary air mover (propeller or fan) is important to ensure that flight time and/or outright performance are maximised. The task at hand is the design and construction of a dynamometer that not only characterise the performance of a BLDC but also the propeller that is used to apply the load on the BLDC. Consequently the dynamometer should be able to measure rotational speed, shaft torque as well as the overall thrust produced by the propeller. In addition to the above, the energy input from the source (battery or power supply) to the speed controller and from the speed controller to the BLDC should also be measured to provide insight into the thermal losses associated with the complete drive train. Consolidating the above measurements into an application (App) that can run on a computer or mobile device is desirable. Provision should also be made for the testing of more complex drive-train arrangements such as contra-rotating props.

Key modules (Scientific and Engineering Knowledge):
Electrical Drive Systems (324) | Electronics (315) | Experimental Techniques (414)

Development of an airfoil performance characteristic demonstrator

Reference	:	MO1
Supervisor	:	Prof Michael Owen
Can be completed in first semester	:	Yes
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The Large Wind Tunnel is one of Department of Mechanical and Mechatronic Engineering's key facilities and the demonstration of airfoil performance is one of the common uses of this facility. The current demonstrator involves a large airfoil with static pressure taps along its top and bottom surfaces. These taps connect to water manometers, which allow for demonstration (and calculation) of the pressure distribution over the airfoil during operation at different angles of attack (which is manually adjusted). The demonstrator is used for public engagement (e.g. Engineering Open Day) and practicals for students within and outside of the Department.
In this project, the student will be required to design a modernized demonstrator. Specifically, changing the angle of attack and measuring the airfoil performance should be automated. The aim is to design a system allows for realtime control, data collection, analysis and presentation. The project includes design of the hardware and software (including data processing) aspects of this device.
The scope of the project includes detailed design and theoretical demonstration. Simulation of airfoil behaviour using e.g. CFD will be used to support the mechanical design process. Component manufacture, sub-system assembly and demonstration outside of the wind tunnel are expected. Operational demonstration in the wind tunnel is not included since the facility will not be available due to the refurbishment of the laboratories.
Mechanical Engineering students are encouraged to take Numerical Fluid Dynamics 414 (CFD).

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Machine Design B (344) | Experimental Techniques (414)

Enhancement of electric vehicle aerodynamics

Reference	:	MO2
Supervisor	:	Prof Michael Owen
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
Electric vehicles (EVs) are a key contributor in the global move towards cleaner energy and transport systems. One of the major obstacles faced by EVs is limited driving range, and one approach to increasing range is aerodynamic enhancement to reduce drag. EVs come in a variety of shapes and sizes from small three-wheeled last-mile transport vehicles, to large trucks for long distance freight transport. This project involves the aerodynamic enhancement of an EV. The student may choose the type of EV to focus on, informed by a detailed review of the literature to identify types and scenarios of maximum impact. A validated numerical (CFD) model must then be developed, used to investigate the aerodynamic characteristics of the vehicle and identify opportunities for aerodynamic improvement. Aerodynamic enhancements must be designed and evaluated in CFD and a "best" solution should be proposed for the given vehicle and operating context. Optimization or parametric analysis may be included. Aerodynamic improvement must be extrapolated to quantify the benefit in terms of range.

Key modules (Scientific and Engineering Knowledge):
Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344)

Design and analysis of a solar cooling system for chilled water at a typical soft-drink facility in South Africa

Reference	:	MO3
Supervisor	:	Prof Michael Owen
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Cooling is a major contributor to energy consumption in many industrial processes. Cooling is typically provided by electrically powered chillers (vapor compression refrigeration cycles). Solar cooling involves using solar energy to replace grid electricity (largely from fossil fuels) as the energy source for industrial cooling systems, thus reducing the emissions footprint of cooling processes. Two classes of solar cooling exist: photovoltaic (PV) and thermal systems. This project focuses on the latter, where solar heat is collected and used in a thermally driven sorption cooling process. Solar thermal cooling is a relatively novel concept that requires case-specific techno-economic analysis to establish its overall attractiveness.
This project involves the conceptual design of a solar thermal system retrofit in an existing soft-drink facility in South Africa. A process model of the system must be developed and an annual simulation of system performance conducted. In addition, financial modelling is required to allow for assessment of the economic impact of the retrofit.
The project will be co-supervised by Dr Francois Rozon, a post-doctoral research fellow at the Centre for Renewable and Sustainable Energy Studies.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414)

Development of a validated tool for determining water flow distribution in a cooling tower spray manifold

Reference	:	MO4
Supervisor	:	Prof Michael Owen
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
Evaporative cooling towers ranging from large natural draft cooling towers for thermal power plants, to small rooftop mechanical draft cooling towers for HVAC systems, are commonly used in industry. The performance of these cooling towers is highly dependent on a uniform water distribution over the heat exchange media, which is realized by means of set of distribution pipes fitted with relative closely spaced spray nozzles. Due to the proximity of the nozzles, the branch dividing loss coefficient for each nozzle may not be independent. In addition, the volume flow from each branch will be influenced by the volume flow and velocity past each nozzle, reduction in static head due to friction, varying pipe and nozzle diameters along the length of the pipe. The effect of the above variables on the volume flow from each nozzle is not well understood and there is a need for a simplified software tool for analyzing the flow in a distribution pipe for a given set of input parameters.
in this project, the student will use a combination of fundamental principles and numerical simulation (CFD and / or 1D process modelling) to analyze the flow distribution in a cooling tower spray manifold. A validated, user-friendly analytical tool for solving the flow distribution in a distribution pipe under specific conditions must be developed.

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Numerical Fluid Dynamics (414)

Investigation on the effect of mixing surfaces in hybrid cooling towers

Reference	:	MO5
Supervisor	:	Prof Michael Owen
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
The widespread use of artificial intelligence and the ever-increasing demand for cloud data storage has resulted in more and more data centers being constructed worldwide. These data centres are often cooled using an evaporative process that produces a visible plume (cloud) above the cooling towers. Regulations prohibit visible plumes in many locations (notably in Europe) and hybrid cooling towers (a combination of dry and evaporative systems) are used for the purpose of plume abatement.
In a hybrid cooling tower, air is simultaneously drawn through a counterflow wet cooling section and a crossflow air-cooled heat exchanger. These two air streams are then mixed before entering the axial flow fan situated at the outlet of the cooling tower. Mixing surfaces are sometimes used to enhance mixing and plume abatement. The effectiveness and effect of these surfaces on the fan inlet air flow conditions are not well documented. Inlet air flow conditions are a dominant factor affecting fan performance and vibration.
This project involves the sizing of a hybrid cooling tower (using information from literature and thermodynamic calculations) and numerical simulation of a simplified system using computational fluid dynamics. The objective of the work is to quantify the impact of mixing surfaces on plume abatement and on the operating conditions experienced by the axial flow fan.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Numerical Fluid Dynamics (414)

Simulation of air-cooled condenser performance and associated annual power plant output

Reference	:	HP1
Supervisor	:	Prof Hannes Pretorius
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Modern thermal power plants in arid and semi-arid locations employ water conserving dry cooling technologies to reject the required heat from the cycle to the environment. Among these technologies are mechanical draft air-cooled condensers (ACCs), which employ a multitude of large diameter axial flow fans to force airflow across heat exchanger bundles.
This topic will involve the thermo-fluid simulation of the performance of an ACC designed for a specific power generation application. By coupling the ACC to a turbine characteristic and using meteorological data for a selected site as input, the energy generation of the power plant will then be determined over a typical year.
The project will include the development of a 0-D ACC simulation model using your choice of non-commercial programming package, and designing an appropriate system size to achieve the required design point performance. Thereafter the model will be used to evaluate the annual performance of the power plant which uses this ACC, for two different heat exchanger finned-tube designs. Site meteorological data will need to be sourced and processed, and the effect of wind will need to be approximated in the model.
As 3 students may select this topic, each will be tasked to size and simulate an ACC for a different power generation application (50 MWe concentrated solar power, 600 MWe (200 MWe steam component) combined cycle gas turbine, or 800 MWe coal-fired application).

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Simulation of natural draft air-cooled condenser performance and associated annual power plant output

Reference	:	HP2
Supervisor	:	Prof Hannes Pretorius
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Modern thermal power plants in arid and semi-arid locations employ water conserving dry cooling technologies to reject the required heat from the cycle to the environment. Among these technologies are natural draft air-cooled condensers (NDACCs), which employ large dry cooling towers to induce airflow across heat exchanger bundles.
This topic will involve the thermo-fluid simulation of the performance of an NDACC designed for a specific power generation application. By coupling the NDACC to a turbine characteristic and using meteorological data for a selected site as input, the energy generation of the power plant will then be determined over a typical year.
The project will include the development of a 0-D NDACC simulation model using your choice of non-commercial programming package, and designing an appropriate system size to achieve the required design point performance. Thereafter the model will be used to evaluate the annual performance of the power plant which uses this NDACC, for three different heat exchanger finned-tube designs. Site meteorological data will need to be sourced and processed, and the effect of wind will need to be approximated in the model.
As 3 students may select this topic, each will be tasked to size and simulate an NDACC for a different power generation application (50 MWe concentrated solar power, 600 MWe (200 MWe steam component) combined cycle gas turbine, or 800 MWe coal-fired application).

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414)

Hospital based robotic assistant (competition)

Reference	:	HP3
Supervisor	:	Prof Hannes Pretorius
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The successful candidate will be responsible for the design, build and control implementation of a floor-bound, obstacle avoiding, pathfinding robot. The robotic assistant should function as a mobile information kiosk, capable of interacting with patients via a custom HMI and leading patients to a designated ward or physician office.
Performance will be measured through various requirements, including but not limited to, speed, efficiency, reliability and safety. A full set of customer requirements from the sponsoring industry partner will be provided at the commencement of the project.
You will be given the chance to compete against other students with the same topic after the final year projects have been examined. Note that each student will have to adhere to a strict set of rules to ensure the integrity of the competition and fairness towards other participants. Examination of the final year project and competitive performance assessment will be mutually exclusive.
Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Electronics (315) | Mechatronics (424)

Design and optimisation of a closed loop evaporative cooling tower for liquid cooling

Reference	:	HR1
Supervisor	:	Prof Hanno Reuter
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
Closed loop evaporative cooling systems are generally used as refrigerant condensers in the HVAC industry. In a specific configuration, these cooling systems can also be used effectively for the cooling of liquids. This project entails the development of a computational and CFD model of such a cooling system and to conduct a parametric study for the optimisation of such cooling systems.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Numerical Fluid Dynamics (414)

Design and optimisation of a closed loop liquid cooling system with a wet-cooling tower

Reference	:	HR2
Supervisor	:	Prof Hanno Reuter
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
This project entails the development of a computational model of such a cooling system and to conduct a parametric study for the optimisation of such cooling systems.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Numerical Fluid Dynamics (414)

Design and optimisation of closed loop dry air-cooled heat exchanger systems using different types of finned tubes

Reference	:	HR3
Supervisor	:	Prof Hanno Reuter
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
This project entails the development of a computational models of such cooling systems and to conduct a parametric study for the optimisation of such cooling systems.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Thermofluid design and evaluation of a reverse cycle air conditioning unit for residential space cooling and heating with the aid of simulation

Reference	:	PGR1
Supervisor	:	Prof Pieter Rousseau
Can be completed in first semester	:	No
Students allowed for this topic	:	3
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
The aim of this project is to conduct the thermofluid system design of a reverse cycle wall split air conditioning unit for residential space cooling and heating with a given nominal cooling capacity, and then to demonstrate its year-round performance with the aid of simulation. The design and evaluation will, amongst others, include the following steps: Understanding the layout and operation of existing residential air conditioning systems, analysing the relevant room loads and weather data to determine suitable design requirements, selecting nominal cycle operating parameters, analysing the refrigeration cycle to determine suitable component design specifications, selecting a suitable off-the-shelf compressor, designing suitable condenser and evaporator heat exchanger layouts, sourcing suitable fans, developing process models for each of the components and integrating these into the cycle simulation, applying the integrated simulation to evaluate the year-round performance of the system using hourly weather data. All analyses and models will require programming in Python and there will be no physical experimentation. The project will allow the participation of three students, each independently developing and analysing their system for a different cooling capacity and climatic region, typified by the weather data for Cape Town, Johannesburg, and Durban, respectively. Each student will also evaluate the year-round performance of their system in Upington to gain insight into the effect of climate variations on their specific design.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Thermofluid Dynamics (344)

Development and simulation of adiabatic cooling system for a sCO2 Brayton recompression cycle

Reference	:	MTC1
Supervisor	:	Mr Carl Tshamala
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Hybrid dry/wet cooling systems are mainly cooling systems that utilized both dry and wet cooling approaches to address the issue of wet cooling excessive water consumption and dry cooling power consumption in order to enhance the overall power plant performance. This is particularly important in arid regions where water resources are scarce and the dry bulb temperature relatively high making the heat rejection challenging. In this study, an adiabatic dry/wet cooling system is investigated for cooling of a concentrated solar power (CSP) plant using a supercritical carbon dioxide recompression Brayton cycle. The student will make use of a pre-built recompression cycle and, will insert an adiabatic cooling system that can insure a relatively stable compressor inlet temperature. The complete 1-D model will be built in Flownex.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414)

Design and simulation of a deluged bare tubes heat exchanger bundle of a sCO2 Brayton recompression cycle

Reference	:	MTC2
Supervisor	:	Mr Carl Tshamala
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Hybrid dry/wet cooling systems are mainly cooling systems that utilized both dry and wet cooling approaches to address the issue of wet cooling excessive water consumption and dry cooling power consumption in order to enhance the overall power plant performance. This is particularly important in arid regions where water resources are scarce and the dry bulb temperature relatively high making the heat rejection challenging. In this study, a deluged cooling system is investigated for cooling of a concentrated solar power (CSP) plant using a supercritical carbon dioxide recompression Brayton cycle. The student will make use of a pre-built recompression cycle and, will insert a deluged cooling system that can insure a relatively stable compressor inlet temperature accounting for variable ambient conditions. The complete 1-D model will be built in Flownex.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414)

Cooling performances enhancement of an air-cooler of an electric power generator

Reference	:	MTC3
Supervisor	:	Mr Carl Tshamala
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
This project focuses on enhancing the cooling performance of an air-cooler used in an electric power generator through the design of an improved finned-tube heat exchanger. The student will determine the cooling requirements and develop an adequate finned-tube configuration considering heat transfer, airflow, and pressure drop constraints. The proposed air-cooler will be modelled and simulated in Flownex to evaluate its thermal and hydraulic performance under realistic operating conditions. Simulation results will be used to assess temperature reduction capability and overall effectiveness, leading to an optimized air-cooler design that improves generator cooling and operational reliability.

Key modules (Scientific and Engineering Knowledge):
Electrical Drive Systems (324) | Heat Transfer A (414) | Modelling (334)

Design and simulation of shell-and-tubes heat exchanger for sCO2 Brayton recompression cycle

Reference	: MTC4
Supervisor	: Mr Carl Tshamala
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
For many years, steam Rankine cycles have been considered to the most effective systems for the power generation industry ranking higher than Brayton cycles. One of the reasons for lower Brayton cycles efficiencies has always been due to the relatively high back work ratio. It is believed that using sCO2 as working fluid and ensuring relatively average compression temperature could significantly contribute to increasing the overall Brayton cycle efficiency. Amongst the many cycles sCO2 configurations known, this study will focus on the Brayton recompression and a full design of a water cooled shell-and-tube heat exchanger is suggested in order to ensure a stable and low compressor inlet temperature (30 and 33°C). The candidate will build 1-D simulation model in Flownex and integrate it into a Brayton recompression cycle. Particular attention shall be drawn to the overall cycle performances in arid region with high solar resources such as Upington.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Modelling (334)

Design and simulation of an air-cooled heat exchanger for sCO2 Brayton recompression cycle

Reference	: MTC5
Supervisor	: Mr Carl Tshamala
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
For many years, steam Rankine cycles have been considered to the most effective systems for the power generation industry ranking higher than Brayton cycles. One of the reasons for lower Brayton cycles efficiencies has always been due to the relatively high back work ratio. It is believed that using sCO2 as a working fluid and ensuring relatively average compression temperature could significantly contribute to increasing the overall Brayton cycle efficiency. This study will focus on the design and simulation of an air-cooled heat exchanger suggested to operate as sCO2 pre-cooler and its integration into the Brayton recompression. The sCO2 outlet temperature is expected to be ranging between 30 - 33°C. The candidate will build 1-D simulation model in Flownex and integrate it into a Brayton recompression cycle. Particular attention shall be drawn to the overall cycle performances in arid region with high solar resources such as Upington.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Modelling (334)

Feasibility study of a sCO2 recompression cycle cooled by reversed Brayton cycle

Reference	:	MTC6
Supervisor	:	Mr Carl Tshamala
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
For many years, steam Rankine cycles have been considered to the most effective systems for the power generation industry ranking higher than Brayton cycles. One of the reasons for lower Brayton cycles efficiencies has always been due to the relatively high back work ratio. It is believed that using sCO2 as working fluid and ensuring relatively average compression temperature could significantly contribute to increasing the overall Brayton cycle efficiency. However, challenges of relatively high ambient temperatures recorded in high-solar resources regions (such as Upington) have been identified as major barrier to achieving higher cycles efficiency. In study, the candidate will be required to investigate the possibility of using a refrigeration loop to ensure cooling, and its challenges. The candidate will build 1-D simulation model in Flownex and integrate it into a Brayton recompression cycle. Particular attention shall be drawn to the overall cycle performances in arid region with high solar resources such as Upington.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414) | Thermofluid Dynamics (344)

Design of a natural draft wet cooling tower for concentrated solar powered sCO2 recompression power block

Reference	:	MTC7
Supervisor	:	Mr Carl Tshamala
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Traditionally, wet cooling is considered as the primary option for heat rejection in power generation. This is motivated by its ability for effective cooling due to water thermal properties. In this project, effective cooling of the working fluid (sCO2) is required to ensure overall cycle efficiency. This project targets 30 - 33°C for the carbon dioxide at the outlet of the cooling tower. The candidate is required to build a 1-D simulation model of the cooling tower in Flownex environment and integrate it to an existing sCO2 Brayton recompression cycle. This project intends to investigate the cycle's overall response variable weather conditions. Water consumption and cycle efficiency are critical for this project, hence the simulation should consider annual profile of the weather conditions in Upington.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Heat Transfer A (414)

Design of a Temperature Control Mechanism for an Atlas Copco ZR 630 Compressor to Ensure Reliable Air Supply in Copper Lixiviation Operations

Reference	: MTC8
Supervisor	: Mr Carl Tshamala
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Lixiviation, or leaching, is a key hydrometallurgical step in copper processing in which chemical solutions dissolve copper from crushed ore for subsequent recovery. At the Shituru copper plant in the Democratic Republic of Congo, this process relies on a steady supply of compressed air produced by an Atlas Copco ZR 630 oil-free compressor. During summer, the compressor frequently trips due to excessively high discharge air temperatures, exceeding the allowable limit designed to protect downstream rubber piping and connectors from thermal damage. These shutdowns disrupt operations, reduce efficiency, and compromise the stability of the lixiviation circuit. This project requires the student to design an effective control mechanism that maintains the compressor's discharge temperature within a safe operating range under varying ambient conditions. The proposed solution should address thermal loading, improve cooling performance, and ensure continuous, reliable compressor operation to support uninterrupted lixiviation at the Shituru plant.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Electrical Drive Systems (324) | Modelling (334)

Design and Simulation of a Redundant Water-Cooled Heat Exchanger System for an Atlas Copco ZR 630 Compressor at the Shituru Copper Plant

Reference	: MTC9
Supervisor	: Mr Carl Tshamala
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechanical
Required elective	: None

Description:
This project focuses on the design and performance simulation (Using Flownex) of a water-cooled heat exchanger system intended to cool the discharge air of an Atlas Copco ZR 630 oil-free compressor operating at the Shituru copper processing plant in the Democratic Republic of Congo. The existing factory-installed coolers regularly experience fouling and reduced heat transfer efficiency, leading to overheating events and operational interruptions during critical plant activities such as copper lixiviation. To improve reliability and ensure uninterrupted air supply during maintenance of the original heat exchangers, the student will design a fully functional redundant water-cooled exchanger sized to handle full compressor load and compatible with existing piping and plant cooling water infrastructure. The project includes thermal design calculations, selection of appropriate heat exchanger geometry, evaluation of pressure drop, and dynamic simulation of the integrated system to assess temperature control performance under varying compressor load and cooling water conditions. The final solution should offer high thermal reliability, ease of integration, and improved operational resilience for the Shituru air compression system.

Key modules (Scientific and Engineering Knowledge):
Heat Transfer A (414) | Thermofluid Dynamics (344)

Upgrade of the Ysterplaat Air Force Museum flight simulator prototype

Reference	:	AV1
Supervisor	:	Dr Adam Venter
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The Ysterplaat Air Force Museum is continuing its initiative to enhance its visitor experience through the development of an immersive flight simulator built around an authentic helicopter cockpit. Last year's project delivered the first prototype, including the physical cockpit setup, sensor integration for the collective and anti-torque controls, and a basic virtual environment.
This year's project will focus on evaluating and improving the existing prototype to move it closer to a robust, visitor-ready system. The student will assess current hardware and software performance to identify shortcomings related to reliability, ergonomics, and overall effectiveness, and will implement upgrades that improve realism, stability, and ease of use. A key requirement for this year is the development and integration of the missing cyclic control system. Additional tasks may include refining cockpit instrumentation and feedback, upgrading visual display systems, and enhancing overall simulator comfort.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314)

Optimisation of a desktop wind tunnel for STEM education and demonstration

Reference	:	AV2
Supervisor	:	Dr Adam Venter
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
Last year's project established the first prototype of a desktop-sized educational wind tunnel designed as a hands-on demonstration tool for STEM learning. The prototype enables basic visualization of airflow around objects and introduces students to key aerodynamic and fluid-dynamic principles.
This year's project will focus on optimising the existing design to improve accuracy, usability, and teaching effectiveness. The student will evaluate the current wind tunnel's performance, identify limitations in flow quality, measurement capabilities, and structural design, and implement improvements accordingly. This may include redesigning and rebuilding components such as the flow conditioning system, test section, or fan arrangement; enhancing measurement and data visualisation capabilities. The aim is to advance the wind tunnel toward a robust, classroom-ready educational tool that provides clear demonstrations, reliable data, and an engaging learning experience for students.
(Available for both Mechatronic & Mechanical students; however Mechanical students will be required to do CFD)

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314) | Experimental Techniques (414)

Characterizing the influence of attached LPS25HB pressure sensors on fan blade aerodynamic characteristics and performance

Reference	:	AV3
Supervisor	:	Dr Adam Venter
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
This project aims to explore the impact of attaching LPS25HB pressure sensors to the surfaces of low-pressure axial flow fans, with a focus on understanding how this affects the fan's aerodynamic characteristics and performance. The LPS25HB sensors are compact, low-power pressure transducers that can be easily mounted on the outer surfaces of axial fans, offering a cost-effective solution for measuring fan performance. However, their influence on fan blade aerodynamics, particularly regarding flow patterns, pressure distribution, and overall blade performance, is not yet widely understood. This study seeks to provide insight into these potential effects, and to further provide guidance on the optimal configuration/placement of these sensors for accurate pressure profiling.

Key modules (Scientific and Engineering Knowledge):
Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344)

Hospital based robotic assistant (competition)

Reference	: AV4
Supervisor	: Dr Adam Venter
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
The successful candidate will be responsible for the design, build and control implementation of a floor-bound, obstacle avoiding, pathfinding robot. The robotic assistant should function as a mobile information kiosk, capable of interacting with patients via a custom HMI and leading patients to a designated ward or physician office. Performance will be measured through various requirements, including but not limited to, speed, efficiency, reliability and safety. A full set of customer requirements from the sponsoring industry partner will be provided at the commencement of the project. Successful candidates are expected to conduct themselves in a respectful and professional manner whenever they interact with the sponsoring industry partner as you will be representing the department and the university.
You will be given the chance to compete against other students with the same topic after the final year projects have been examined. Note that each student will have to adhere to a strict set of rules to ensure the integrity of the competition and fairness towards other participants. Examination of the final year project and competitive performance assessment will be mutually exclusive.
Disclaimer: Given the fact that the Mechanical and Mechatronic Engineering department is under renovation and laboratory spaces are limited, we cannot guarantee a space to work in. Candidates must be willing to work on this project at home. No use of AI allowed.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Mechatronics (424)

Design and development of a mechanical casualty recovery system for NSRI Rescue 2

Reference	:	AV5
Supervisor	:	Dr Adam Venter
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	None

Description:
Gemini Legend, callsign Rescue 2, is a 6.5 m rigid inflatable boat (RIB) operated by NSRI Station 2 in Bakoven and is routinely deployed in rapid-response rescues where swift patient retrieval is essential. However, recovering injured or unconscious individuals from the water presents significant challenges, particularly on small rescue craft with limited deck space, high sides, and dynamic sea states. Current retrieval methods rely heavily on physical strength and coordinated manual effort, placing both rescuers and casualties at risk.
The aim of this project is to design, develop, and test a mechanical casualty retrieval system tailored to the constraints and operating environment of Rescue 2. The system should enable rescuers to recover casualties onto the vessel more easily, effectively, and safely, while accounting for the practical limitations of coastal rescue operations.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Machine Design B (344) | Modelling (334)

Autonomous navigation and control of a model vessel

Reference	:	AV6
Supervisor	:	Dr Adam Venter
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project focuses on developing and demonstrating autonomous navigation capabilities for a scaled ship model. The student will design and build a small-scale model equipped with the necessary sensing, actuation, and control hardware. The goal is to implement and test autonomous guidance, navigation, and control algorithms in calm-water conditions, including basic waypoint finding and collision avoidance. The project is undertaken in collaboration with engineers from the CSIR.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Machine Design A (314)

Design of a low-cost, mobile honey extractor

Reference	:	JVDS1
Supervisor	:	Prof Johan van der Spuy
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The extraction of honey from beehive frames using centrifugal forces is an acceptable standard in the beekeeping industry. Such a honey extractor can however be bulky and difficult to store and transport. It therefore requires a relatively permanent installation that is not always available.
The student is required to design an alternative, low cost, honey extraction mechanism. The mechanism must be compact and easy to store and transport. It should have sufficient capacity to handle two standard size bee frames. It is known that the viscosity of honey is highly sensitive to heat and that the addition of heat might simplify the extraction process considerably. The concept should therefore have the ability to incorporate a small amount of heat addition. To prevent damage to the wax, however, the heating should be controllable.
The concept should be designed, constructed and shown to operate using a mock-up, if original filled bee frames are not available.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Machine Design A (314)

Reverse-engineering and calculating the performance of the Junkers Jumo 004 turbojet.

Reference	:	JVDS2
Supervisor	:	Prof Johan van der Spuy
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
The SAAF museum at Ysterplaat AFB has access to a sectioned full-scale Junker Jumo 004 B1 turbojet engine. This engine is significant in that it was the world's first production turbojet engine. SU has signed an MOU with the SAAF museum at Ysterplaat, by which SU students will be able to access specific hardware that are currently in the museum's possession.
For this project the student will be required to reverse engineer the flow path of the engine that is currently on display in the museum. A previous student has already compiled a CAD model of the existing engine. The performance of the engine will subsequently be modelled using an in-house mean-line code, CFTurbo, GasTurb and Cadence/FINE/Turb and compared to published data for the engine.
The focus of the work will be to replicate the engine's performance and not to design the engine.

Key modules (Scientific and Engineering Knowledge):
Energy Systems M (434) | Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344)

Development of a pressure sensor for wireless measurement on a fan blade

Reference	:	JVDS3
Supervisor	:	Prof Johan van der Spuy
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The department has significant experience in the development of components for air-cooled condensers (ACCs). These condensers are particularly popular in South Africa where water resources are limited. Recently a 24 ft. fan and system test facility has been constructed at the department. The facility is referred to as the MinwaterCSP facility. For this project, the student will design a remote sensing pressure measuring device that can be used to measure the pressure on a rotating fan blade. The lay-out and data processing of such a set-up has been developed by a previous student and is available to this project.
The student will design a pressure sensing device that he will build and compare to results from values measured on an existing device. The device will be tested on a moving vehicle in a freestream environment.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (245) | Mechatronics (424)

Development of a planetarium projector for an existing dome

Reference	:	JVDS4
Supervisor	:	Prof Johan van der Spuy
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical and/or Mechatronic
Required elective	:	None

Description:
The SAAF museum at Ysterplaat AFB has an existing planetarium dome that is currently not being utilised. The dome was originally built to train pilot in night navigation but has a tremendous potential as attraction for teaching and display to visitors of the museum. The existing projector is certified as a museum display and may no longer be used for active planetarium shows. This project will therefore look at replacing the existing projector with a more user-friendly, digital projector and display system.
The project will require the student to make contact with the curator of the museum and to set up the user-requirements for the system. The student must then analyse the design and function of the existing projector system and consequently build a complete stand and integrated digital projector system to replicate the original projector. The outcome of the system must be that it can be used as-is during museum days by appropriately trained personnel.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (245) | Mechatronics (424)

Proof of Concept for a Windmill Design with Improved Water Pumping Efficiency

Reference	:	JVDS5
Supervisor	:	Prof Johan van der Spuy
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	26644428
Direction	:	Mechanical
Required elective	:	None

Description:
The task of the skripsie is a "proof of concept". In other words, the performance of a windmill should be determined for a sample application with given wind patterns and borehole depth (by determining the total efficiency of the windmill at different wind speeds, since the windmill will not be operating at the best efficiency point all the time): A design of the system for maximum efficiency should be made by matching torque vs rotational speed curves of the wheel and pump in a way that both operate as close to the best efficiency point as possible by choosing a suitable pump and selecting the right gear ratio for the bevel gear pair. For the pump, the data needed to do this can be calculated from readily available pump characteristic curves, but similar data is not necessarily available for the aerofoils. The data for the wheel should be obtained by building a power transmission system and testing the wheel to find its torque versus rotational speed at different wind speeds.

Key modules (Scientific and Engineering Knowledge):
Machine Design A (314) | Machine Design B (344) | Thermofluid Dynamics (344)

Development of a unique air filter system

Reference	:	JVDS6
Supervisor	:	Prof Johan van der Spuy
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
A recent investigation at Stellenbosch University has identified the tremendous detrimental effect of smoke inhalation coming of in-house fireplaces. A recent need has been identified to develop an effective in-house filter systems that can clean living areas from air-borne particles, as well as any volatile organic compounds.
An potential solution has already been identified. The student is required to assemble and commission a test facility where different filter materials can be tested and their performance quantified to a level of scientific publication.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Machine Design A (314) | Mechatronics (424)

Quantifying the performance of a unique fan blade tip appendage

Reference	:	JVDS7
Supervisor	:	Prof Johan van der Spuy
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechanical
Required elective	:	Computational Fluid Dynamics (CFD)

Description:
A unique fan blade tip appendage has been developed by a previous student. This work will consider the further testing and development of the tip appendage into an implementable product. The performance of the appendage will be quantified using experimental and numerical techniques. Specific attention will be paid to the noise characteristics of the appendage.

Key modules (Scientific and Engineering Knowledge):
Numerical Fluid Dynamics (414) | Thermofluid Dynamics (344) | Vibration and Noise (354)

2 Mechatronic Topics at EE

Sorting machine for Lego pieces

Reference	:	BB1
Supervisor	:	Prof Bernard Bekker
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Many houses have boxes full of unsorted Lego pieces, ranging from small Technic gears and axles to Lego bricks, plates, and minifigures. This project will build a sorting machine that accepts a load of unsorted Lego pieces as input, separates the load into individual pieces on a conveyor belt that passes through an identification system, and then sorts the individual pieces into multiple bins based on the identification.
The primary focus of the project will be on the sorting machine itself, with success criteria such as the consistency and accuracy of individual-piece separation and the speed from separation to bin. As the identification system in this sorting machine is the focus of a second M&M final-year project, this project will only need to place the individual Lego piece into the correct bin based on the bin number received from the identification system.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Mechatronics (424)

Intelligent irrigation system

Reference	:	JCB1
Supervisor	:	Mr Nelius Bekker
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project focuses on the design and development of an intelligent irrigation system that optimises water usage through automated monitoring and control. The system should integrate environmental sensing, data processing, and decision-making algorithms to determine precise irrigation requirements. The system's capabilities should also include remote monitoring and data logging.
The project encompasses hardware design, embedded programming, control algorithm development, and system testing, culminating in a fully functional prototype capable of autonomously managing irrigation based on real-world conditions.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (245) | Mechatronics (424)

Automated Tea Maker

Reference	:	JCB2
Supervisor	:	Mr Nelius Bekker
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project involves the design and development of an automated tea-making system capable of preparing consistent, high-quality tea with minimal human intervention. The system automates key steps in the tea preparation process—including water heating, temperature regulation, steeping time control, and dispensing—to ensure repeatable results tailored to user preferences. The project covers hardware design, embedded electronics, control algorithm implementation, mechanical design of the brewing mechanism, and the development of a prototype that reliably automates the tea-making process.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Mechatronics (424)

Smart inverter control for voltage rise mitigation

Reference	: JBK1
Supervisor	: Dr Johan Beukes
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Operational small-scale embedded generators cause line voltage to rise, due to the direction of power flow. This rise currently limits the use of renewable energy on residential feeders to 25% of capacity before voltage limits are exceeded. If voltage can be controlled on these systems, this limit can be increased to higher levels. Inverters can control active and reactive power independently by controlling phase angle and voltage respectively. Smart inverters have the capability to change operating conditions based on communication signals. This project will start with investigating and quantifying the problem by acquiring information and data. An existing smart inverter will be installed and a communication link to an external computer will be established. A control strategy will be developed, tested in Matlab Simulink and then implemented using the assembled hardware.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Electronics (315)

Microwave Sensor Loaded With Complementary Curved Ring Resonator for Material Permittivity Detection

Reference	:	LB1
Supervisor	:	Dr Leanne Bodenstein
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project focuses on the development of a microwave sensor utilizing a complementary curved ring resonator designed for the precise detection of material permittivity. By leveraging the unique electromagnetic properties of the curved ring resonator, the sensor aims to enhance sensitivity and accuracy in measuring dielectric constants across various materials. The sensor operates by analyzing changes in resonance frequency and bandwidth in response to the introduction of different substances, enabling real-time monitoring and characterization. Applications of this technology span multiple fields, including materials science, food quality assessment, and biomedical diagnostics, where understanding material properties is crucial. This innovative approach promises to advance non-destructive testing methods and improve the efficiency of material analysis processes. The student will have to design the sensor, manufacture and test the sensor and analize the performance of the sensor.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (245) | Electronics (315)

Finite element solver for sensitivity analysis of electro-mechanical devices

Reference	:	MMB1
Supervisor	:	Prof Matthys Botha
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Sensitivity analysis with respect to structural elements can be done efficiently with the finite element method (FEM). The objective of this project is to investigate its use, for electro-mechanical devices, such as transformers and electric machines. It will be be based on 2D magneto-static analysis of such geometries. Existing tools should be leveraged where applicable (e.g. MATLAB). Once the base FEM solver is verified to function correctly, the focus will shift to sensitivity analysis. This project will require that magneto-static field theory, the FEM, and sensitivity analysis be studied and realised in the noted application context. With a working solver, various application case studies could be undertaken, within the time constrains of the module.

Key modules (Scientific and Engineering Knowledge):
Electrical Drive Systems (324) | Modelling (334)

Design of an electric scooter for urban micro-mobility

Reference	:	SB1
Supervisor	:	Dr Stefan Botha
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Design a complete stand-on electric micro-mobility scooter using mostly off-the-shelf drivetrain components. The full system architecture, including motor, controller, battery, braking, and frame design with CAD models The work should include drivetrain sizing calculations (torque, speed, range, gradeability), battery and BMS considerations.

Key modules (Scientific and Engineering Knowledge):
Electronics (245) | Mechatronics (424) | Modelling (334)

Development of a digital twin for a micro-mobility scooter

Reference	:	SB2
Supervisor	:	Dr Stefan Botha
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Development of a digital twin for a micro-mobility scooter, incorporating drive cycles and detailed electrical and mechanical behaviour. The study must evaluate charging requirements, range, and duty cycles, and the digital twin must be calibrated and experimentally validated using measured ride data. The final objective is to produce an assessment of the viability, cost-effectiveness, and sustainability of micro-mobility scooters for commuter use in urban sub-Saharan Africa

Key modules (Scientific and Engineering Knowledge):
Electrical Drive Systems (324) | Mechatronics (424) | Modelling (334)

Design of a Humanoid Robotic Hand

Reference	:	WDD1
Supervisor	:	Dr William Duckitt
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The "Design of a Humanoid Robotic Hand" project aims to create an advanced mechatronic system that mimics the dexterity and functionality of the human hand. This innovative endeavor involves the integration of mechanical, electrical, and software engineering principles to develop a highly capable and versatile robotic hand. The project will focus on achieving precise and coordinated movements, allowing the robotic hand to perform intricate tasks with a level of agility comparable to the human hand.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Electronics (245) | Mechatronics (424)

Design of pick and place robotic arm

Reference	:	WDD2
Supervisor	:	Dr William Duckitt
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The design of a pick-and-place robotic arm focuses on developing a mechanical and control architecture capable of accurately moving objects between predefined locations in a repeatable and efficient manner. Key considerations include the kinematic structure (such as the number and configuration of joints), actuator selection to balance torque, speed, and cost, as well as end-effector design tailored to the target objects and tasks. The control system must provide precise trajectory planning, real-time feedback, and safe interaction with the work environment, often integrating sensors for position, force, or vision-based guidance. Ultimately, the goal is to achieve high reliability, robustness, and operational throughput while minimizing system complexity and overall manufacturing cost.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Mechatronics (424)

Optimization of the topology of FR4 based interdigitated electrodes for electrochemical biosensing applications

Reference	:	TE1
Supervisor	:	Dr Taskeen Ebrahim
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This topic will include COMSOL based multiphysics simulations of gold interdigitated electrodes on FR4 (PCB) substrate. The student will send designs for fabrication, surface treat the samples in the lab, and take measurements of the electrical response of the sensors represented in the form of Bode Plots. The final step will be to use and develop circuit fitting techniques to obtain the equivalent circuit model of the electrodes.

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Electronics (245) | Experimental Techniques (414)

Embedded Reinforcement Learning

Reference	: HAE1
Supervisor	: Prof Herman Engelbrecht
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Reinforcement learning (RL) is a field of machine learning concerned with agents learning how to take actions in an environment in order to maximise a cumulative reward. Deep Reinforcement Learning (DRL) combines deep learning and reinforcent learning to solve complex tasks.
The purpose of this topic is to implement and train an embedded RL agent to solve scenarios in the Miniworld (https://miniworld.farama.org/) virtual environment (an environment simulator for reinforcement learning and robotics research supported by the Farama Foundation). The topic will first involve implementing and training an DRL agent on desktop computer using images obtained from the virtual environment. The purpose is to train a policy network and obtain the baseline performance on a desktop computer. After succesfully training and evaluating an DRL agent, the next step is to compress the policy network so that it can be deployed onto an embedded microcontroller unit (MCU) such as the Raspberry Pi 2 Pico W. An additional challenge is to use a Raspberry Pi camera to obtain the images from a computer screen.
A solid foundation in programming and mathematics is needed for this project. The student will gain experience in reinforcement learning, deep learning and implementing algorithms from these fields on MCUs using Python, C/C++ and the PyTorch deep learning framework.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (214) | Computer Systems (245) | Design (E) (314)

Cooperative Collision Avoidance for Autonomous Vehicles using Horizontal Manoeuvres

Reference	:	JAAE1
Supervisor	:	Dr Japie Engelbrecht
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
A cooperative collision prediction and avoidance system for multiple autonomous vehicles must be designed and verified in simulation. The collision avoidance must be performed using horizontal manoeuvres only. The vehicles are assumed to have a minimum forward speed, and a maximum turn rate. It is assumed that all vehicles broadcast their current position, velocity and intent to one another. The vehicles must predict imminent collisions and then cooperatively re-plan their paths to avoid the collisions, while minimising both their control effort and their deviation from their original paths, and while obeying their own vehicle motion constraints. If time allows, the system must also be practically demonstrated using physical mobile robots.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Modelling (334)

Automated reversing for a vehicle with a trailer

Reference	:	JAAE2
Supervisor	:	Dr Japie Engelbrecht
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The purpose of the project is to develop, implement, and test a feedback control system which assists with reversing motor vehicles with trailers. The vehicle must be able to reverse into a designated parking area without jack-knifing. An existing mobile robot with a trailer will be available to test the feedback control system. The system must be demonstrated both in simulation and with practical tests.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Modelling (334)

Design and Development of an Electronic Park Brake (EPB) Controller for Emergency Braking Applications

Reference	:	CF1
Supervisor	:	Dr Callen Fisher
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	24774472
Direction	:	Mechatronic
Required elective	:	None

Description:
This project involves designing an Electronic Park Brake (EPB) controller capable of safely performing controlled emergency braking, particularly in vehicles without an Anti-lock Braking System (ABS). Unlike conventional EPBs, which are intended for stationary parking, this system will regulate actuator force while the vehicle is in motion to prevent wheel lock-up and maintain stability. The controller will communicate with the vehicle's Controller Area Network (CAN) bus to monitor parameters such as brake activation and speed, ensuring safe operation under real-time conditions. The design will focus on reliability, safety, and adaptability to future automotive technologies, allowing the EPB to function effectively as both a parking and emergency brake system.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Mechatronics (424)

Development of a probe mechanism to determine drilled hole depth

Reference	:	CF2
Supervisor	:	Dr Callen Fisher
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
In underground mining operations, it is useful to know the depth of the drilled holes to estimate the blast result. The student is required to design a mechanical probe that can attach to a robot arm to probe the hole and determine its depth (up to 3m deep). Additionally, the probe must have sensors to determine the angle of the hole. Mechanical design must be compact, light weight, and accurate to 1cm.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Design (E) (314) | Mechatronics (424)

Infrared remote-controlled toy vehicle with countermeasures

Reference	:	CJF1
Supervisor	:	Prof Coenrad Fourie
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Develop a toy vehicle platform with steering mechanism and at least forward and reverse drive options. Use the minimum number of electrical motors, and develop and implement necessary gear systems to allow drive and steering. System must be battery-operated, and use a mix of digital and analogue electronics to receive and decode frequency-modulated infrared (IR) signals to operate the vehicle. The infrared transmission must work in darkness and bright daylight, and distinguish between at least 4 signals to control the toy vehicle. Countermeasures must be implemented to prevent signal jamming (IR) from rendering the toy vehicle inoperable. The vehicle may also deploy jamming techniques to incapacitate other toy vehicles in its theater of operations. A strong grasp of analogue filtering will be required.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (245) | Electronics (315)

Physics-Based Ray-Tracing Engine for Integrated Circuit Visualisation (Delphi Implementation)

Reference	: CJF2
Supervisor	: Prof Coenrad Fourie
Can be completed in first semester	: No
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Develop a compact but advanced ray-tracing engine in Delphi to render realistic 3D images of integrated circuit (IC) structures. Traditional IC visualisation tools rely on 2D layout or basic extrusions, which makes it difficult to illustrate the true structure and material interactions in modern devices. A physics-based rendering approach can provide highly realistic images by simulating the behaviour of light as it reflects and refracts through the various metal and dielectric layers found in semiconductor manufacturing.
For this project, design and implement the core components of a ray tracer, including camera models, ray–geometry intersection, shading, and acceleration structures to improve performance. The application must handle physically-motivated material models, with particular focus on: • Dielectric refraction using Snell's law for materials such as passivation layers and silicon oxide / silicon dioxide isolation layers. • Metallic reflectivity, incorporating Fresnel effects and surface roughness to model gold, niobium and aluminium thin-films, and silver/gold interconnects or bond pads. • Dual-surface interactions, such as clearcoat or thin water films over a substrate, requiring reflection and refraction at multiple interfaces.
For higher marks, the Delphi-based engine should compile for Windows, macos and Linux.
A Delphi-based GUI will enable users to construct and render simplified IC structures, define material properties, adjust lighting and camera settings, and export images.
This project is suited for a mechatronics student with strong programming interest, as it combines numerical methods, physics-based modelling, computer graphics, and practical software development. It offers a valuable opportunity to explore modern rendering techniques in an engineering context relevant to microelectronics visualisation.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (214) | Computer Systems (245)

Design of a Synchronous Condenser using Novel Wound Rotor Synchronous Generator for Network Stability

Reference	: KSG1
Supervisor	: Dr Karen Garner
Can be completed in first semester	: Yes
Students allowed for this topic	: 2
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Renewable sources of energy are proliferating in many of the world's distributed generation networks. They bring with them both a novel solution to the ubiquitous energy demand but their reliance on converters presents novel problems in the form of their integration with the existing alternating current networks. A synchronous generator is designed for use as a synchronous condenser; to contribute synchronous electrical inertia to a simulated energy network susceptible to frequency and voltage deviations. The machine's physical design is digitally modelled, with its electromagnetic and electromechanical performance determined via finite element analysis. The chosen, satisfactory machine is further modelled in a simulated network constructed power systems engineering simulation software.

Key modules (Scientific and Engineering Knowledge):
Electrical Drive Systems (324) | Modelling (334)

Evaluation of a Reverberation Chamber using Ray Tracing

Reference	:	JG1
Supervisor	:	Dr Jacki Gilmore
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
A reverberation chamber (RC) is a screened room (Faraday cage) used for electromagnetic investigations. Since an RC is effectively a high-Q cavity resonator, mode stirrers are used to lower the possibility of standing waves developing within the chamber. This project aims to employ ray tracing (as implemented in MATLAB, FEKO or CST) to evaluate the effectiveness of mode stirrers in scattering the EM waves in a reverberation chamber. Results can be verified using the Reverberation Chamber and RF measurement equipment in the microwave laboratory at E&E.

Key modules (Scientific and Engineering Knowledge):

A Cost Effective Touch and Heat Feedback Solution for a Prosthetic Hand

Reference	: LGROOT1
Supervisor	: Mr Lanche Grootboom
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Prosthetics are used as an assisting mechanism for people who have lost limbs for various reasons. They are used to conceal disabilities and provide functionality. While significant progress has been made in prosthetic movement, feedback mechanisms have only recently gained attention from developers, doctors, and engineers. A well known psycological effect of those people who have lost limbs is the phantom hand syndrome, where the individual experiences the feeling of the limb, without it being present. Incorporating feedback into prosthetics offers benefits such as improved everyday use, better control over applied force, and enhanced overall control. However, cost remains a significant challenge. This project aims to design and build a cost-effective Force and Temperature feedback system for a prosthetic hand without compromising feedback quality. The student will be required to conduct research on enhancing feedback affordably, utilizing phantom limb syndrome and optimal placement of FSR and NTC sensors. The system can potentially be tested with test subjects to assess whether they could determine the location of force and temperature applied to the prosthetic hand without visual input. A cost breakdown and comparison/study must also be provided to evaluate the total cost for such a device.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Design (E) (314) | Electronics (245)

Design of an Electrically Excited Synchronous Generator

Reference	:	NG1
Supervisor	:	Dr Nkosinathi Gule
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project focuses on the design and analysis of a rotor for an Electrically Excited Synchronous Generator (EESG). The work involves developing a rotor that efficiently incorporates field windings to generate a controllable excitation field, enabling improved torque production and operational flexibility. Key tasks include the analytical design and optimization of the rotor for minimal losses, and ensuring effective thermal management of the excitation windings. Finite Element Analysis (FEA) will be used to evaluate electromagnetic performance and flux distribution. The outcome of the project will be a robust, high-efficiency EESM rotor design suitable for modern electric generator applications.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Electrical Drive Systems (324) | Vibration and Noise (354)

Development of a High-Power Audio Amplifier with an Automated Thermal Management System

Reference	:	TH1
Supervisor	:	Ms Tessa Hall
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project involves the design and construction of an audio amplifier capable of delivering at least 75 WRMS into an 8 Ω load. The amplifier should feature integrated short-circuit protection for enhanced reliability and safety and a 3-channel graphic equaliser to allow precise control over bass, midrange, and treble frequencies. The thermal management system should monitor the thermal behaviour of the audio system and implement predictive control to cool the system when required. A real-time user interface for the system should be implemented.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Electronics (245) | Electronics (315)

CubeSat Attitude Determination Using Laser Retroreflector via Ground-Based Laser Ranging

Reference	:	HWJ1
Supervisor	:	Prof Willem Jordaan
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project focuses on developing a specialized Laser Retroreflector Array (LRA) to be mounted on a CubeSat, thereby overcoming the size and power constraints of traditional on-board attitude sensing by utilizing precise ground-based Satellite Laser Ranging (SLR) stations; the core innovation involves designing the LRA geometry—optimizing the placement and orientation of individual retroreflector elements—to ensure that the differential time-of-flight and distinct return signal patterns from the ground station can be processed by a computational algorithm to accurately infer the CubeSat's 3-axis attitude, thus providing a passive, highly accurate, and non-cooperative method for attitude determination critical for post-failure analysis or active debris removal missions. The project entails the development of the reflector/receiver system along with a laboratory experiment to demonstrate the working of the algorithms for attitude determination.

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Design (E) (314) | Experimental Techniques (414)

Optical Tracking for High-Altitude Balloon Communication

Reference	:	HWJ2
Supervisor	:	Prof Willem Jordaan
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This engineering project aims to develop an optical system for actively controlling a ground-based tracking antenna to maintain robust communication with a high-altitude balloon (HAB) during the critical initial ascent phase of its mission, when rapid movement and complex atmospheric conditions pose significant challenges to traditional tracking methods. The project involves designing and implementing a vision-based tracker that utilizes real-time image processing of the balloon's visual signature to generate precise pointing commands for the communication antenna's pan-tilt mechanism, thereby ensuring continuous data telemetry and command uplink during the dynamic early stages of flight before stable, GPS-aided tracking can reliably take over. This system will enhance mission success rates by mitigating communication dropouts caused by rapid changes in balloon position and atmospheric obscurants, providing a vital bridge until the HAB reaches its operational altitude.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Data Analytics (Eng) (344) | Design (E) (314)

Correlating brain responses with audio features

Reference	:	HK1
Supervisor	:	Prof Herman Kamper
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project explores the relationship between speech audio features and brain responses using publicly available neural datasets. The candidate will extract pitch and other acoustic features (such as intensity, speech rate, envelope) from recorded speech stimuli and analyse how these features correlate with neural signals such as EEG or MEG. The primary goal is to identify temporal and spatial brain patterns associated with specific speech characteristic variations. The project integrates signal processing (vibration and noise), data analysis and neuroscience concepts to extract meaningful insights from real brain and audio data. The candidate will generate correlation results, visualisations and insights into how the brain processes speech.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Data Analytics (Eng) (344) | Vibration and Noise (354)

Design and evaluation of PM brushless exciter.

Reference	:	MJK1
Supervisor	:	Prof Maarten Kamper
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Brushless exciters play an important role in synchronous electrical machines where they provide power to the rotor windings. In this project we look at the design and evaluation of a 6-kW permanent magnet (PM) brushless exciter for a 200-kW, slow-rotating synchronous machine. The exciter must be analyzed by basic analysis and simulated with a diode rectifier as load using MOTORCAD finite element software. A small existing exciter must be tested in the laboratory and compared with MOTOR-CAD results.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electrical Drive Systems (324)

Implementation of a fast EMF safety analysis software utility

Reference	:	DLUDIC1
Supervisor	:	Dr Danie Ludick
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
EMF safety analysis is becoming very important as the number of mobile base stations (MBS) is expanded (especially with the increased rollout of 5G technology). As part of an international effort to ensure health and safety, groups such as ICNIRP and the FCC have set out clear regulations on non-ionizing radiation. This project will focus on developing a fast and accurate electromagnetic field (EMF) analysis tool that can calculate the power density (normalised to the corresponding safety limit) around an arbitrary number of MBS antennas. The software utility, written in an object-oriented language such as C++, will receive as input the radiation patterns of the MBS antennas and then, using suitable formulas, calculate the field strength (and corresponding power densities) on a defined 2D planar grid. Visualization of this field distribution on the 2D grid also forms part of this project. In addition, multi-core processing could be used to ensure that results are generated efficiently. The results will be verified against those generated by commercial computational electromagnetic tools such as FEKO and IXUS.

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Modelling (334)

Automatic Building/Terrain Profile Modelling for RF/EMF Assessments

Reference	:	DLUDIC2
Supervisor	:	Dr Danie Ludick
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Radio Frequency (RF) safety compliance assessments require simulating the antennas installed at a specific site (e.g., rooftop, tower, pole, etc.), and accounting for their transmitted power and electrical characteristics (such as antenna patterns and electrical tilt configurations). One of the most time-consuming steps in this process is creating a 3D model of the building, including the terrain profile and accurately placing the antennas.
The goal of this project is to streamline this step by automatically extracting building geometry (and nearby structures) from web-based data sources such as OpenStreetMap, the Google Earth SDK, and similar APIs. The project will involve developing a software utility that takes a GPS coordinate as input and then retrieves available building and terrain data for that location. Once the site's antennas are specified, the system will support performing an RF safety assessment using the imported spatial data.

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Modelling (334)

Yield optimization of microwave structures

Reference	:	PM1
Supervisor	:	Prof Petrie Meyer
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Microwave devices are typically machined or etched, and suffer from dimensional errors which affect the electrical performance. In this project, a machine learning numerical model must be used in an optimization loop to optimize structures for yield and not only optimal nominal performance. The project is very mathematical, but will require only MATLAB programming.

Key modules (Scientific and Engineering Knowledge):
Data Analytics (Eng) (344) | Modelling (334)

A study of methods for passivity enhancement in grid-connected voltage source inverters.

Reference	:	FM1
Supervisor	:	Dr Fred Mwaniki
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Inverter control design based on passivity properties ensure that the interconnected converter-grid system is robust towards grid variation and stability uncertainties. Passivity enhancement involves ensuring that the grid and the converter are passive. The converter, however, is an active device and hence non-passive. This project aims at identifying the sources on non-passivity and investigating methods of enhancing passivity in grid-connected inverters to ensure converter-grid stability.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Electronics (315) | Modelling (334)

Light sabre

Reference	:	TRN1
Supervisor	:	Prof Thomas Niesler
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Design, build and demonstrate a prototype light sabre, using an addressable LED strip to generate the light effects of the blade and an acoustic transducer to generate the accompanying sound effects. A triaxial accelerometer or other inertial sensor can be used to sense motion and modulate the blade and sound appropriately. The controlling microelectronics and power source should be in the handle of the sabre. There are many ways in which this project can be extended. However, before you choose it, note that although humorous, it is also much more difficult than it sounds. You must have enjoyed and done well at electronic design, and be comfortable both designing and programming micro-controller based systems.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (214) | Computer Systems (245) | Design (E) (314)

Design, Build, and Test an Automated Precision Powder Dispenser

Reference	: AJR1
Supervisor	: Dr Arnold Rix
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:

Accurate and repeatable powder dispensing systems are essential in applications ranging from laboratory sample preparation and additive manufacturing to reloading and precision dosing of granular materials. Recent commercial devices—such as the V4 AutoTrickler and the SuperTrickler—demonstrate that achieving high accuracy at high dispensing speeds requires an intelligent integration of mechanical design, material flow physics, sensors, and control algorithms. This project challenges a final-year mechatronic student to design, prototype, and experimentally validate an automated precision powder dispenser that matches or exceeds the performance characteristics of leading devices in this class.

The system must be capable of dispensing small quantities of fine powder (e.g., grains, metallic powders, or similarly sized particulates) to a target mass with high precision and repeatability. Students must investigate suitable mechanical feed mechanisms—such as trickle tubes, vibration-assisted feed, auger-based drives, or dual-stage coarse-fine dispensing—and determine how geometry, material selection, and actuation influence flow characteristics.

A key component of the project is the development of an intelligent control strategy. This may include predictive algorithms, adaptive control, or machine-learning-assisted tuning to optimise dispensing speed without overshoot, even as powder properties vary. Students must design the embedded electronics, microcontroller interface, and software to handle sensor data acquisition, filtering, calibration, and actuation control. Safety, robustness, and ease of calibration must be built into the system from the outset.

The prototype must be fully tested and evaluated. Students are expected to characterise dispensing accuracy, response time, repeatability, long-term stability, and sensitivity to environmental factors such as vibration, humidity, and powder type. A full experimental plan should be designed and executed, with results compared to benchmark performance from leading commercial systems where possible.

Deliverables include a functional prototype, CAD drawings, an electronics and software design package, a control-system analysis, a comprehensive test report, and a

user-focused operations manual. The final product should demonstrate the student's ability to combine mechanical, electronic, and control engineering principles in a cohesive, high-performance mechatronic system.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Electronics (315)

Design, Build, and Test an Electromagnetic Force Compensation Precision Scale

Reference	: AJR2
Supervisor	: Dr Arnold Rix
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Electromagnetic Force Compensation (EMFC) scales represent the gold standard in precision mass measurement, offering microgram-level resolution through active balancing of gravitational force with an electromagnetically generated counterforce. These systems form the backbone of laboratories, calibration facilities, and high-end industrial instrumentation. This project challenges a final-year mechatronic student to design, prototype, and validate a fully functional EMFC scale that demonstrates a deep integration of mechanical design, sensing, actuation, and advanced control systems.
The core principle of EMFC involves suspending or supporting a weighing pan via a precision flexure mechanism and maintaining it in a fixed position through feedback-controlled electromagnetic force. When a mass is placed on the pan, the resulting displacement is detected by a high-resolution position sensor—typically an optical sensor, Hall-effect sensor, or capacitive displacement sensor. A coil–magnet actuator then generates a counteracting electromagnetic force to restore the pan to its nominal position. The current required to achieve this equilibrium is directly proportional to the applied mass.
Students will be required to design the mechanical suspension system using flexures or a low-friction parallel-guidance mechanism to ensure minimal hysteresis and drift. The design must isolate the weighing element from environmental vibrations while remaining compact and manufacturable. The electromagnetic actuator must be optimally dimensioned, with careful attention to magnetic field uniformity, coil construction, and thermal effects that influence long-term stability.
A crucial part of the project is the development of a fast, stable closed-loop control system. Students must implement precise sensor readout, signal conditioning, filtering, and a current-control loop capable of maintaining sub-micrometre positional stability. Considerations such as noise reduction, electromagnetic interference, temperature compensation, and linearity must be addressed through both hardware and software design.
The prototype must be thoroughly tested under controlled conditions. Students

will characterise sensitivity, linearity, repeatability, drift, bandwidth, step response, and the system's noise floor. Calibration procedures—potentially using standard masses—should be developed, documented, and validated.
Expected deliverables include a fully functioning EMFC scale prototype, detailed CAD models, electromagnetic simulations, control-system design documentation, embedded software, a calibration methodology, and a complete experimental performance evaluation. The final system should demonstrate the student's ability to design and integrate high-precision electromechanical subsystems and illustrate a professional understanding of metrology-grade instrumentation.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Electronics (315)

Design, Build, and Test a Smart Pulse-Driven Measurement and Control System with Integrated Mobile Application

Reference	: AJR3
Supervisor	: Dr Arnold Rix
Can be completed in first semester	: Yes
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Modern portable measurement and control devices increasingly rely on seamless integration between embedded hardware and smartphone-based user interfaces. The SG Pulse Pro—a compact, pulse-driven instrument used for real-time sensing, triggering, and data recording—exemplifies this new class of smart tools that combine high-speed signal processing with intuitive mobile software. This project challenges a final-year mechatronics student (or team) to design, prototype, and validate an innovative pulse-based sensing and control device paired with a dedicated smartphone application for configuration, data visualisation, and advanced functionality.
The hardware component should generate, measure, and process electrical or mechanical pulses depending on the application scenario—such as timing signals, vibration pulses, event triggers, switch responses, or precision counting events. Students must design the embedded electronics, selection of microcontroller/SoC, high-speed signal conditioning, and appropriate sensor interfaces. The device must support wireless communication (Bluetooth Low Energy is recommended) to connect seamlessly to a mobile application. Attention must be paid to minimising noise, ensuring repeatable pulse detection, optimising battery life, and delivering robust real-time responsiveness.
The smartphone application forms a critical part of this project. Students will develop a custom Android app capable of device discovery, secure pairing, configuration management, real-time data acquisition, and user-friendly visualisation of pulse characteristics. Features may include adjustable thresholds, pulse-rate analysis, event logging, waveform preview, firmware update tools, and cloud-based data export if desired. A clean, intuitive user interface—built using modern mobile development frameworks—must complement the hardware's capabilities and prioritise usability under field conditions.
Control algorithms and signal-processing techniques should be implemented on the embedded system, the mobile app, or a combination of both. Students must justify

where each computational layer resides, balancing responsiveness, power consumption, and data throughput. The system must be rigorously tested to characterise accuracy, latency, noise immunity, wireless reliability, and user experience.
Deliverables include a working hardware prototype, circuit schematics, enclosure design, firmware, mobile application code, a communication protocol, and a complete evaluation report. The final outcome should demonstrate full end-to-end integration between a precision pulse-processing instrument and a modern smartphone platform, illustrating the student's capability to design high-performance mechatronic systems enhanced by intelligent digital interfaces.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Electronics (315)

Rally detection and automatic video capture for spikeball

Reference	:	JCS1
Supervisor	:	Dr JC Schoeman
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Spikeball (or roundnet) is an emerging sport that is mostly played at the amateur level. Much like volleyball, two teams of two players each have up to three touches to save, set and spike a round ball. The key differences are (a) that the ball is much smaller, (b) that the ball is bounced off of the net instead of hit over it, and (c) that there are no boundary lines or team-specific sides after the serve.
Video recordings of sport (especially at professional level) is very common, where some benefits include live broadcasting, making refereeing decisions, post-match reviewing for coaching staff, and simply making entertaining videos, to name a few. However, most of these solutions are very expensive, both due to the required hardware as well as the need to store large amounts of video data (and manually filter through it afterwards).
The aim of this project is to developed a low-cost system that can automatically detect spikeball rallies and record them when prompted by the user. More specifically, a smartphone app and/or Raspberry Pi setup needs to be developed that constantly stores (and overwrites) the last rally in a video buffer, and only saves it to permanent memory when the user indicates this desire, either by pressing a button (on the phone or a watch), giving a voice command or indicating otherwise using for example a hand signal. This is quite a challenge, since rallies are not always the same length and the configuration of the players can vary a lot, potentially even blocking certain camera angles. The solution will rely quite heavily on image processing and app/software development, as well as other peripherals as necessary.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (214) | Data Analytics (Eng) (344) | Design (E) (314)

Autonomous racing on unseen tracks using reinforcement learning

Reference	:	JCS2
Supervisor	:	Dr JC Schoeman
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
RoboRacer is an international community performing autonomous systems research through building and racing of 1:10 scale vehicles. The common goal is to race around known tracks as fast as possible – usually first in simulation and then in the real world. This provides a platform for addressing many research and systems engineering questions and solving both theoretical and practical problems. The current approaches to autonomous racing ranges from more conventional control approaches (e.g., model predictive control) to data-driven solutions involving machine learning (e.g., reinforcement learning and imitation learning).
Almost exclusively, however, current research focuses on known tracks, where the robot can either use an exact representation to plan its trajectory or can repeatedly interact with the track in simulation. This does not provide a clear and automatic way to extend these frameworks to settings where the vehicle can race on a track it has not seen before. In contrast, this would be quite natural for an expert human driver. This project therefore aims to extend reinforcement learning algorithms to RoboRacer on unknown tracks. This will first be done in simulation before applying the solution to a physical vehicle.
Reinforcement Learning (RL) is a subfield of machine learning where an agent learns to make optimal decisions through trial-and-error interactions with an environment to maximize its total cumulative reward. Unlike supervised learning, it does not rely on labeled data but gathers its own experience in a dynamic feedback loop. RL has seen growing popularity in recent years; successful applications include agents capable of beating grandmaster-level human players at the games of Chess or Go. This project therefore provides an opportunity to learn about state-of-the-art techniques at the intersection of machine learning and robotics. Although the scope might initially seem large, this project builds on the successful work of a recent MEng graduate, which provides a solid foundation.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Data Analytics (Eng) (344)

Develop an automated waveguide VSWR tuner

Reference	:	WES1
Supervisor	:	Dr Werner Steyn
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Waveguides are used as high frequency transmission lines in high power applications. In these systems maximum power transfer to a complex load is crucial. The Voltage Standing Wave Ratio (VSWR), which is measured with a Vector Network Analyzer (VNA) in the lab, is an indication of the magnitude of the signal that is reflected from the load. The student will design a system that reads the VSWR from the VNA and automatically adjusts tunable elements in the waveguide to obtain minimum VSWR and maximum power transfer to a load.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Electrical Drive Systems (324)

Develop a roll over azimuth positioner system for high frequency antenna measurements.

Reference	:	WES2
Supervisor	:	Dr Werner Steyn
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Many types of antenna measurements require roll over azimuth positioner systems. At high frequencies the antennas are relatively small and light-weight, but the accuracy and precision of the positioner becomes very important. It is also important to use plastic components as far as possible. The student will develop such a positioner with a matlab or python GUI for control.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Electrical Drive Systems (324)

A Lithium-Ion cell test chamber

Reference	:	JMS1
Supervisor	:	Dr Johann Strauss
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project involves the design, construction and evaluation of a lithium-ion cell test chamber to perform aging tests on these cells with specific focus on the temperature control in the chamber. The project will first involve evaluating an existing chamber to determine its effectiveness and identify deficiencies. The next step will be to improve/redesign the heating and cooling system of the chamber, as well as designing the electronic controls to control the temperature. This project will require basic cooling and heating system design (requiring knowledge of heat transfer and typical cooling cycles), as well as analog and digital electronic system design to control the temperature (including microcontroller firmware design).

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Electronics (315) | Mechatronics (424)

Automated feeding system for individual animals in a feedlot

Reference	:	JMS2
Supervisor	:	Dr Johann Strauss
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The animal science department at Agricultural Sciences has the need to control the feeding of individual animals as part of their research projects. To be able to do this, the animals must be individually identified when they go to the feeding troughs and the feed that the specific animal gets must be automatically provided according to a predetermined schedule. For this project, it will be necessary to identify animals using RFID tags when they come to a specific feeding trough and then to automatically provide the necessary feed to the animal using weighers and/or conveyors and/or turnouts. So, the design will involve analog and digital electronic design and microprocessor software design, as well as basic mechanical design of the necessary systems (including electric motors) to provide the feed.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Electrical Drive Systems (324) | Mechatronics (424)

Scale system for automated feeding of individual animals in a feedlot

Reference	:	JMS3
Supervisor	:	Dr Johann Strauss
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The Animal Sciences department at Agricultural Sciences has the need to control the feeding of individual animals as part of their research projects. To do this, they need, among other things, to accurately weigh the feed provided to an animal. One method of doing this is to weigh the decrease in weight of the central silo as the feed is removed.
For this project, it will be necessary to design a scale using load cells to accurately determine the weight of the silo and to design and build a motorized automatic feed dispenser. So, the design will involve analog and digital electronic design and microprocessor software design, as well as basic mechanical design of the dispenser to provide the feed.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Electrical Drive Systems (324) | Mechatronics (424)

Test Rig for Developing Thermal Models of High Voltage Overhead Power Lines or Medium Voltage Cables

Reference	:	AGS1
Supervisor	:	Dr Andrew Swanson
Can be completed in first semester	:	Yes
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Development of a test rig is for testing of high voltage conductors and/or medium voltage cables. The design will involve development of a high current power supply for the test rig using a phase angle controller and a PLC, and should allow for temperature cycling. Acquisition and logging of the current and temperature is required. The development of thermal models of the conductors and/or cables can be developed based on the results.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Heat Transfer A (414)

Partial Discharge Data Acquisition System

Reference	:	AGS2
Supervisor	:	Dr Andrew Swanson
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Partial discharge occurs in ageing high voltage systems. Partial discharge is a very small current with a short duration (ns-us) that needs to be measured with an appropriately sensitive tool. The student is to design a data acquisition system for partial discharge that accounts for the measurement, amplification, noise and filtering, and corresponding phase angle of the signal. The system should display the partial discharge patterns on an appropriate platform such as Matlab. (Note that a significant portion of the project does not require the use of high voltage equipment as it involves small signals that can be recreated in an electronics lab - but the equipment is available when required).

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Electronics (315) | Experimental Techniques (414)

Development of a Stereoscopic Calibration and Surface Reconstruction System for 3D Deformation Analysis

Reference	:	RPT1
Supervisor	:	Dr Rensu Theart
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project aims to develop a stereoscopic imaging system capable of precise 3D surface reconstruction for deformation analysis. Using two synchronised cameras and a well-characterised calibration plate, the system will establish accurate stereo geometry to recover real-world depth information. Speckle-patterned samples will be imaged to enable correspondence tracking between views, allowing the extraction of a 3D surface mesh that represents the object's geometry under load. The project will focus on stereo calibration, rectification, depth mapping, and mesh generation, providing a foundation for future Digital Image Correlation (DIC) and strain analysis. The resulting system will offer a low-cost, open-source tool for high-precision, non-contact measurement in experimental mechanics.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Modelling (334) | Experimental Techniques (414)

Subsea buoyancy drifter

Reference	:	JV1
Supervisor	:	A/Prof Jaco Versfeld
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
There is a need for a subsea drifting buoy, that can adjusts its depth by changing its buoyancy. The project entails the development of the mechanical as well as software control of such a system. The system must be small (less than 1.5 meters) and should be able to be deployed by one or two persons.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Design (E) (314)

Drifting buoy design for minimal low frequency noise.

Reference	:	JV2
Supervisor	:	A/Prof Jaco Versfeld
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
We have several applications where we want to deploy drifting buoys in the ocean for short periods (3 - 6 hours) and record underwater sounds - mainly for our research on whale and dolphin vocalisations. During deployment, these buoys record underwater sounds using hydrophones. The aim of this project is to develop these buoys so that they have minimal low frequency noise (noise in the bands 10 Hz - 100 Hz). The buoys should be light weight, and be able to fit on a small boat. The design should be based on spar buoys, which counter the effect of waves. If time allows, the project can be expanded, where three or more buoys are deployed, and work together as an array. The aim here would be to determine the location of underwater sounds, using localisation techniques or Direction of Arrival techniques.
We have a number of Audiomoths - (acoustic recorders), as well as hydrophones that can be used in this project. The audiomoths can be modified to use GPS timing information, in order to do synchronised recording. LoRa devices should be integrated into the buoys in order to help with retrieval of the buoys.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (214) | Data Analytics (Eng) (344) | Design (E) (314)

Thermal Analysis of an Axial-Flux Permanent-Magnet Machine: Evaluation of Advanced Cooling Techniques for Enhanced Power Density

Reference	: RJW1
Supervisor	: Prof. Rong-Jie Wang
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
Axial-flux permanent-magnet (PM) machines are increasingly adopted in electric vehicles, drones, and other compact drives because of their high torque and power density. However, their flat, compact geometry presents significant challenges for heat removal. Conventional air or water-jacket cooling often limits continuous performance, motivating the development of more effective thermal management solutions. Recent studies have investigated advanced techniques such as direct oil cooling, dielectric-fluid immersion, and heat-pipe-assisted cooling to enhance heat extraction from windings and magnets while maintaining lightweight construction.
The aim of this project is to model and compare the thermal behaviors of a YASA-type axial-flux PM machine under both conventional and advanced cooling methods, in order to identify the configuration that achieves the highest sustainable power density.

Key modules (Scientific and Engineering Knowledge):
Electrical Drive Systems (324) | Thermofluid Dynamics (344)

Enhanced Electrodynamic Levitation with Transverse-Flux Linear Induction Drive

Reference	:	RJW2
Supervisor	:	Prof. Rong-Jie Wang
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Electrodynamic levitation (EDL) systems use electromagnetic forces to provide contactless support and propulsion of a moving body. These systems are increasingly attractive for applications such as rail transportation, precision handling, and automated production lines. Transverse-flux linear induction motors (TF-LIMs) are particularly suitable for EDL because they can generate high lift and thrust forces at low speeds, while offering compact geometry and modular design potential.
Achieving stable and efficient levitation, however, requires careful optimization of electromagnetic design parameters and effective control of the dynamic interactions between lift, thrust, and drag forces. This project aims to enhance both the electromagnetic design and control strategies of a small-scale TF-LIM-based EDL system to achieve reliable, energy-efficient levitation and smooth bidirectional motion.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Electrical Drive Systems (324) | Electronics (315)

Detection of Energy Inefficiencies in Mobile Cellular Network Base Stations

Reference	:	RW1
Supervisor	:	Prof Riaan Wolhuter
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
This project aims to develop a robust algorithm capable of detecting both sudden and gradual increases in energy consumption at mobile network base stations. Real-world energy meter data is available for approximately 5,000 sites, recorded at 30-minute intervals over a full year. Due to differences in site equipment and configurations, each location has a unique consumption profile. The algorithm must adapt to these variations and identify abnormal consumption trends by analysing historical data patterns.
The project will involve data cleaning and pre-processing, time series analysis, and the application of machine learning techniques. The final solution will be applied at scale and validated on real-world data. Sites flagged by the algorithm will be further investigated by the mobile network operator for potential energy inefficiencies.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Data Analytics (Eng) (344) | Design (E) (314)

Reinforcement Learning for Energy-Efficient Temperature Control in Mobile Cellular Network Base Station Containers

Reference	: RW2
Supervisor	: Prof Riaan Wolhuter
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
This project aims to develop a robust algorithm capable of detecting both sudden and gradual increases in energy consumption at mobile network base stations. Real-world energy meter data is available for approximately 5,000 sites, recorded at 30-minute intervals over a full year. Due to differences in site equipment and configurations, each location has a unique consumption profile. The algorithm must adapt to these variations and identify abnormal consumption trends by analysing historical data patterns.
The project will involve data cleaning and pre-processing, time series analysis, and the application of machine learning techniques. The final solution will be applied at scale and validated on real-world data. Sites flagged by the algorithm will be further investigated by the mobile network operator for potential energy inefficiencies.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Data Analytics (Eng) (344) | Mechatronics (424)

Detection of leaf wetness in cultivated crops

Reference	:	RW3
Supervisor	:	Prof Riaan Wolhuter
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
The level of leaf wetness in agriculture is important in terms of plant health and potential disease. From references it appears that IR absorbtion by the plant is a good indication of leaf wetness. The proposal is to design and build a sensor device that would utilise IR spectrum sensing and level measurement, to evaluate the degree of leaf wetness, or the amount of water directly on leaves. The student will have to investigate whether an active IR source should be used, or natural IR radiation would be sufficient. Active testing and evaluation of the completed device will be required.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (245) | Design (E) (314) | Electronics (315)

Design and Implementation of a Motor Controller and Smart Battery Management System for an Electric Skateboard

Reference	:	AADUP1
Supervisor	:	Dr Armand du Plessis
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Electric personal mobility devices such as e-skateboards, e-bikes and scooters rely on sophisticated power-electronic systems to ensure performance, safety, and battery longevity. While hobbyist solutions exist, they typically use off-the-shelf ESCs (Electronic Speed Controllers) and BMS (Battery Management System) units with limited configurability, sub-optimal protection, and no engineering transparency.
This project aims to develop a custom high-power Motor Controller and a Smart Battery Management System for a LiFePO4 battery pack, specifically configured for an electric skateboard. The final system will demonstrate motor control, protection, and battery health estimation, transforming a hobby platform into an academically rigorous engineering design.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Electrical Drive Systems (324) | Electronics (315)

Open-world object tracking in video footage

Reference	:	CVD1
Supervisor	:	Dr Corné van Daalen
Can be completed in first semester	:	Yes
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
In order for an intelligent mobile robot to operate reliably in the real world, it has to be able to detect and track moving objects in its environment. However, most deep-learning-based object tracking approaches use closed-world object detection, which means that they can only detect (and track) a set of predefined object classes (or types); this limits their applicability in general and unusual scenarios. The aim of this project is to detect and track objects in a sequence of images without knowing the types of objects being tracked.
The idea is to first divide each image into a number of smaller regions called superpixels, and then use probabilistic reasoning techniques to group superpixels that belong to the same moving object together, and then to track the moving objects across subsequent images. The probabilistic reasoning techniques will make use of image processing techniques such as optical flow and feature detection to figure out which superpixels are moving, which superpixels belong together, and how superpixels move from one image to the next.
This is a challenging project that would be suitable for a student willing to immerse themselves in image processing, probabilistic reasoning, and computer programming. The open-source computer vision library, OpenCV, is available for Python and C/C++, and an in-house library for probabilistic reasoning is available for C++ on Linux.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Data Analytics (Eng) (344)

Identification System for a Lego-Piece Sorting Machine

Reference	:	CVS1
Supervisor	:	Mrs Chantelle van Staden
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Many households own large collections of mixed Lego pieces that vary widely in colour, size, and geometry, from standard bricks and plates to Technic gears, axles, and minifigure accessories. This project focuses on developing an automated identification system capable of accurately recognising individual Lego pieces as they move along a conveyor belt within a sorting machine.
The aim is to design and implement an identification module that receives a single, isolated Lego piece from the mechanical feed system, captures visual or sensor data in real time, and classifies the piece according to attributes such as colour, shape/-geometry, and size. The module must then output a bin number or category to the sorting system, which physically places the piece into the correct bin.
Because training a recogniser using thousands of real Lego pieces may be impractical, the project may explore alternative approaches, including:
Training with synthetic data generated from digital Lego models, public CAD libraries, or images extracted from standard Lego instruction manuals.
Feature-based classification, such as outline/contour matching, dimension-based clustering, and colour extraction, requires little or no pre-training.
Unsupervised or semi-supervised learning where the system groups visually similar pieces together and "learns as it goes."
Adaptive modes such as: Colour-blind mode (only shape/geometry considered), Shape-insensitive mode (colour-only classification), Full classification mode (colour + geometry + size).
Success criteria include the accuracy and repeatability of identification, robustness to lighting variations and conveyor movement, and the speed at which a piece can be reliably classified before it reaches the sorting stage. The project will also evaluate the performance of different training methods, sensing approaches (camera, structured lighting, or depth sensors), and classification techniques, and select a final design that provides the best trade-off between simplicity, speed, and accuracy for Lego-piece identification in a real-world sorting machine.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Data Analytics (Eng) (344) | Design (E) (314)